
How to Provide Best WiFi Services?
HPE Aruba Networking
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25+ Years of Wi-Fi®



802.11a 802.11b 1999	802.11g 2003	Wi-Fi 4 802.11n 2009	Wi-Fi 5 802.11ac 2013	Wi-Fi 6 802.11ax 2019	Wi-Fi 6E 802.11ax 2021	Wi-Fi 7 802.11be 2024

https://www.wi-fi.org/news-events/newsroom/wi-fi-alliance-celebrates-25-years-of-wi-fi-innovation-and-impact

TYPES OF WI-FI
(WIRELESS STANDARDS)

WI-FI VERSION	STANDARD	MAX SPEED	BANDS
Wi-Fi 1	802.11b	11 Mbps	2.4 GHz
Wi-Fi 2	802.11a	54 Mbps	5 GHz
Wi-Fi 3	802.11g	Up to 600 Mbps	2.4 GHz
Wi-Fi 4	802.11a	Up to 3.5 Gbps	2.4 / 5 GHz
Wi-Fi 5	802.11ac	Up to 9.6 Gbps	2 GHz + 5 GHz
Wi-Fi 6	802.11ax	Up to 9.6 Gbps	2.4 GHz, 5 GHz 6 GHz
Wi-Fi 7	802.11be	Up to 46 Gbps	2,4 GHz, 5 GHz, 6 GHz

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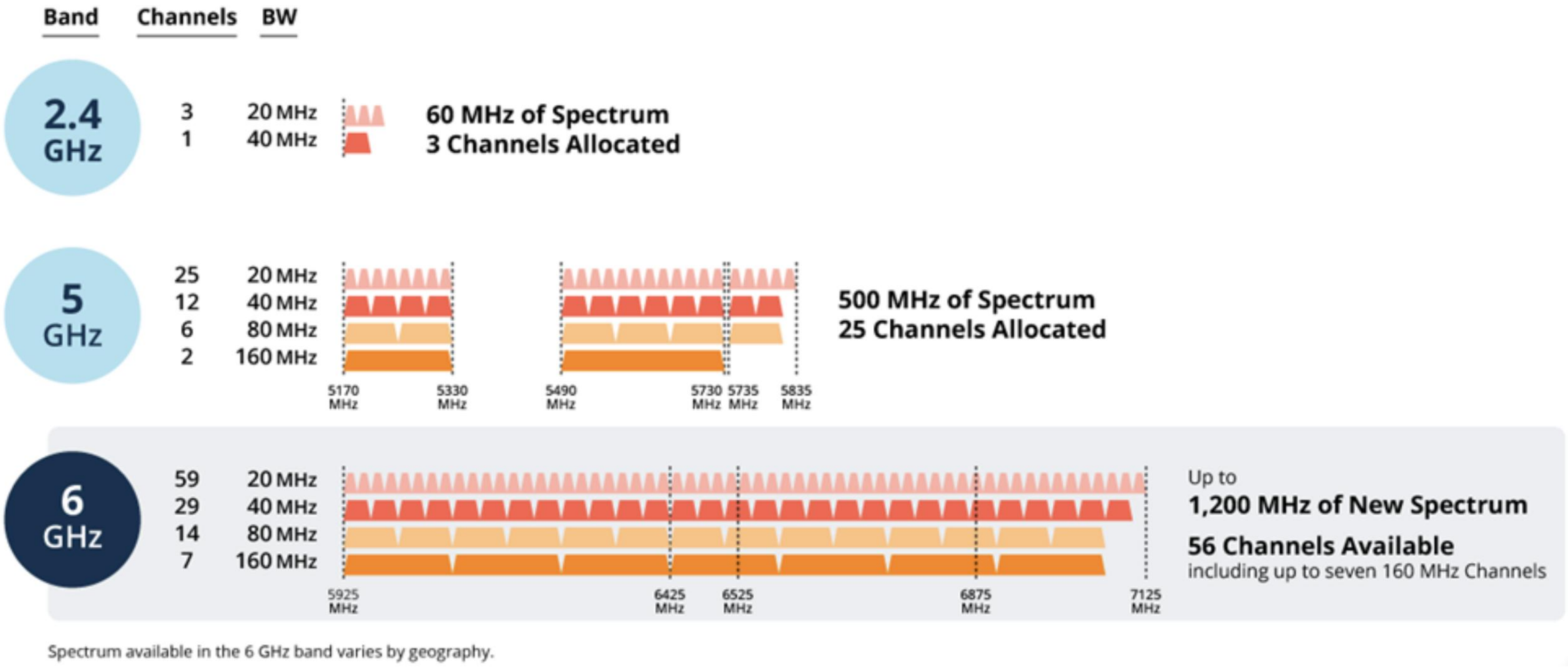
Generational Wi-Fi Naming Conventions

- To help folks identify device generations without understanding the IEEE alphabet soup, the Wi-Fi Alliance created a naming scheme using numbers to identify the various generations.
  - The IEEE organization works on the standard.
  - The Wi-Fi Alliance is responsible for the branding and certification of Wi-Fi products.

IEEE Equivalent	Wi-Fi Alliance Generation
802.11n	Wi-Fi 4
802.11ac	Wi-Fi 5
802.11ax	Wi-Fi 6
802.11ax operating in 6 GHz	Wi-Fi 6E
802.11be	Wi-Fi 7

https://www.wi-fi.org/discover-wi-fi
https://www.wi-fi.org/system/files/Generational_Wi-Fi_User_Guide_202304.pdf

WIFI Spectrum & Channel



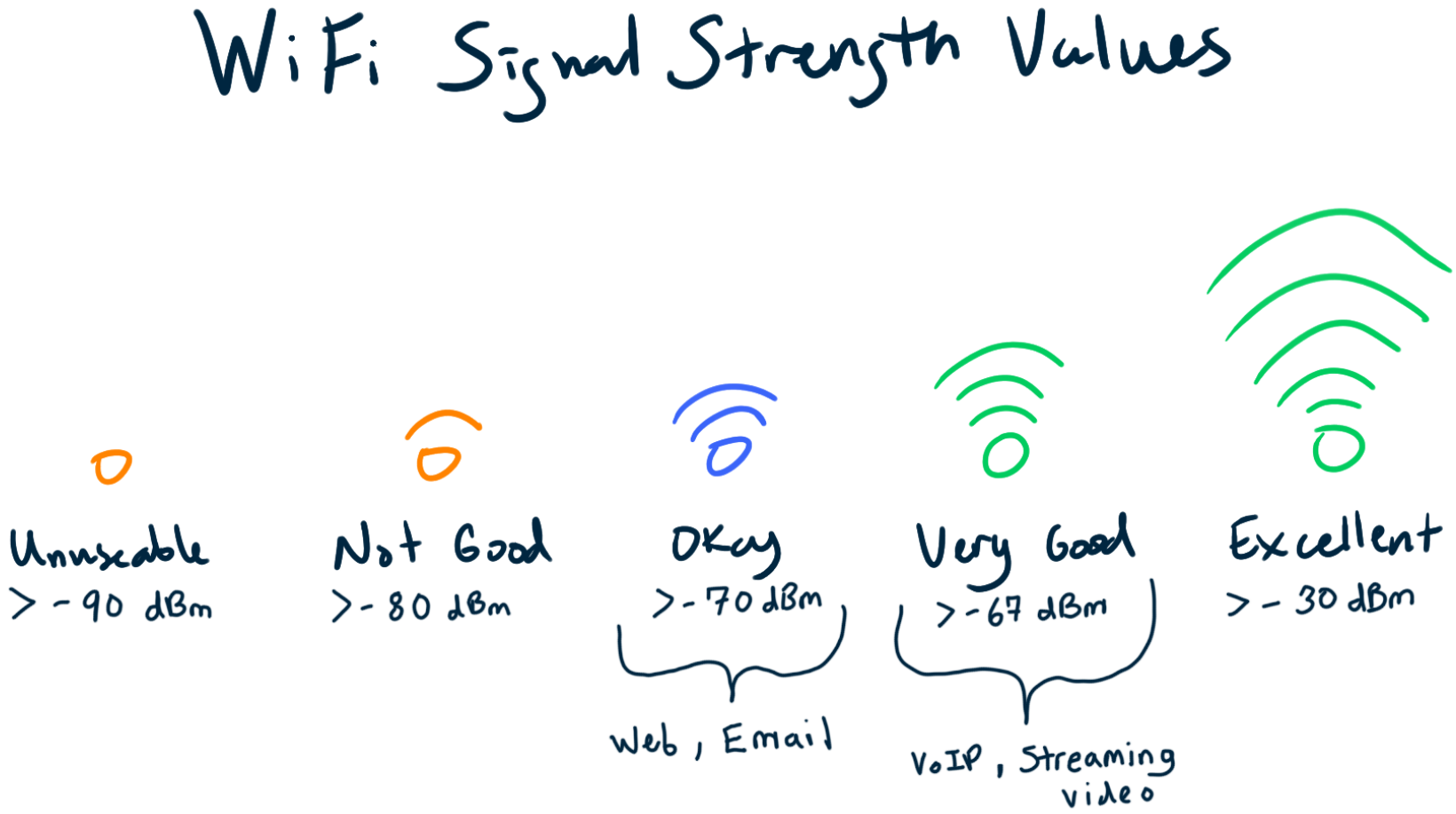
Spectrum available in the 6 GHz band varies by geography.

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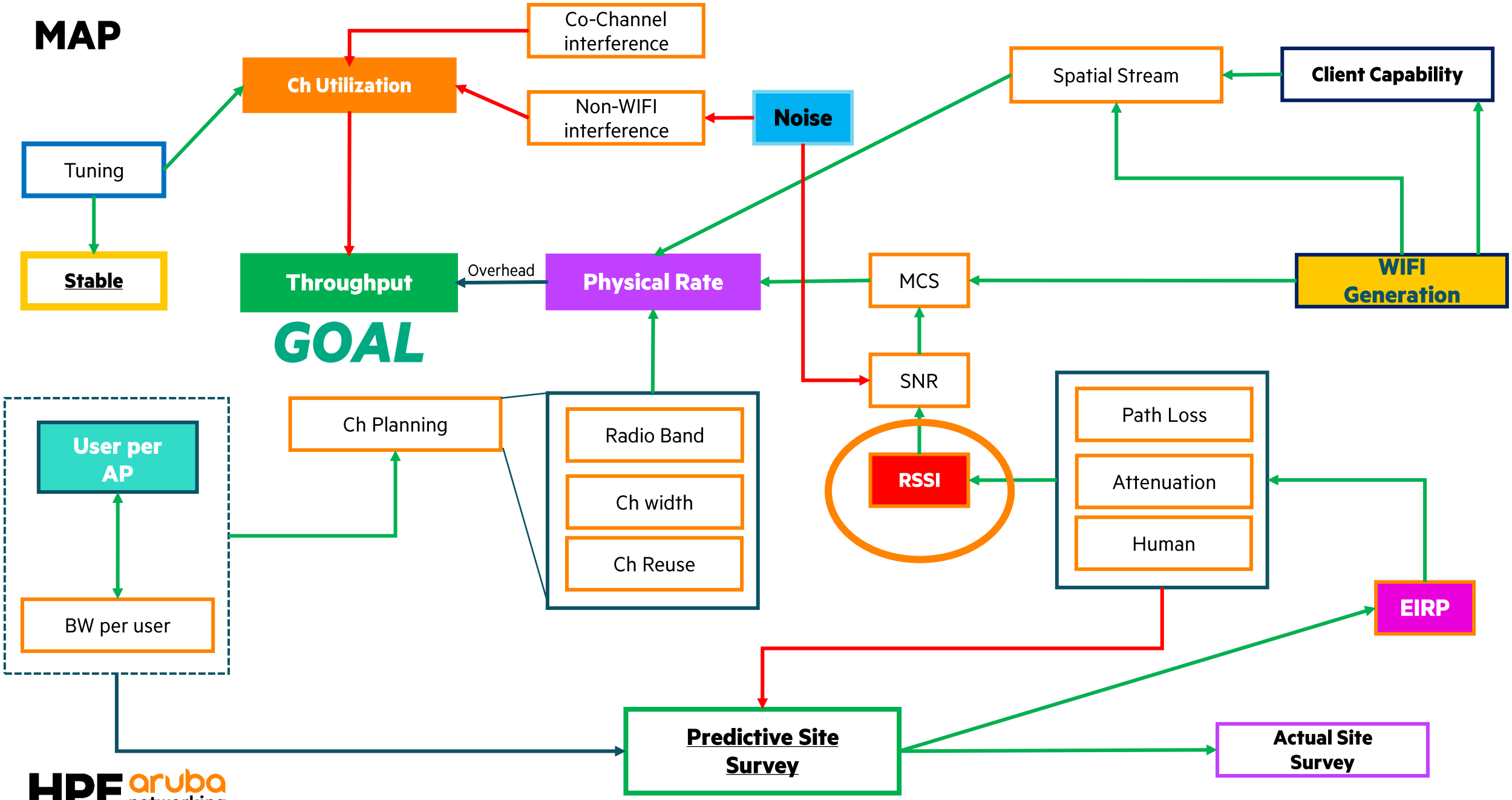
What's Good WIFI ?

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What is Good WIFI ?



Belief vs Truth


MAP
Co-Channel interference
Ch Utilization
Non-WIFI interference
Noise
Spatial Stream
Client Capability
Tuning
Stable
Throughput
Overhead
Physical Rate
MCS
WIFI Generation
GOAL
SNR
Ch Planning
Radio Band
Ch width
Ch Reuse
Path Loss
Attenuation
Human
RSSI
User per AP
BW per user
EIRP
Predictive Site Survey
Actual Site Survey
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เป้าหมาย (GOAL) คือ Throughput (อัตราการรับส่งข้อมูลที่ใช้งานได้จริง, ใน 1 วินาที ผู้ใช้ส่งหรือรับข้อมูลจริงได้กี่บิต)

Wi-Fi Generation ⟶ MCS (Modulation and Coding Scheme) ,

SNR ⟶ MCS MCS ที่ใช้จริงถูกจำกัดด้วย SNR , Wi-Fi Generation มีผลต่อ MCS

EIRP ⟶ Path Loss+ Attenuation + Human ⟶ RSSI (Received Signal Strength Indicator)

Spatial Stream คือ จำนวน Stram ของข้อมูล ที่ใช้ส่งพร้อมกันด้วย MIMO

Physical Rate ≠ Throughput เพราะว่ามี Overhead เช่น MAC address , PHY overhead, ACK ...

Channel Utilization คือ ช่องสัญญาณนั้นถูกใช้งานมากน้อยแค่ไหน

Co-Channel Interference (AP หลายตัวใช้ Channel เดียวกัน) , Non-Wi-Fi Interference (อุปกรณ์อื่นที่ใช้ย่านความถี่เดียวกัน)

User per AP และ BW per User
ด้านซ้ายล่างเป็นเรื่อง Capacity Planning
ดังนั้นการออกแบบ Wi-Fi ไม่ใช่แค่ระดับสัญญาณแรงแค่ไหน แต่ต้องดูด้วยว่า รองรับผู้ใช้กี่คน แต่ละคนต้องการ bandwidth เท่าไร?

Predictive Site Survey ใช้แบบจำลองก่อนติดตั้งจริง → Actual Site Survey เดินสำรวจจริงด้วยเครื่องมือ

Throughput ของ Wi-Fi ไม่ได้ขึ้นกับ Physical Rate เพียงอย่างเดียว แต่เกิดจากความสามารถของ Client + MCS + Spatial Stream + Channel Width + คุณภาพสัญญาณ (RSSI/SNR) + Interference + Channel Utilization + จำนวนผู้ใช้ และการวางแผนช่องสัญญาณ

WIFI Foundation : Signal & Modulation

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LAN Comparison: Wired vs. Wireless

Wired	Wireless
Transmits data on wires (bounded medium)	Transmits date using RF (unbounded medium)
Fixed location (due to cable)	Mobile (within RF area)
Contention based access • CSMA/CD • Can transmit and listen at same time • Only 2 devices on cable (client and switch)	Contention based access • CSMA/CA • Cannot listen while transmitting • Many devices on same channel • Shared medium

CSMA/CD was used in now-obsolete shared-medium Ethernet variants (10BASE5, 10BASE2), and in the early versions of twisted-pair Ethernet, which used repeater hubs. Modern Ethernet networks, built with switches and full-duplex connections, no longer need to use CSMA/CD, because each Ethernet segment, or collision domain, is now isolated. CSMA/CD is still supported for backwards compatibility and for half-duplex connections. The IEEE 802.3 standard, which defines all Ethernet variants, for historical reasons still bore the title "Carrier sense multiple access with collision detection (CSMA/CD) access method and physical layer specifications" until 802.3-2008, which uses new name "IEEE Standard for Ethernet".

HUB vs Switch

Hub	Switch
Broadcast device	Multicast device
Connects devices in the same network	Connects devices in the same network
Only one device can send data at a time	Multiple devices can send data at the same time
Does not store any device information	Stores and uses MAC addresses to transfer data

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CSMA/CA

- Wi-Fi networks use CSMA/CA
  - (CSMA/CA = Carrier Sense Multiple Access with Collision Avoidance)

- Wi-Fi networks must sense if the channel is busy before transmitting (CCA)
  - (CCA = Clear Channel Assessment)

- Wi-Fi is a shared, half-duplex environment
  - Only one device can transmit at a time
  - Devices must wait their turn to transmit
  - More devices = more wait to transmit
  - More wait = slower overall throughput



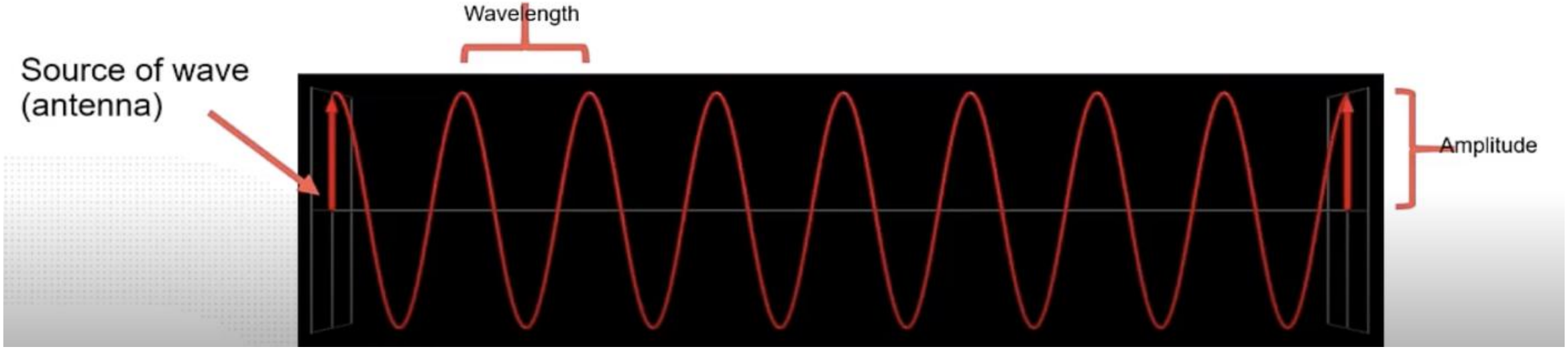
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Transmitting data (ones and zeros)

- All packets and frames are transmitted as bits (ones and zeroes)

- Wired networks send electrical signals across wires
  - Changing electrical voltage is used to represent ones and zeroes

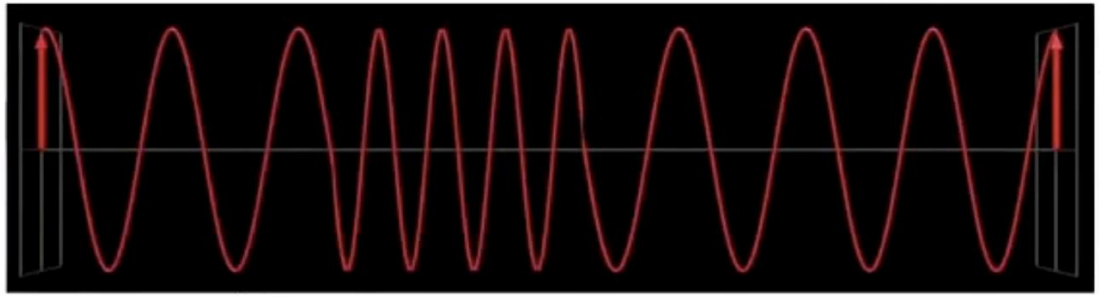
- Wireless networks transmit radio waves
  - Changing properties of RF waves is used to represent ones and zeroes



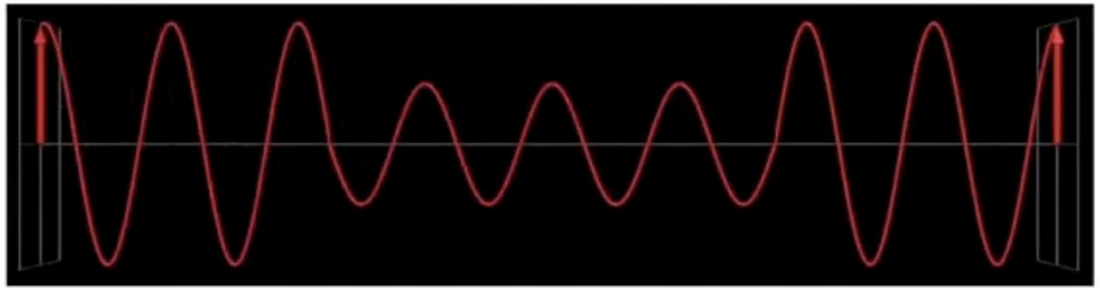
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Data modulation examples

- Frequency Shift Keying (FSK)
  - Frequency is changed to represent ones and zeroes



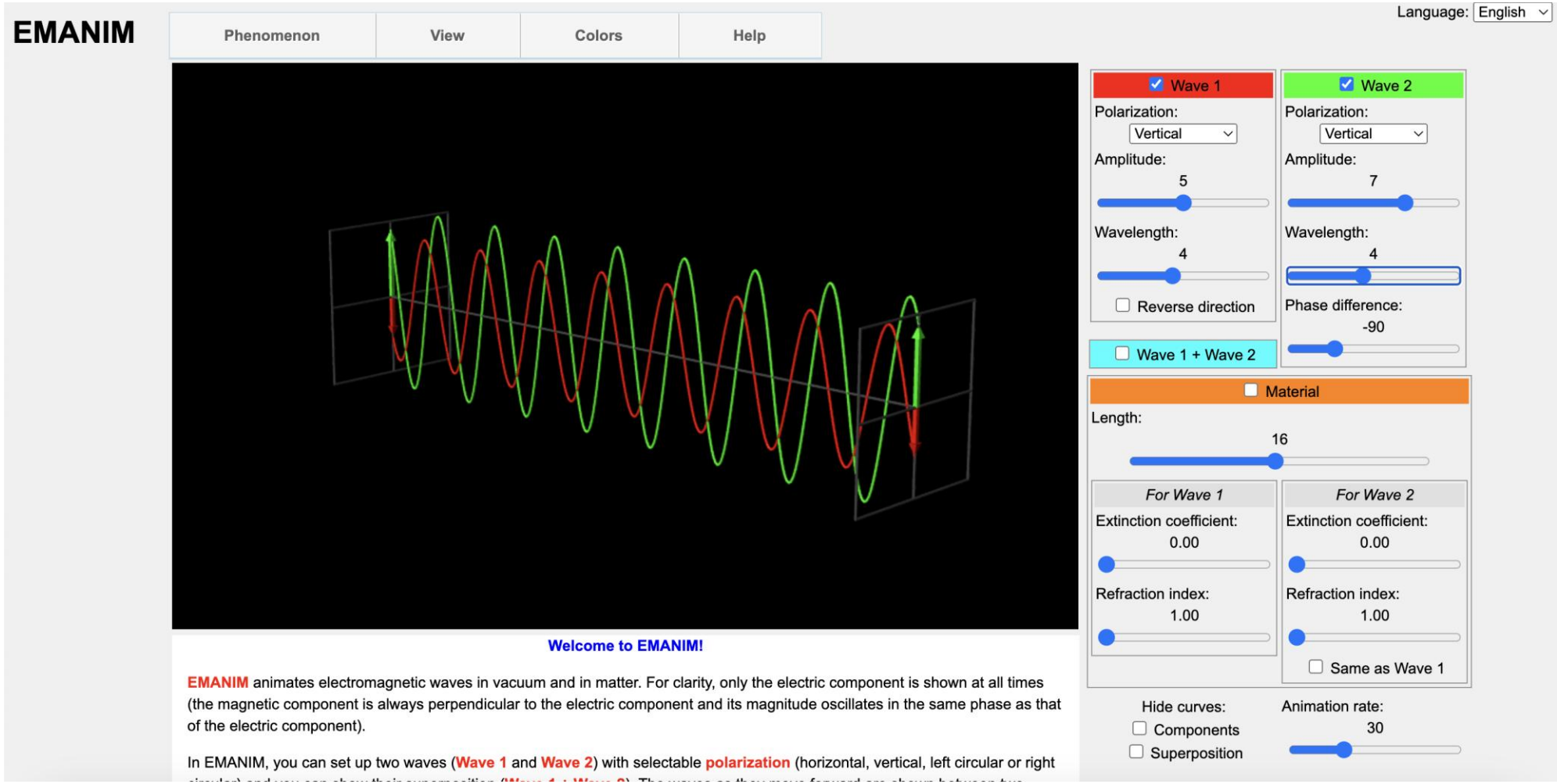
- Amplitude Shift Keying (ASK)
  - Amplitude is changed to represent ones and zeroes



- Phase Shift Keying (PSK)
  - Phase is change to represent ones and zeroes



Sample Wave Form



https://emanim.szialab.org/index.html

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WIFI Modulation Technique

MCS Index				Spatial Stream	Modulation	Coding
HT	VHT	HE	EHT			
0	0	0	0	1	BPSK	1/2
1	1	1	1	1	QPSK	1/2
2	2	2	2	1	QPSK	3/4
3	3	3	3	1	16-QAM	1/2
4	4	4	4	1	16-QAM	3/4
5	5	5	5	1	64-QAM	2/3
6	6	6	6	1	64-QAM	3/4
7	7	7	7	1	64-QAM	5/6
	8	8	8	1	256-QAM	3/4
	9	9	9	1	256-QAM	5/6
		10	10	1	1024-QAM	3/4
		11	11	1	1024-QAM	5/6
			12	1	4096-QAM	3/4
			13	1	4096-QAM	5/6

BPSK : Binary Phase Shift Keying

QPSK : Quadrature Phase Shift Keying

16QAM : 16 Quadrature Amplitude Modulation

64QAM : 64 Quadrature Amplitude Modulation

256QAM : 256 Quadrature Amplitude Modulation

1024QAM : 1024 Quadrature Amplitude Modulation

WIFI4	**HT**
WIFI5	**VHT**
WIFI6	**HE**
WIFI7	**EHT**

High Throughput

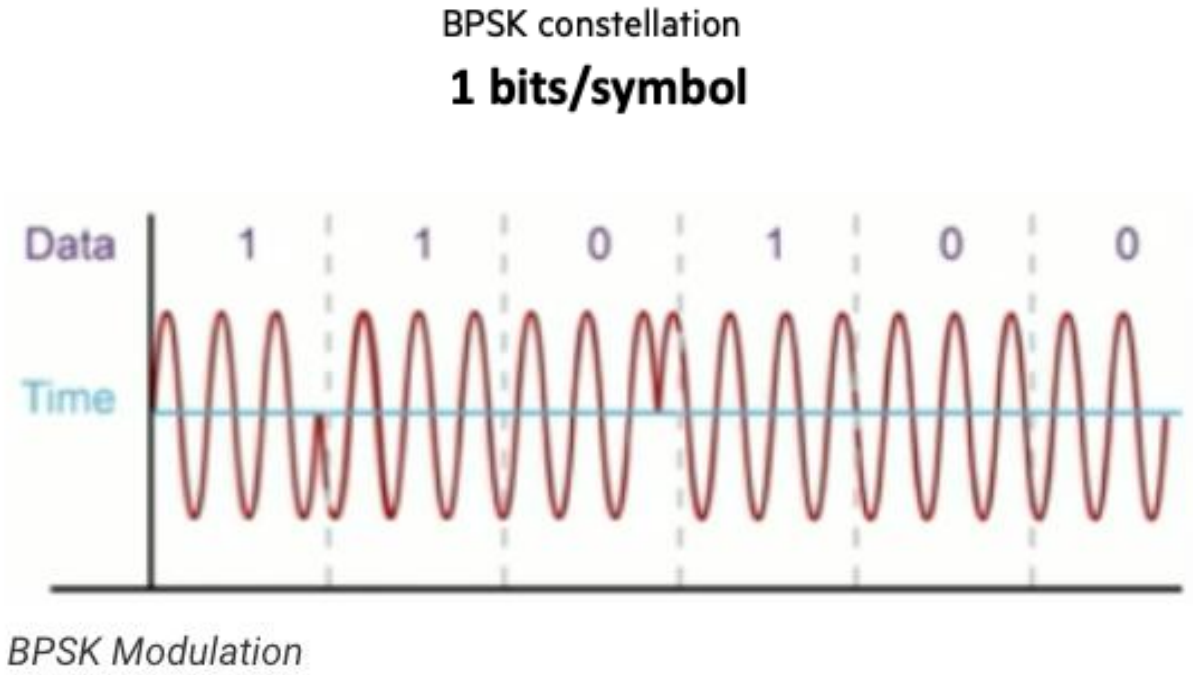
Very High Throughput

High Efficiency

Extremely High Throughput

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BPSK



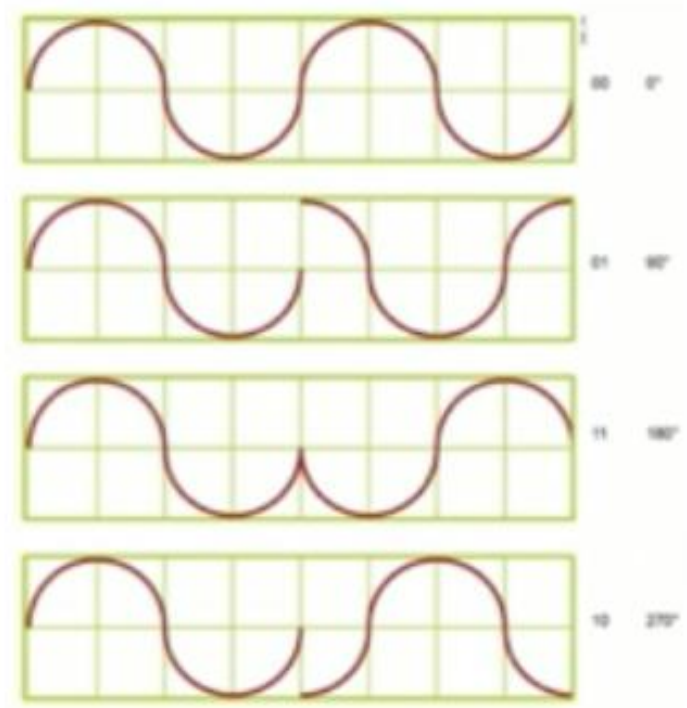
BPSK Modulation

Any RF signal can be decomposed into its in-phase and quadrature components. Hence, we can map them on an IQ constellation plane as shown above.

QPSK

QPSK constellation

2 bits/symbol

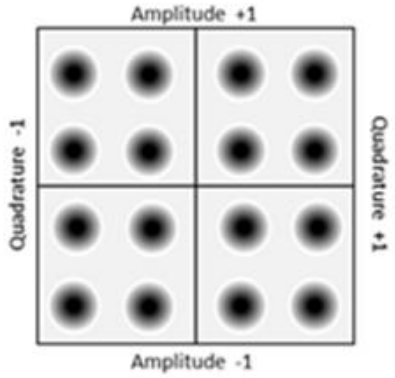


QPSK Modulation

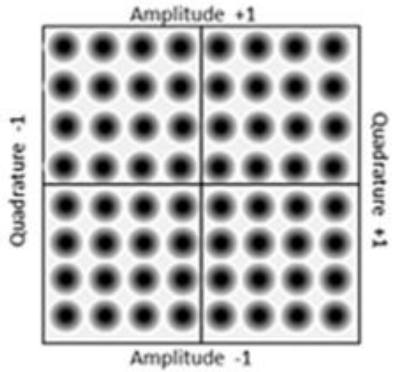
Spec	Encoding	Modulation	Data Rate	Channel Width
802.11	Barker chipping	BPSK	1 Mbps	22 Mhz
802.11	Barker chipping	QPSK	2 Mbps	22 Mhz
802.11b	CCK-16 codes	QPSK	5.5 Mbps	22 Mhz
802.11b	CCK-128 codes	QPSK	11 Mbps	22 Mhz

PSK Rates with Different Encoding

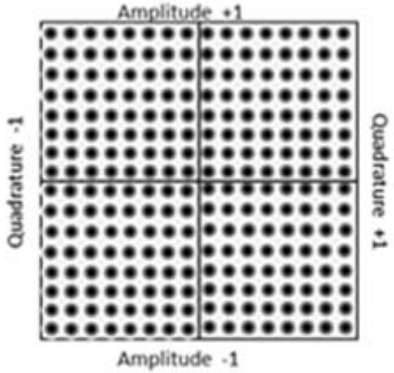
QAM: Quadrature amplitude modulation



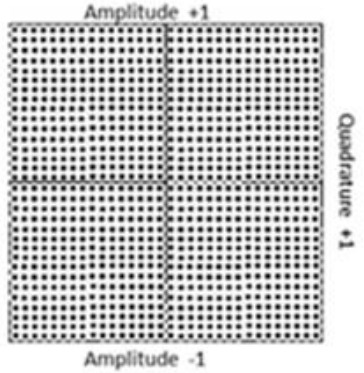
16-QAM constellation



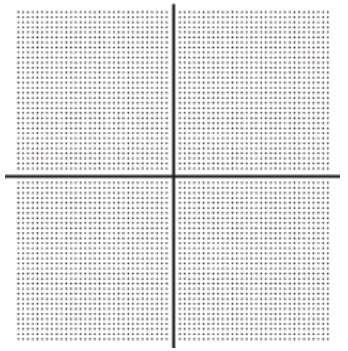
64-QAM constellation

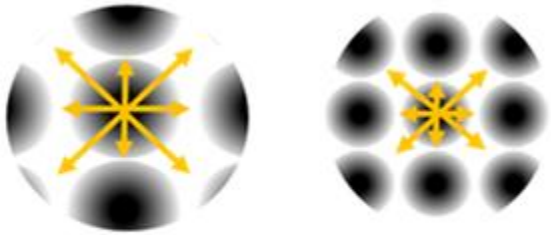


256-QAM constellation



1024-QAM constellation


4096-QAM constellation


Decreased error margin

Bits per tone

BPSK	1-Bit
QPSK	2-Bit
16 QAM	4-Bit
64 QAM	6-Bit
256 QAM	8-Bit
1024 QAM	10-Bit
4096 QAM	12-Bit

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QAM: Quadrature amplitude modulation (16-QAM example)

- Consider a digital data stream

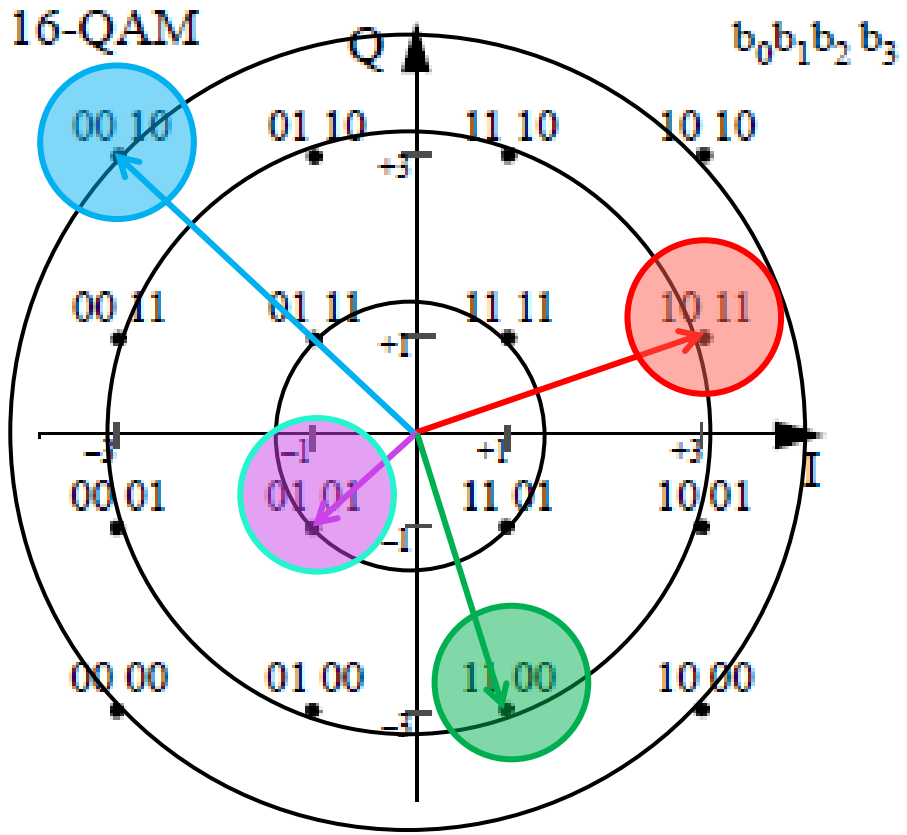
1011001001011100

- Modulating using 16-QAM (2^4 → 4 bits per symbol)

1011001001011100

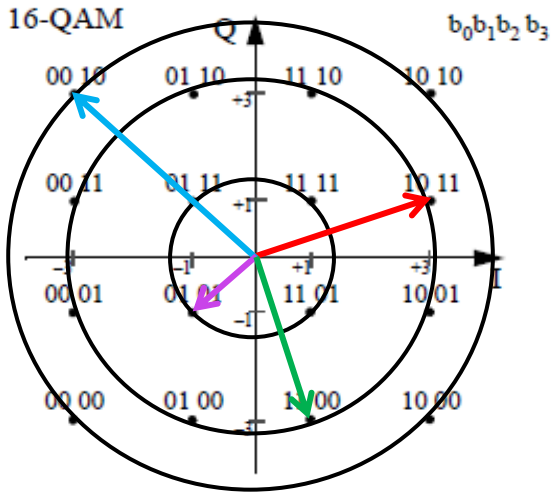
Symbol	Magnitude (V)	Phase
1011	0.74536	22.5°
0010	1.00000	135°
0101	0.33333	225°
1100	0.74536	292.5°

*Example shown here is of only four sub-carriers

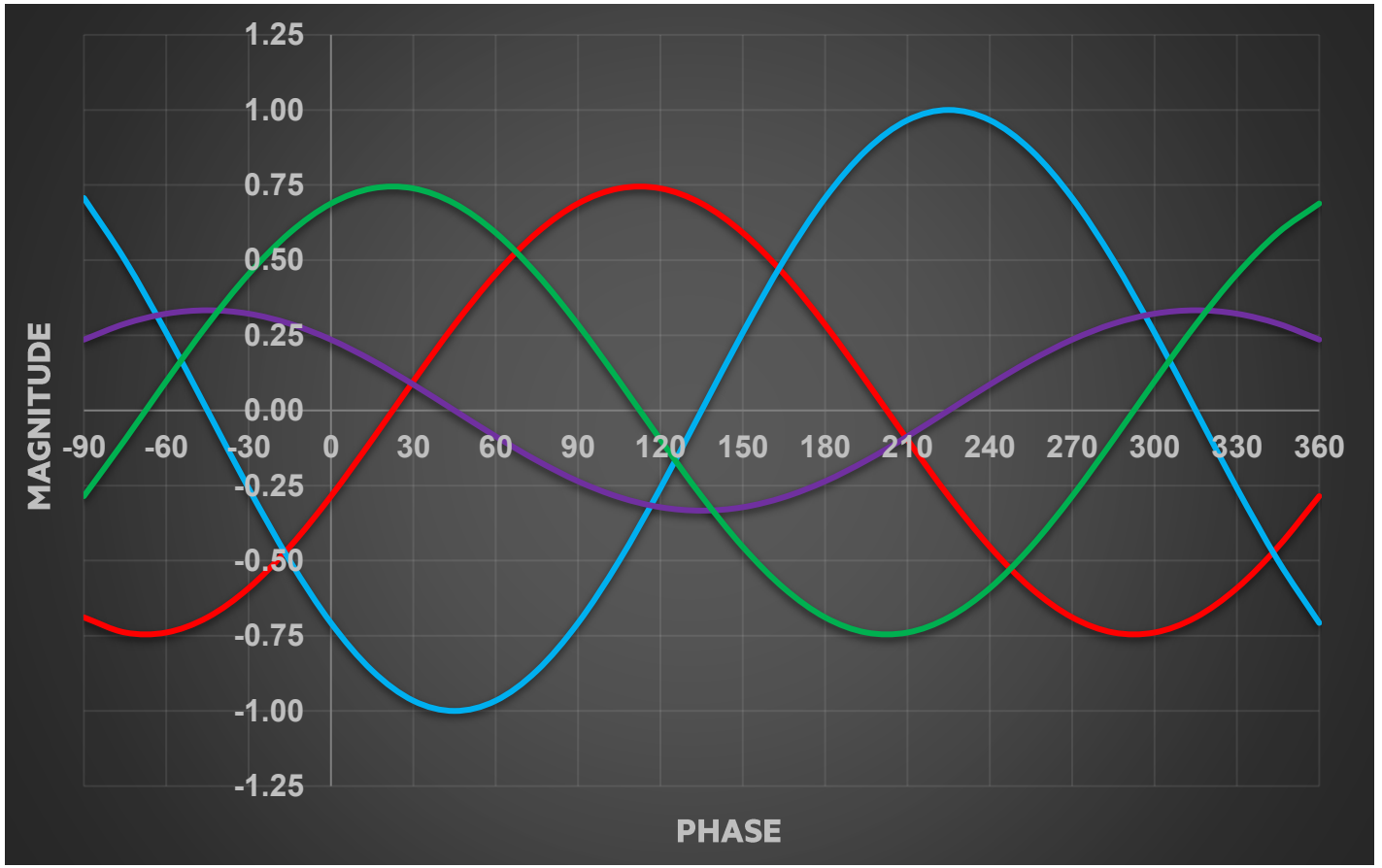


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QAM: Quadrature amplitude modulation

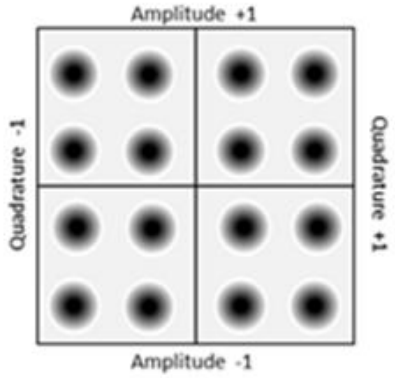


Symbol	Magnitude (V)	Phase
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1100	0.74536	292.5°

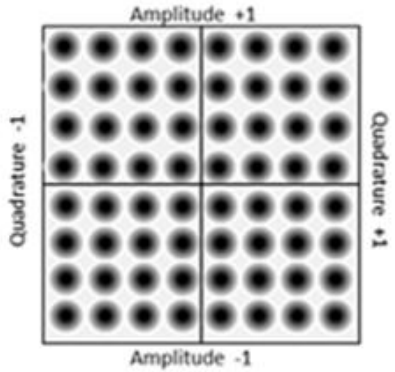


- All sub-carriers combine to form a time domain OFDM signal which gets transmitted after amplification
- Receiver applies FFT on this time domain signal to determine various sub-carriers
- Sub-carriers' phase and magnitude in turn determines received bits

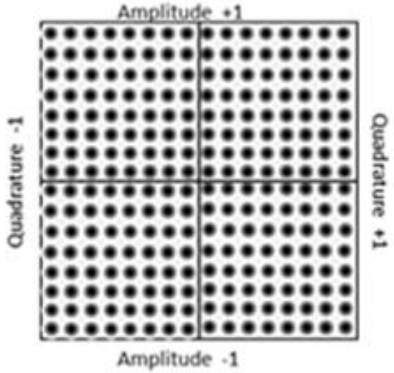
QAM: Quadrature amplitude modulation



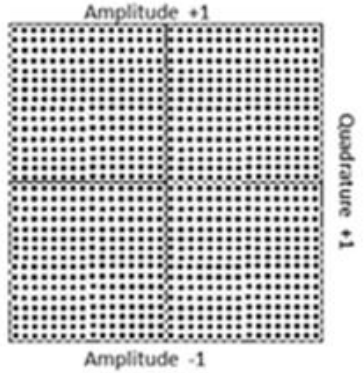
16-QAM constellation



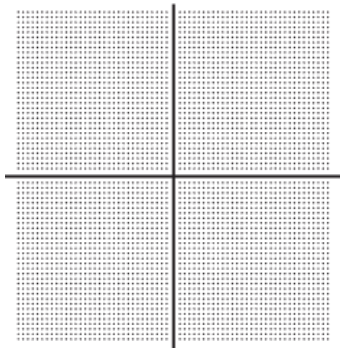
64-QAM constellation

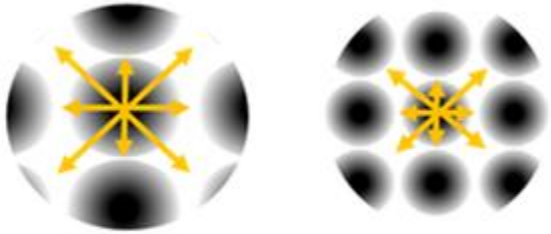


256-QAM constellation



1024-QAM constellation


4096-QAM constellation


Decreased error margin

Bits per tone

BPSK	1-Bit
QPSK	2-Bit
16 QAM	4-Bit
64 QAM	6-Bit
256 QAM	8-Bit
1024 QAM	10-Bit
4096 QAM	12-Bit

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Signal Power & Modulation

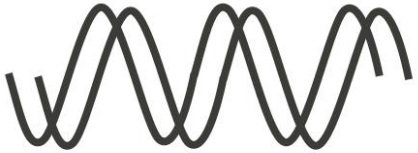
Modulation	Rate (R)	Minimum sensitivity (20 MHz PPDU) (dBm)	Maximum sensitivity (40 MHz PPDU) (dBm)	Minimum Sensitivity (80 MHz PPDU) (dBm)
BPSK	1/2	-82	-79	-76
QPSK	1/2	-79	-76	-73
QPSK	3/4	-77	-74	-71
16-QAM	1/2	-74	-71	-68
16-QAM	3/4	-70	-67	-64
64-QAM	2/3	-66	-63	-60
64-QAM	3/4	-65	-62	-59
64-QAM	5/6	-64	-61	-58
256-QAM	3/4	-59	-56	-53
256-QAM	5/6	-57	-54	-51

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PPDU = PLCP Protocol Data Unit
Packet ในระดับ Physical Layer ของ Wi-Fi

Distance, Performance and MCS (Modulation and Coding Scheme)

Nice, sharp waveforms support complex MCS
Higher data rate







AP

-46 dBm

Degraded waveforms require simpler MCS
Lower data rate





-76 dBm

Distance increases, signal weakens, waveforms degrade, error rates increase

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HT, VHT, HE, EHT

WIFI4	HT
WIFI5	VHT
WIFI6	HE
WIFI7	EHT

MCS Index				Spatial Stream	Modulation	Coding
HT	VHT	HE	EHT			
0	0	0	0	1	BPSQ	1/2
1	1	1	1	1	QPSK	1/2
2	2	2	2	1	QPSK	3/4
3	3	3	3	1	16-QAM	1/2
4	4	4	4	1	16-QAM	3/4
5	5	5	5	1	64-QAM	2/3
6	6	6	6	1	64-QAM	3/4
7	7	7	7	1	64-QAM	5/6
	8	8	8	1	256-QAM	3/4
	9	9	9	1	256-QAM	5/6
		10	10	1	1024-QAM	3/4
		11	11	1	1024-QAM	5/6
			12	1	4096-QAM	3/4
			13	1	4096-QAM	5/6

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MCS Index

MCS Index				Spatial Stream	Modulation	Coding	OFDM (Prior 11ax)								OFDM & OFDMA (Starting with 11ax)																				
							20MHz		40MHz		80MHz		160MHz		26-tone RU			52-tone RU			106-tone RU			242-tone RU / 20MHz			484-tone RU / 40MHz			996-tone RU / 80MHz			2x996-tone RU / 160MHz			4x996-tone RU / 320 MHz		
HT	VHT	HE	EHT				0.8µs GI	0.4µs GI	0.8µs GI	0.4µs GI	0.8µs GI	0.4µs GI	0.8µs GI	0.4µs GI	0.8µs GI	1.6µs GI	3.2µs GI	0.8µs GI	1.6µs GI	3.2µs GI	0.8µs GI	1.6µs GI	3.2µs GI	0.8µs GI	1.6µs GI	3.2µs GI	0.8µs GI	1.6µs GI	3.2µs GI	0.8µs GI	1.6µs GI	3.2µs GI	0.8µs GI	1.6µs GI	3.2µs GI	0.8µs GI	1.6µs GI	3.2µs GI
0	0	0	0	1	BPSQ	1/2	6.5	7.2	13.5	15	29.3	32.5	58.5	65	0.9	0.8	0.8	1.8	1.7	1.5	3.8	3.5	3.2	8.6	8.1	7.3	17.2	16.3	14.6	36.0	34.0	30.6	72.1	68.1	61.3	144.1	136.1	122.5
1	1	1	1	1	QPSK	1/2	13	14.4	27	30	58.5	65	117	130	1.8	1.7	1.5	3.5	3.3	3.0	7.5	7.1	6.4	17.2	16.3	14.6	34.4	32.5	29.3	72.1	68.1	61.3	144.1	136.1	122.5	288.2	272.2	245.0
2	2	2	2	1	QPSK	3/4	19.5	21.7	40.5	45	87.8	97.5	175.5	195	2.6	2.5	2.3	5.3	5.0	4.5	11.3	10.6	9.6	25.8	24.4	21.9	51.6	48.8	43.9	108.1	102.1	91.9	216.2	204.2	183.8	432.4	408.3	367.5
3	3	3	3	1	16-QAM	1/2	26	28.9	54	60	117	130	234	260	3.5	3.3	3.0	7.1	6.7	6.0	15.0	14.2	12.8	34.4	32.5	29.3	68.8	65.0	58.5	144.1	136.1	122.5	288.2	272.2	245.0	576.5	544.4	490.0
4	4	4	4	1	16-QAM	3/4	39	43.3	81	90	175.5	195	351	390	5.3	5.0	4.5	10.6	10.0	9.0	22.5	21.3	19.1	51.6	48.8	43.9	103.2	97.5	87.8	216.2	204.2	183.8	432.4	408.3	367.5	864.7	816.7	735.0
5	5	5	5	1	64-QAM	2/3	52	57.8	108	120	234	260	468	520	7.1	6.7	6.0	14.1	13.3	12.0	30.0	28.3	25.5	68.8	65.0	58.5	137.6	130.0	117.0	288.2	272.2	245.0	576.5	544.4	490.0	1,152.9	1,088.9	980.0
6	6	6	6	1	64-QAM	3/4	58.5	65	121.5	135	263.3	292.5	526.5	585	7.9	7.5	6.8	15.9	15.0	13.5	33.8	31.9	28.7	77.4	73.1	65.8	154.9	146.3	131.6	324.3	306.3	275.6	648.5	612.5	551.3	1,297.1	1,225.0	1,102.5
7	7	7	7	1	64-QAM	5/6	65	72.2	135	150	292.5	325	585	650	8.8	8.3	7.5	17.6	16.7	15.0	37.5	35.4	31.9	86.0	81.3	73.1	172.1	162.5	146.3	360.3	340.3	306.3	720.6	680.6	612.5	1,441.2	1,361.1	1,225.0
	8	8	8	1	256-QAM	3/4	78	86.7	162	180	351	390	702	780	10.6	10.0	9.0	21.2	20.0	18.0	45.0	42.5	38.3	103.2	97.5	87.8	206.5	195.0	175.5	432.4	408.3	367.5	864.7	816.7	735.0	1,729.4	1,633.3	1,470.0
	9	9	9	1	256-QAM	5/6	N/A	N/A	180	200	390	433.3	780	866.7	11.8	11.1	10.0	23.5	22.2	20.0	50.0	47.2	42.5	114.7	108.3	97.5	229.4	216.7	195.0	480.4	453.7	408.3	960.8	907.4	816.7	1,921.6	1,814.8	1,633.3
		10	10	1	1024-QAM	3/4									13.2	12.5	11.3	26.5	25.0	22.5	56.3	53.1	47.8	129.0	121.9	109.7	258.1	243.8	219.4	540.4	510.4	459.4	1,080.9	1,020.8	918.8	2,161.8	2,041.7	1,837.5
		11	11	1	1024-QAM	5/6									14.7	13.9	12.5	29.4	27.8	25.0	62.5	59.0	53.1	143.4	135.4	121.9	286.8	270.8	243.8	600.5	567.1	510.4	1,201.0	1,134.3	1,020.8	2,402.0	2,268.5	2,041.7
			12	1	4096-QAM	3/4									15.9	15.0	13.5	31.8	30.0	27.0	67.5	63.8	57.4	154.9	146.3	131.6	309.7	292.5	263.3	648.5	612.5	551.3	1,297.1	1,225.0	1,102.5	2,594.1	2,450.0	2,205.0
			13	1	4096-QAM	5/6									17.6	16.7	15.0	35.3	33.3	30.0	75.0	70.8	63.8	172.1	162.5	146.3	344.1	325.0	292.5	720.6	680.6	612.5	1,441.2	1,361.1	1,225.0	2,882.4	2,722.2	2,450.0
8	0	0	0	2	BPSQ	1/2	13	14.4	27	30	58.5	65	117	130	1.8	1.7	1.5	3.5	3.3	3.0	7.5	7.1	6.4	17.2	16.3	14.6	34.4	32.5	29.3	72.1	68.1	61.3	144.1	136.1	122.5	288.2	272.2	245.0
9	1	1	1	2	QPSK	1/2	26	28.9	54	60	117	130	234	260	3.5	3.3	3.0	7.1	6.7	6.0	15.0	14.2	12.8	34.4	32.5	29.3	68.8	65.0	58.5	144.1	136.1	122.5	288.2	272.2	245.0	576.5	544.4	490.0
10	2	2	2	2	QPSK	3/4	39	43.3	81	90	175.5	195	351	390	5.3	5.0	4.5	10.6	10.0	9.0	22.5	21.3	19.1	51.6	48.8	43.9	103.2	97.5	87.8	216.2	204.2	183.8	432.4	408.3	367.5	864.7	816.7	735.0
11	3	3	3	2	16-QAM	1/2	52	57.8	108	120	234	260	468	520	7.1	6.7	6.0	14.1	13.3	12.0	30.0	28.3	25.5	68.8	65.0	58.5	137.6	130.0	117.0	288.2	272.2	245.0	576.5	544.4	490.0	1,152.9	1,088.9	980.0
12	4	4	4	2	16-QAM	3/4	78	86.7	162	180	351	390	702	780	10.6	10.0	9.0	21.2	20.0	18.0	45.0	42.5	38.3	103.2	97.5	87.8	206.5	195.0	175.5	432.4	408.3	367.5	864.7	816.7	735.0	1,729.4	1,633.3	1,470.0
13	5	5	5	2	64-QAM	2/3	104	115.6	216	240	468	520	936	1040	14.1	13.3	12.0	28.2	26.7	24.0	60.0	56.7	51.0	137.6	130.0	117.0	275.3	260.0	234.0	576.5	544.4	490.0	1,152.9	1,088.9	980.0	2,305.9	2,177.8	1,960.0
14	6	6	6	2	64-QAM	3/4	117	130	243	270	526.5	585	1053	1170	15.9	15.0	13.5	31.8	30.0	27.0	67.5	63.8	57.4	154.9	146.3	131.6	309.7	292.5	263.3	648.5	612.5	551.3	1,297.1	1,225.0	1,102.5	2,594.1	2,450.0	2,205.0
15	7	7	7	2	64-QAM	5/6	130	144.4	270	300	585	650	1170	1300	17.6	16.7	15.0	35.3	33.3	30.0	75.0	70.8	63.8	172.1	162.5	146.3	344.1	325.0	292.5	720.6	680.6	612.5	1,441.2	1,361.1	1,225.0	2,882.4	2,722.2	2,450.0
	8	8	8	2	256-QAM	3/4	156	173.3	324	360	702	780	1404	1560	21.2	20.0	18.0	42.4	40.0	36.0	90.0	85.0	76.5	206.5	195.0	175.5	412.9	390.0	351.0	864.7	816.7	735.0	1,729.4	1,633.3	1,470.0	3,458.8	3,266.7	2,940.0
	9	9	9	2	256-QAM	5/6	N/A	N/A	360	400	780	866.7	1560	1733.3	23.5	22.2	20.0	47.1	44.4	40.0	100.0	94.4	85.0	229.4	216.7	195.0	458.8	433.3	390.0	960.8	907.4	816.7	1,921.6	1,814.8	1,633.3	3,843.1	3,629.6	3,266.7
		10	10	2	1024-QAM	3/4									26.5	25.0	22.5	52.9	50.0	45.0	112.5	106.3	95.6	258.1	243.8	219.4	516.2	487.5	438.8	1,080.9	1,020.8	918.8	2,161.8	2,041.7	1,837.5	4,323.5	4,083.3	3,675.0
		11	11	2	1024-QAM	5/6									29.4	27.8	25.0	58.8	55.6	50.0	125.0	118.1	106.3	286.8	270.8	243.8	573.5	541.7	487.5	1,201.0	1,134.3	1,020.8	2,402.0	2,268.5	2,041.7	4,803.9	4,537.0	4,083.3
			12	2	4096-QAM	3/4									31.8	30.0	27.0	63.5	60.0	54.0	135.0	127.5	114.8	309.7	292.5	263.3	619.4	585.0	526.5	1,297.1	1,225.0	1,102.5	2,594.1	2,450.0	2,205.0	5,188.2	4,900.0	4,410.0
			13	2	4096-QAM	5/6									35.3	33.3	30.0	70.6	66.7	60.0	150.0	141.7	127.5	344.1	325.0	292.5	688.2	650.0	585.0	1,441.2	1,361.1	1,225.0	2,882.4	2,722.2	2,450.0	5,764.7	5,444.4	4,900.0
16	0	0	0	3	BPSQ	1/2	19.5	21.7	40.5	45	87.8	97.5	175.5	195	2.6	2.5	2.3	5.3	5.0	4.5	11.3	10.6	9.6	25.8	24.4	21.9	51.6	48.8	43.9	108.1	102.1	91.9	216.2	204.2	183.8	432.4	408.3	367.5
17	1	1	1	3	QPSK	1/2	39	43.3	81	90	175.5	195	351	390	5.3	5.0	4.5	10.6	10.0	9.0	22.5	21.3	19.1	51.6	48.8	43.9	103.2	97.5	87.8	216.2	204.2	183.8	432.4	408.3	367.5	864.7	816.7	735.0
18	2	2	2	3	QPSK	3/4	58.5	65	121.5	135	263.3	292.5	526.5	585	7.9	7.5	6.8	15.9	15.0	13.5	33.8	31.9	28.7	77.4	73.1	65.8	154.9	146.3	131.6	324.3	306.3	275.6	648.5	612.5	551.3	1,297.1	1,225.0	1,102.5
19	3	3	3	3	16-QAM	1/2	78	86.7	162	180	351	390	702	780	10.6	10.0	9.0	21.2	20.0	18.0	45.0	42.5	38.3	103.2	97.5	87.8	206.5	195.0	175.5	432.4	408.3	367.5	864.7	816.7	735.0	1,729.4	1,633.3	1,470.0
20	4	4	4	3	16-QAM	3/4	117	130	243	270	526.5	585	1053	1170	15.9	15.0	13.5	31.8	30.0	27.0	67.5	63.8	57.4	154.9	146.3	131.6	309.7	292.5	263.3	648.5	612.5	551.3	1,297.1	1,225.0	1,102.5	2,594.1	2,450.0	2,205.0
21	5	5	5	3	64-QAM	2/3	156	173.3	324	360	702	780	1404	1560	21.2	20.0	18.0	42.4	40.0	36.0	90.0	85.0	76.5	206.5	195.0	175.5	412.9	390.0	351.0	864.7	816.7	735.0	1,729.4	1,633.3	1,470.0	3,458.8	3,266.7	2,940.0
22	6	6	6	3	64-QAM	3/4	175.5	195	364.5	405	N/A	N/A	1579.5	1755	23.8	22.5	20.3	47.6	45.0	40.5	101.3	95.6	86.1	232.3	219.4	197.4	464.6	438.8	394.9	972.8	918.8	826.9	1,945.6	1,837.5	1,653.8	3,891.2	3,675.0	3,307.5
23	7	7	7	3	64-QAM	5/6	195	216.7	405	450	877.5	975	1755	1950	26.5	25.0	22.5	52.9	50.0	45.0	112.5	106.3	95.6	258.1	243.8	219.4	516.2	487.5	438.8	1,080.9	1,020.8	918.8	2,161.8	2,041.7	1,837.5	4,323.5	4,083.3	3,675.0
	8	8	8	3	256-QAM	3/4	234	260	486	540	1053	1170	2106	2340	31.8	30.0	27.0	63.5	60.0	54.0	135.0	127.5	114.8	309.7	292.5	263.3	619.4	585.0	526.5	1,297.1	1,225.0	1,102.5	2,594.1	2,450.0	2,205.0	5,188.2	4,900.0	4,410.0
	9	9	9	3	256-QAM	5/6	260	288.9	540	600	1170	1300	N/A	N/A	35.3	33.3	30.0	70.6	66.7	60.0	150.0	141.7	127.5	344.1	325.0	292.5	688.2	650.0	585.0	1,441.2	1,361.1	1,225.0	2,882.4	2,722.2	2,450.0	5,764.7	5,444.4	4,900.0
		10	10	3	1024-QAM	3/4									39.7	37.5	33.8	79.4	75.0	67.5	168.8	159.4	143.4	387.1	365.6	329.1	774.3	731.3	658.1	1,621.3	1,531.3	1,378.1	3,242.6	3,062.5	2,756.3	6,485.3	6,125.0	5,512.5
		11	11	3	1024-QAM	5/6									44.1	41.7	37.5	88.2	83.3	75.0	187.5	177.1	159.4	430.1	406.3	365.6	860.3	812.5	731.3	1,801.5	1,701.4	1,531.3	3,602.9	3,402.8	3,062.5	7,205.9	6,805.6	6,125.0
			12	3	4096-QAM	3/4									47.6	45.0	40.5	95.3	90.0	81.0	202.5	191.3	172.1	464.6	438.8	394.9	929.1	877.5	789.8	1,945.6	1,837.5	1,653.8	3,891.2	3,675.0	3,307.5	7,782.4	7,350.0	6,615.0
			13	3	4096-QAM	5/6									52.9	50.0	45.0	105.9	100.0	90.0	225.0	212.5	191.3	516.2	487.5	438.8	1,032.4	975.0	877.5	2,161.8	2,041.7	1,837.5	4,323.5	4,083.3	3,675.0	8,647.1	8,166.7	7,350.0
24	0	0	0	4	BPSQ	1/2	26	28.9	54	60	117	130	234	260	3.5	3.3	3.0	7.1	6.7	6.0	15.0	14.2	12.8	34.4	32.5	29.3	68.8	65.0	58.5	144.1	136.1	122.5	288.2	272.2	245.0	576.5	544.4	490.0
25	1	1	1	4	QPSK	1/2	52	57.8	108	120	234	260	468	520	7.1	6.7	6.0	14.1	13.3	12.0	30.0	28.3	25.5	68.8	65.0	58.5	137.6	130.0	117.0	288.2	272.2	245.0	576.5	544.4	490.0	1,152.9	1,088.9	980.0
26	2	2	2	4	QPSK	3/4	78	86.7	162	180	351	390	702	780	10.6	10.0	9.0	21.2	20.0	18.0	45.0	42.5	38.3	103.2	97.5	87.8	206.5	195.0	175.5	432.4	408.3	367.5	864.7	816.7	735.0	1,729.4	1,633.3	1,470.0
27	3	3	3	4	16-QAM	1/2	104	115.6	216	240	468	520	936	1040	14.1	13.3	12.0	28.2	26.7	24.0	60.0	56.7	51.0	137.6	130.0	117.0	275.3	260.0	234.0	576.5	544.4	490.0	1,152.9	1,088.9	980.0	2,305.9	2,177.8	1,960.0
28	4	4	4	4	16-QAM	3/4	156	173.3	324	360	702	780	1404	1560	21.2	20.0	18.0	42.4	40.0	36.0	90.0	85.0	76.5	206.5	195.0	175.5	412.9	390.0	351.0	864.7	816.7	735.0	1,729.4	1,633.3	1,470.0	3,458.8	3,266.7	2,940.0
29	5	5	5	4	64-QAM	2/3	208	231.1	432	480	936	1040	1872	2080	28.2	26.7	24.0	56.5	53.3	48.0	120.0	113.3	102.0	275.3	260.0	234.0	550.6	520.0	468.0	1,152.9	1,088.9	980.0	2,305.9	2,177.8	1,960.0	4,611.8	4,355.6	3,920.0
30	6	6	6	4	64-QAM	3/4	234	260	486	540	1053	1170	2106	2340	31.8	30.0	27.0	63.5	60.0	54.0	135.0	127.5	114.8	309.7	292.5	263.3	619.4	585.0	526.5	1,297.1	1,225.0	1,102.5	2,594.1	2,450.0	2,205.0	5,188.2	4,900.0	4,410.0
31	7	7	7	4	64-QAM	5/6	260	288.9	540	600	1170	1300	2340	2600	35.3	33.3	30.0	70.6	66.7	60.0	150.0	141.7	127.5	344.1	325.0	292.5	688.2	650.0	585.0	1,441.2	1,361.1	1,225.0	2,882.4	2,722.2	2,450.0	5,764.7	5,444.4	4,900.0
	8	8	8	4	256-QAM	3/4	312	346.7	648	720	1404	1560	2808	3120	42.4	40.0	36.0	84.7	80.0	72.0	180.0	170.0	153.0	412.9	390.0	351.0	825.9	780.0	702.0	1,729.4	1,633.3	1,470.0	3,458.8	3,266.7	2,940.0	6,917.6	6,533.3	5,880.0
	9	9	9	4	256-QAM	5/6	N/A	N/A	720	800	1560	1733.3	3120	3466.7	47.1	44.4	40.0	94.1	88.9	80.0	200.0	188.9	170.0	458.8	433.3	390.0	917.6	866.7	780.0	1,921.6	1,814.8	1,633.3	3,843.1	3,629.6	3,266.7	7,686.3	7,259.3	6,533.3
		10	10	4	1024-QAM	3/4									52.9	50.0	45.0	105.9	100.0	90.0	225.0	212.5	191.3	516.2	487.5	438.8	1,032.4	975.0	877.5	2,161.8	2,041.7	1,837.5	4,323.5	4,083.3	3,675.0	8,647.1	8,166.7	7,350.0
		11	11	4	1024-QAM	5/6									58.8	55.6	50.0	117.6	111.1	100.0	250.0	236.1	212.5	573.5	541.7	487.5	1,147.1	1,083.3	975.0	2,402.0	2,268.5	2,041.7	4,803.9	4,537.0	4,083.3	9,607.8	9,074.1	8,166.7
			12	4	4096-QAM	3/4									63.5	60.0	54.0	127.1	120.0	108.0	270.0	255.0	229.5	619.4	585.0	526.5	1,238.8	1,170.0	1,053.0	2,594.1	2,450.0	2,205.0	5,188.2	4,900.0	4,410.0	10,376.5	9,800.0	8,820.0
			13	4	4096-QAM	5/6									70.6	66.7	60.0	141.2	133.3	120.0	300.0	283.3	255.0	688.2	650.0	585.0	1,376.5	1,300.0	1,170.0	2,882.4	2,722.2	2,450.0	5,764.7	5,444.4	4,900.0	11,529.4	10,888.9	9,800.0

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https://mcsindex.net/

WIFI6E&7 160MHz

WIFI6E

WIFI7

MCS Index	Spatial Stream	Modulation	Coding	OFDM & OFDMA 2x996-tone RU / 160MHz 0.8µs GI	1.6µs GI	3.2µs GI
0	2	BPSQ	1/2	144.1	136.1	122.5
1	2	QPSK	1/2	288.2	272.2	245.0
2	2	QPSK	3/4	432.4	408.3	367.5
3	2	16-QAM	1/2	576.5	544.4	490.0
4	2	16-QAM	3/4	864.7	816.7	735.0
5	2	64-QAM	2/3	1,152.9	1,088.9	980.0
6	2	64-QAM	3/4	1,297.1	1,225.0	1,102.5
7	2	64-QAM	5/6	1,441.2	1,361.1	1,225.0
8	2	256-QAM	3/4	1,729.4	1,633.3	1,470.0
9	2	256-QAM	5/6	1,921.6	1,814.8	1,633.3
10	2	1024-QAM	3/4	2,161.8	2,041.7	1,837.5
11	2	1024-QAM	5/6	2,402.0	2,268.5	2,041.7
12	2	4096-QAM	3/4	2,594.1	2,450.0	2,205.0
13	2	4096-QAM	5/6	2,882.4	2,722.2	2,450.0

GI = Guard Interval

เป็นช่วงเวลาสั้น ๆ ที่ Wi-Fi เว้นไว้ระหว่าง **OFDM symbols**

GI มีไว้เพื่อป้องกันปัญหา **multipath interference** เพราะสัญญาณ Wi-Fi ไม่ได้เดินทางมาถึง receiver แค่ทางเดียว

I จึงเป็นเหมือน **ช่วงกันชน** เพื่อไม่ให้สัญญาณของ symbol ก่อนหน้าไปรบกวน symbol ถัดไป

3.2uS ทน multipath ได้ดี

WIFI Foundation : Antenna & RSSI

HPE aruba networking

dB vs dBi vs dBm

Three RF measurements—three different meanings

dB
RELATIVE GAIN OR LOSS

Compares two power levels
No fixed reference • Dimensionless

+3 dB ≈ 2× power
−3 dB ≈ ½ power
+10 dB = 10× power

$$dB = 10 \log_{10}\left(\frac{P_2}{P_1}\right)$$

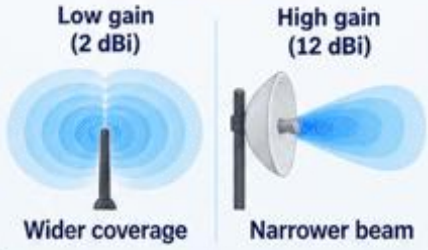


dBi
ANTENNA GAIN

Referenced to an isotropic radiator
Shows how an antenna focuses energy

Higher dBi → narrower beam
Lower dBi → wider coverage

Antennas redirect power—
they do not create it



dBm / dB(mW)
ABSOLUTE POWER

Referenced to 1 milliwatt

0 dBm = 1 mW
10 dBm = 10 mW
20 dBm = 100 mW
30 dBm = 1 W

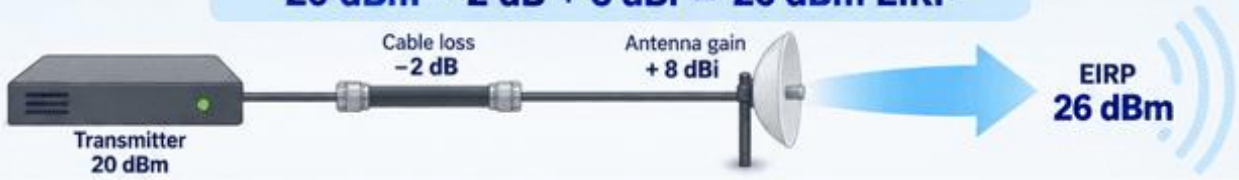
$$dBm = 10 \log_{10}\left(\frac{P}{1\ mW}\right)$$



PUTTING THEM TOGETHER

EIRP (dBm) = Transmitter power (dBm) − Cable loss (dB) + Antenna gain (dBi)

20 dBm − 2 dB + 8 dBi = 26 dBm EIRP



dB changes a level • **dBi** describes an antenna • **dBm** states a power level

Always distinguish antenna gain, system gain/loss and absolute RF power.

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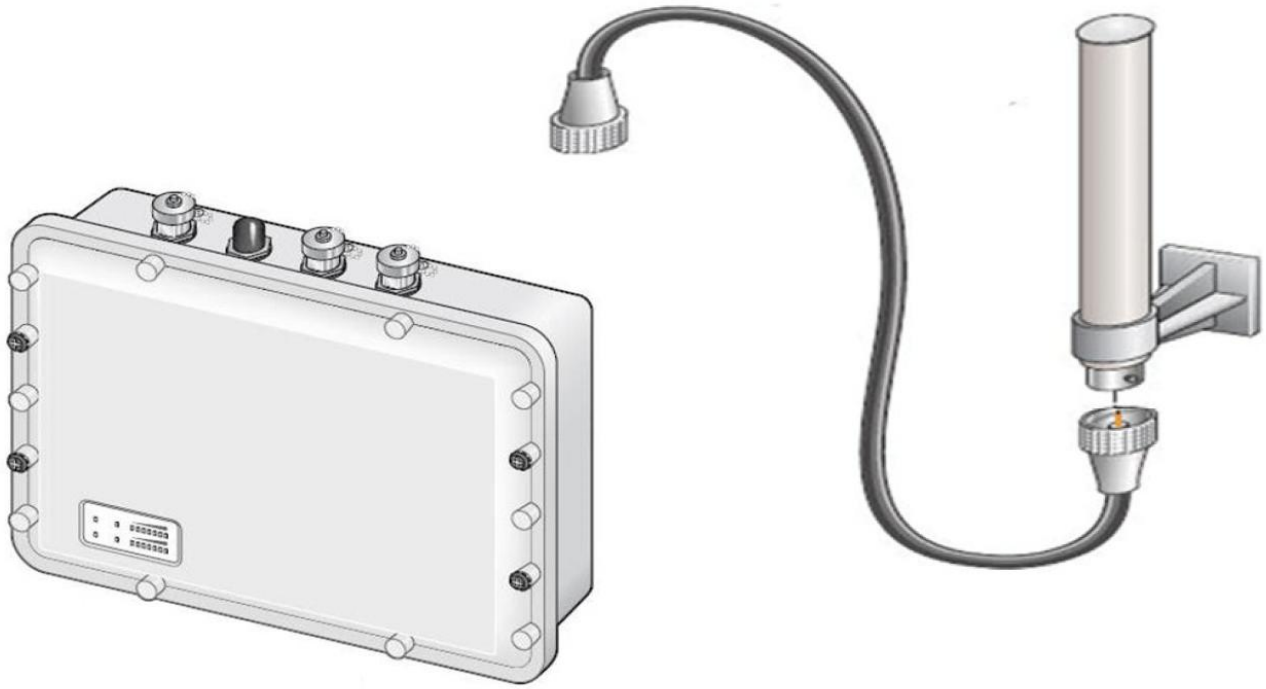
EIRP - *Effective or Equivalent isotropic radiated power*

- กำลังส่งออกอากาศ เทียบเท่าแบบไอโซทรอปิก
- กำลังของสัญญาณ ที่เสมือนว่าออกจากเสาอากาศอุดมคติ ที่แผ่พลังงานเท่ากันทุกทิศทาง

20dBm − **3dB cable and connector loss** + **10dBi antenna gain** = **27dBm EIRP**

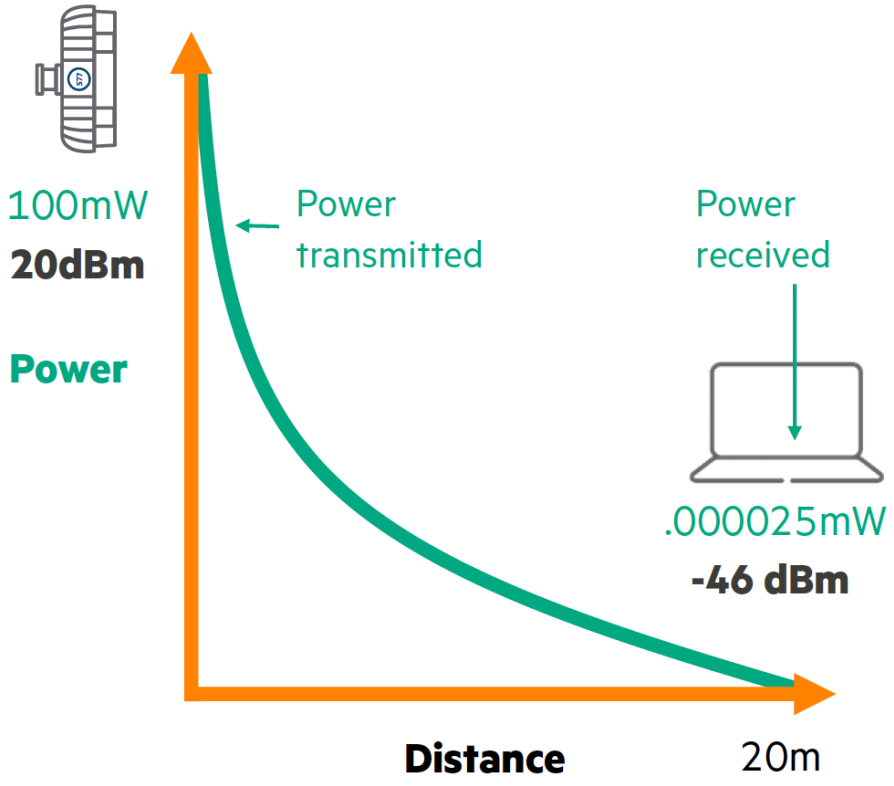
Access Point Tx Power

HPE Aruba Configure EIRP Value



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Milliwatts and dBm



Received Signal Strength

Distance(m)	mW	dBm
	1	0
0	100	20
1	.0098911	-20
10	.0000989	-40
20	.0000247	-46
100	.0000010	-60
1000 (1km)	.0000000099	-80

RF Power is a logarithmic calculation

dBm is much easier to work with than mW

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Thailand - NBTC Maximum Output

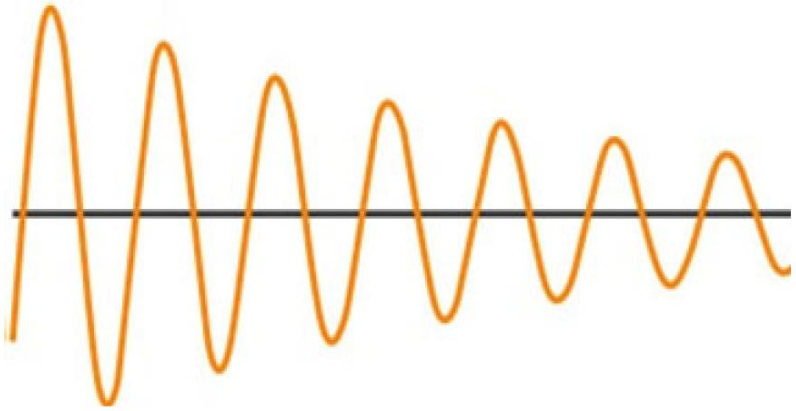
	Frequency (GHz)	Maximum EIRP (mW)	Maximum EIRP (dBm)
2.4 GHz	2.400-2.483	100	20
5 GHz	5.150-5.350	200	23
	5.470-5.725	500	27
6 GHz	5.925-6.425	250	24

$$P_{(\mathrm{mW})} = 1\mathrm{mW} \cdot 10^{(P_{(\mathrm{dBm})}/10)}$$

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Things that negatively affect RF signal

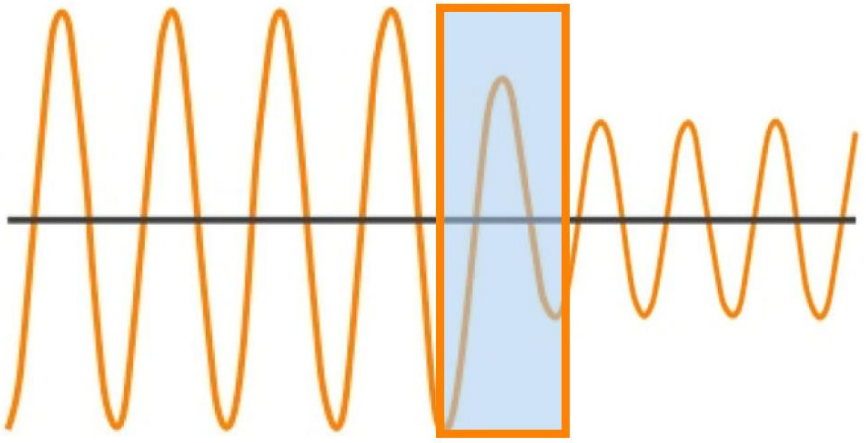
Free Space Path Loss



Distance

Absorption

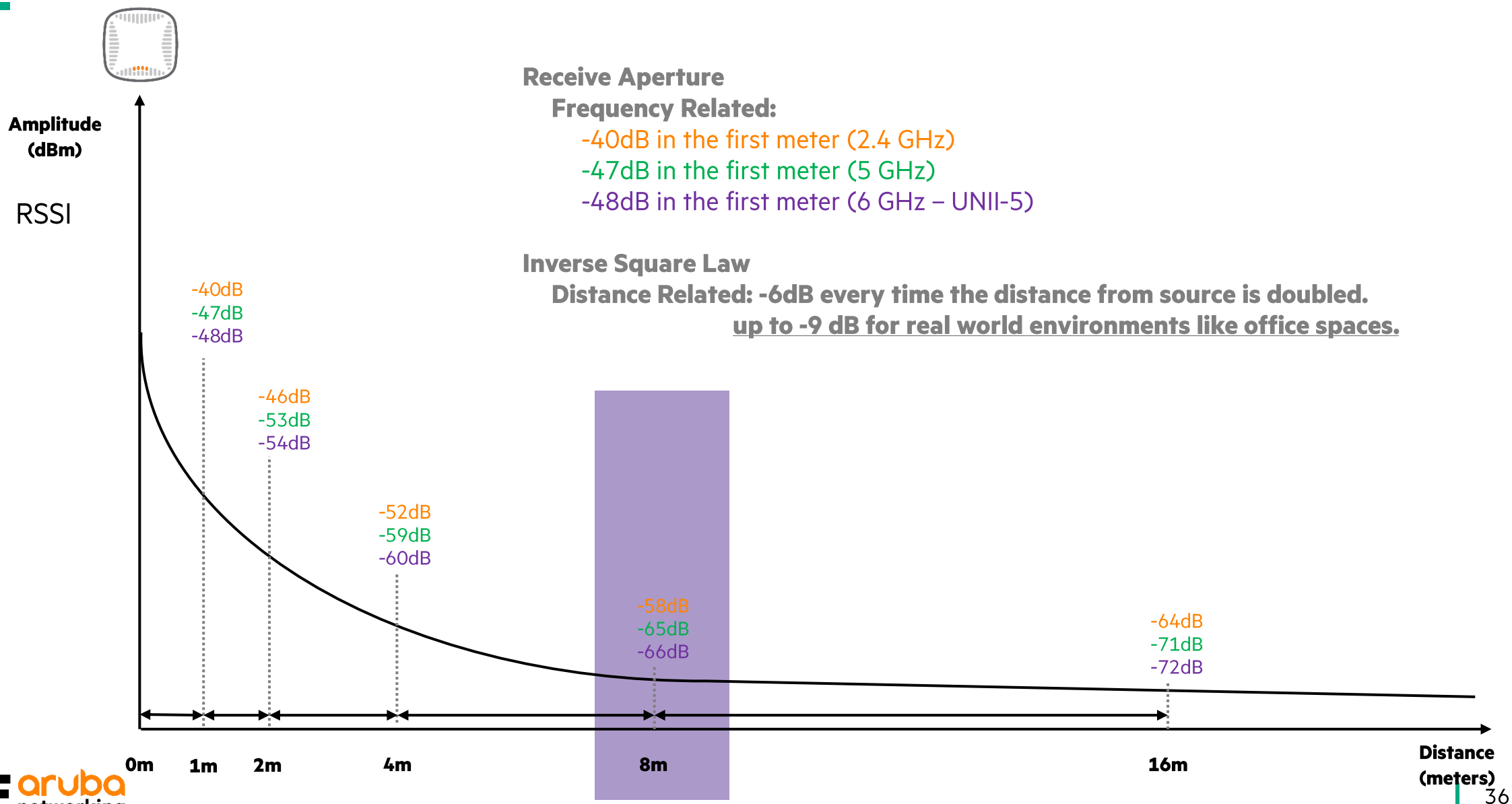
5 and 6GHz bands affected more than 2.4GHz



Signal traveling through an object

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Free Space Path Loss



Receive Aperture

Frequency Related:

-40dB in the first meter (2.4 GHz)

-47dB in the first meter (5 GHz)

-48dB in the first meter (6 GHz – UNII-5)

Inverse Square Law

Distance Related: -6dB every time the distance from source is doubled.

up to -9 dB for real world environments like office spaces.

Free Space Propagation

Propagation is the attenuation of the RF signal as it travels through a medium.

In **free space propagation** (propagation an antenna signal in the air), we do get signal attenuation.

Propagation in the first meter

$$\text{Signal loss: } -20 \times \log_{10}(4 \times \frac{\pi}{\lambda})$$

2.450 GHz: there is a loss of **-40.2dB** in the first meter
5.180 GHz: there is a loss of **-46.7dB** in the first meter
7.125 GHz: there is a loss of **-49.5dB** in the first meter

Free space propagation after the first meter

The physics of free space propagation changes **after** the first meter.

Distances of a factor of 2, result in a 6dB loss

2m = 6dB loss

8m = (2 x 2 x 2)m

8m = 6dB + 6dB + 6dB = 18dB loss

Distances of a factor of 10, result in a 20dB loss.

10m = 20dB loss

100m = (10 x10)m

100m = 20dB + 20dB =40dB loss

Example: *What is the attenuation of the RF signal as it travels **801** meters through free space?*

800m = (**10** x **10** x **2** x **2** x **2**)m, which is two distances of 10, and three distances of 2

800m = 20 dB + 20dB + 6dB +6dB + 6dB = 58 dB of loss

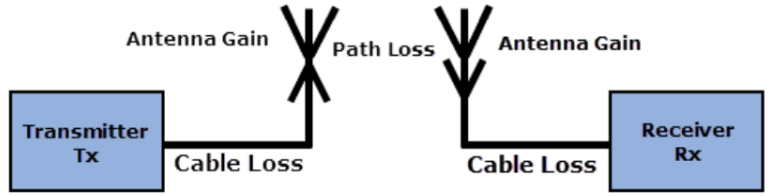
At 2.4 GHz = - 40 dB(1st meter) – 58 dB(800m) = -98 dB of loss

At 5.8 GHz = - 48 dB(1st meter) – 58 dB(800m) = -106 dB of loss

At 7.1 GHz = - 50 dB(1st meter) – 58 dB(800m) = -108 dB of loss

Calculating FSPL (Free-Space Path Loss)

$$FSPL = 20\log_{10}(d) + 20\log_{10}(f) + 20\log_{10}\left(\frac{4\pi}{c}\right) - G_t - G_r$$

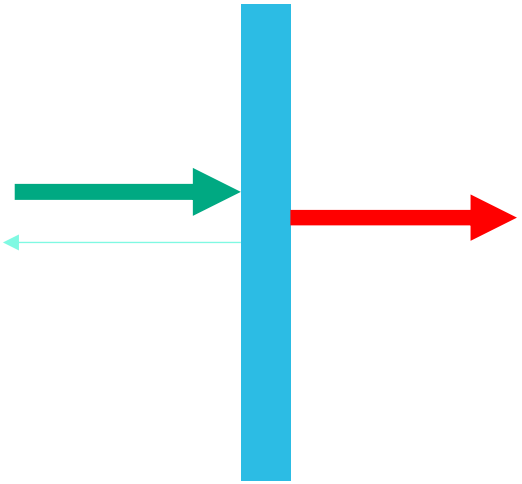


https://www.pasternack.com/t-calculator-fspl.aspx

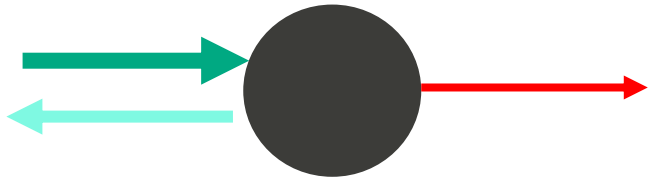
Freq. (GHz)	1M (dB)	5M (dB)	10M (dB)	20M (dB)
2.450	-40.2	-54.2	-60.2	-66.2
5.180	-46.7	-60.7	-66.7	71.7
7.125	-49.5	-63.5	-69.5	???

Material reflection / Absorption

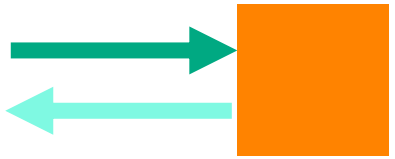
Materials have properties that are different from free space.
All materials will bounce energy.



Drywall



Water



Metal

WIFI Attenuation

Attenuation in 2.4 GHz and 5 GHz

Indoor Environment	Attenuation in 2.4 GHz	Attenuation in 5 GHz
Fabric, blinds, ceiling tiles	Approximately 1 dB	Approximately 1.5 dB
Interior drywall	3–4 dB	3–5 dB
Cubicle wall	2–5 dB	4–9 dB
Wood door (Hollow –Solid)	3–4 dB	6–7 dB
Brick/concrete wall	6–18 dB	10–30 dB
Glass/window (not tinted)	2–3 dB	6–8 dB
Double-pane coated glass	13 dB	20 dB
Steel/fire exit door	13–19 dB	25–32 dB

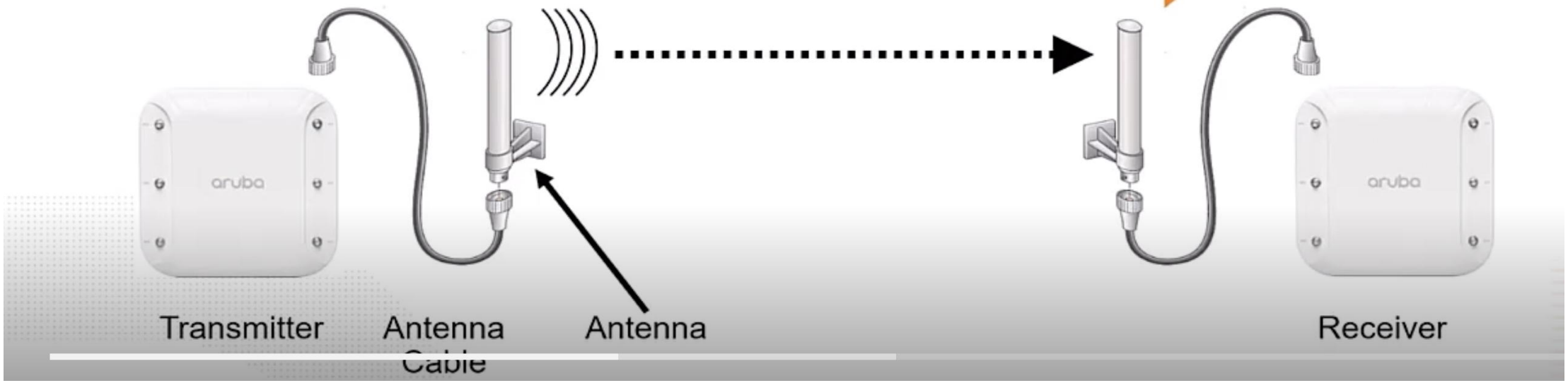
Antenna function

When antenna is connected to a transmitter

- Radiates or directs RF waves away from the antenna
- Receives RF carrier signal from the transmitter

When antenna is connected to a receiver

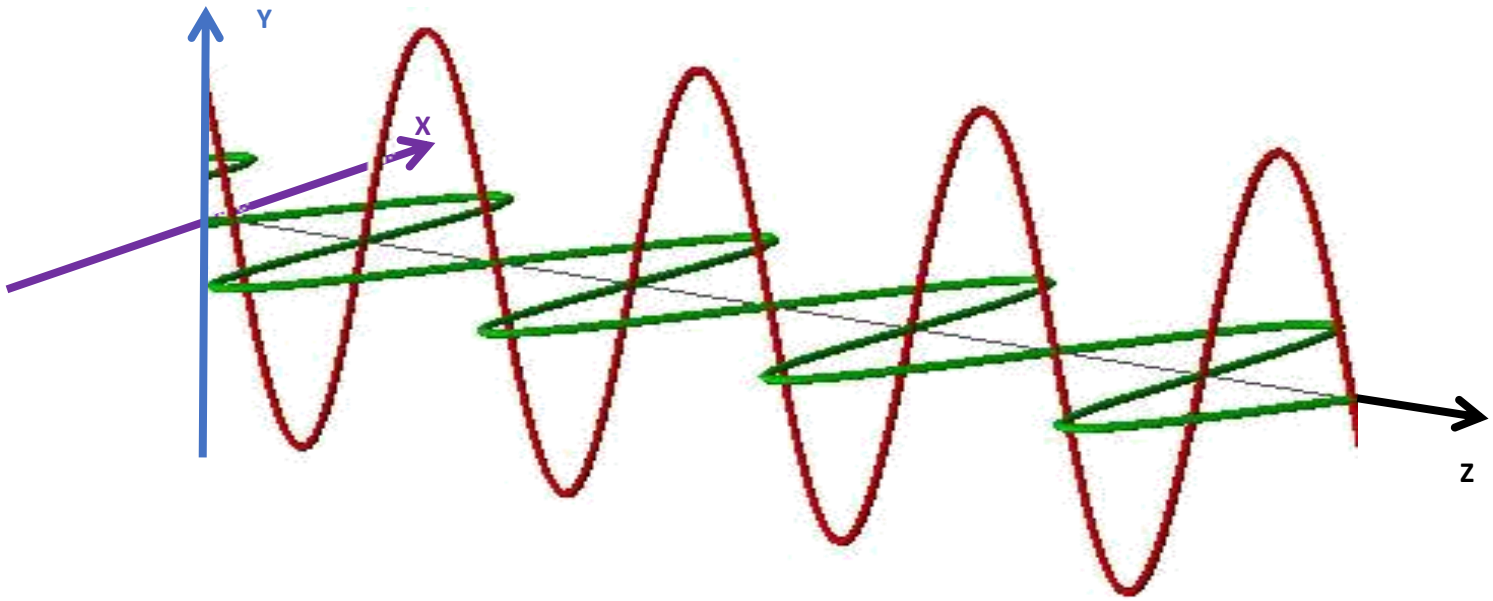
- Takes the RF waves it receives and directs an AC signal to the receiver



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Polarization

Polarization is the horizontal or vertical orientation of a wave.
Polarization is dictated by the **electric field.**



X-Axis: Magnetic field
Y-Axis: Electric field
Z-Axis: Electromagnetic energy

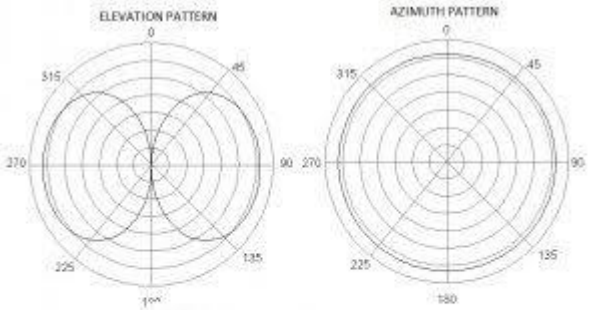
- Needed to understand antenna theory and the RSSI (Received Signal Strength)
- **The RSSI increases when the receiving antenna is polarized the same as the transmitting antenna**

Antenna types

- Antennas direct or focus RF signal
  - Similar to the way a light fixture focuses light
  - The effective output (signal strength) is **EIRP**

❖EIRP = Equivalent Isotropically Radiated Power

- Two primary types of antennas
  - Omnidirectional
  - Directional



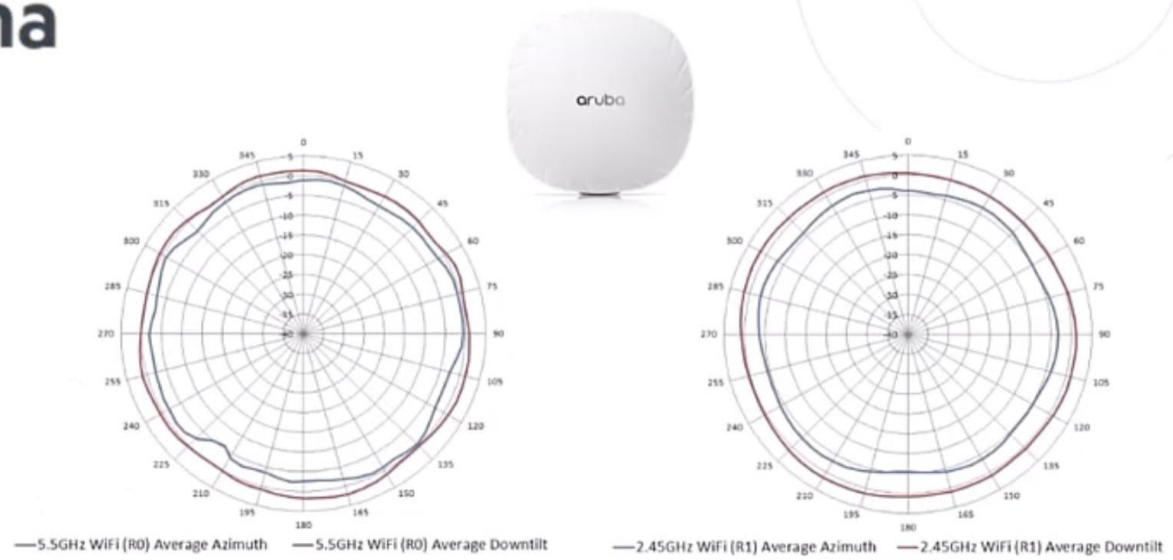
- Omnidirectional antennas
  - Radiates RF signal in all directio

- Directional Antennas
  - Focus/aim RF signal
  - Focus can be narrow (10 degrees)
  - Focus can be broad (180 degrees)

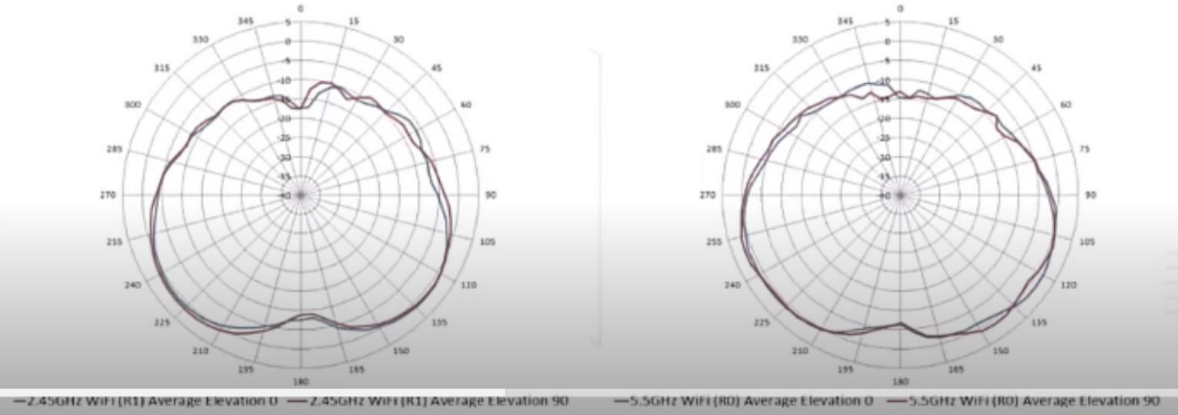
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Omnidirectional antenna

- Azimuth
  - H-Plane, top view, or Horizontal
  - Similar coverage in all directions



- Elevation
  - E-Plane, side view, or Vertical
  - Ceiling mounted AP so more signal radiates below the AP



07:24 / 11:10

5 GHz Radio

2.4 GHz Radio

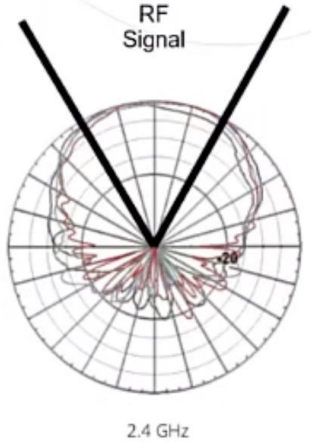
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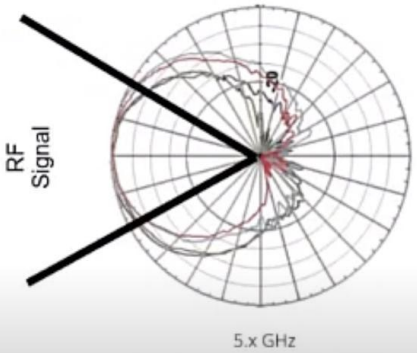
Directional antenna



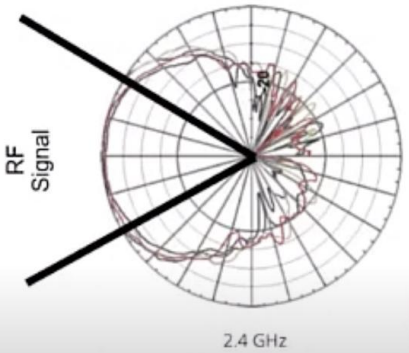
- Azimuth
  - Top view / Horizontal
  - Focused in one direction
  - This antenna = 60-degree horizontal coverage
- Elevation
  - Side view / Vertical
  - Focused vertically too
  - This antenna = 60-degree vertical coverage







5 GHz Radio



2.4 GHz Radio

- Higher gain antennas have a narrower focus
- Lower gain antennas have a broader focus

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Dipole Antenna



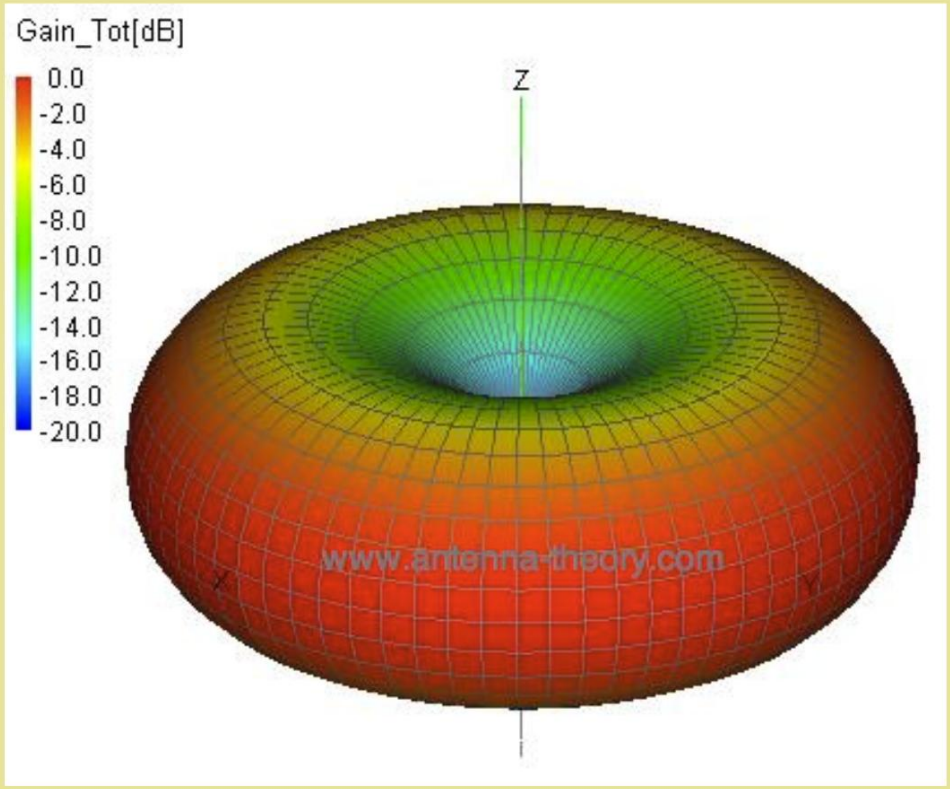
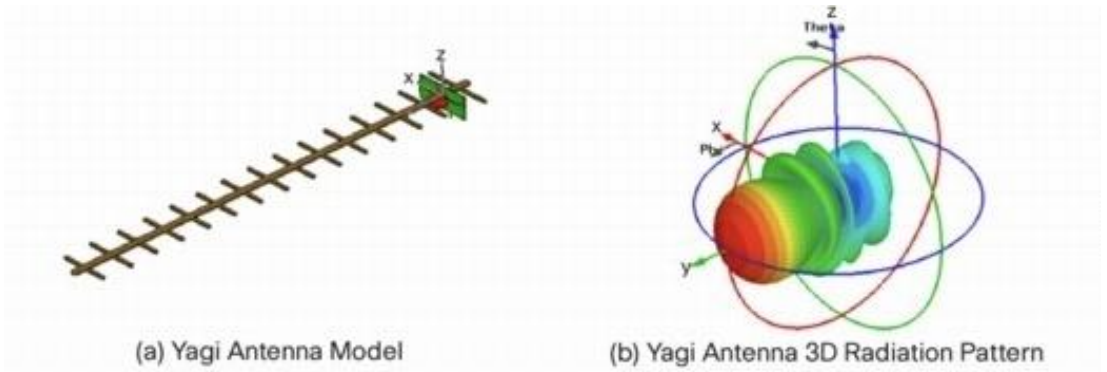
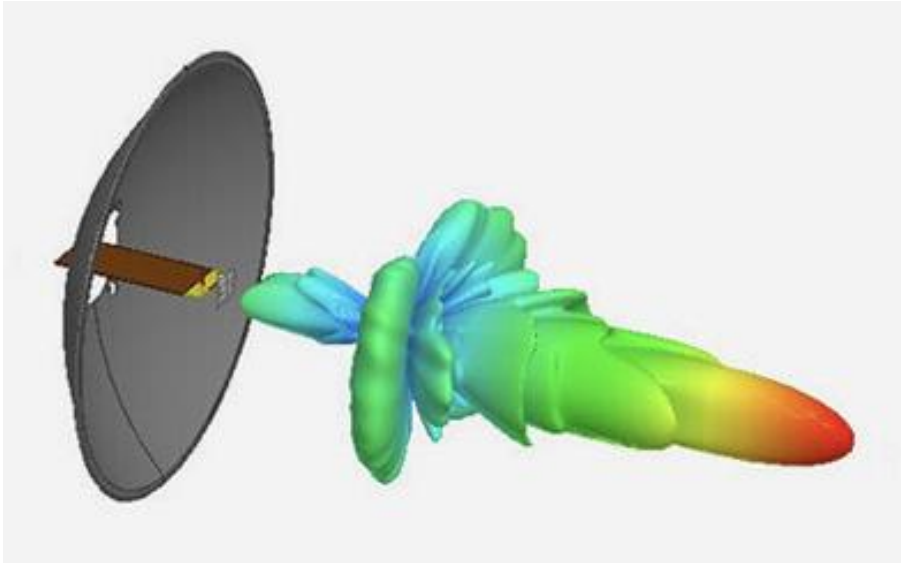


Figure 4. Normalized 3d radiation pattern for the 1-wavelength dipole antenna.

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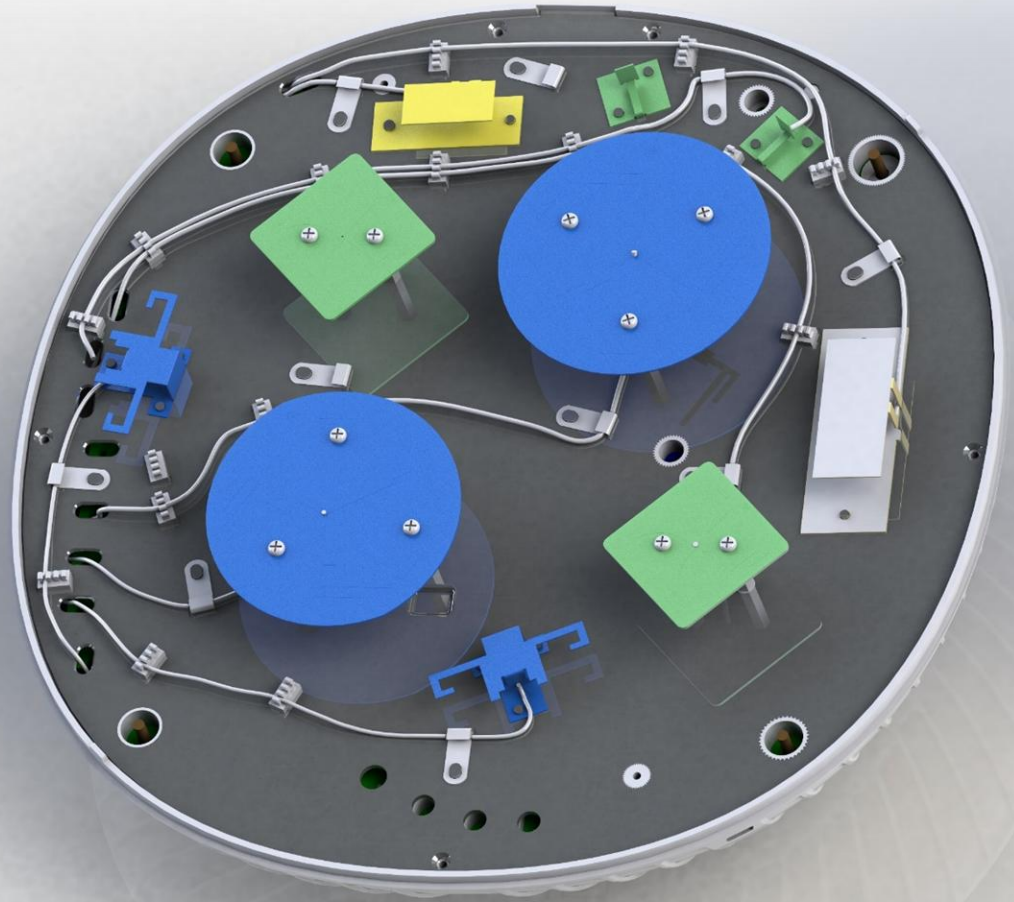
Yaki and Disc Antenna





Antenna sub-system (AP-655)

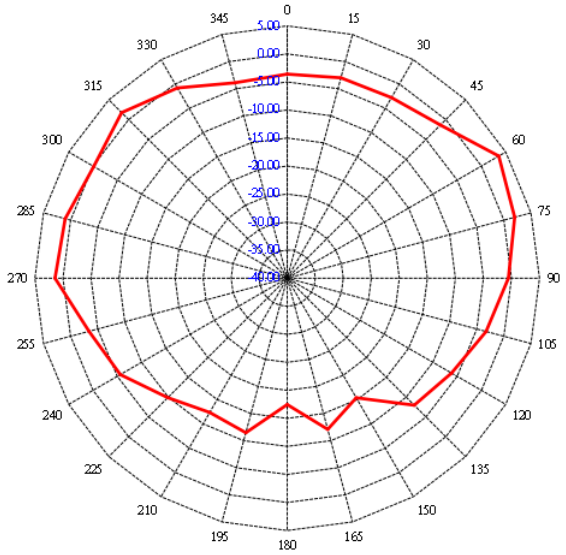
- 2/5 GHz Dual Band Elements
- 6 GHz Single Band Elements
- BLE Single Band Element
- GNSS Single Band Element

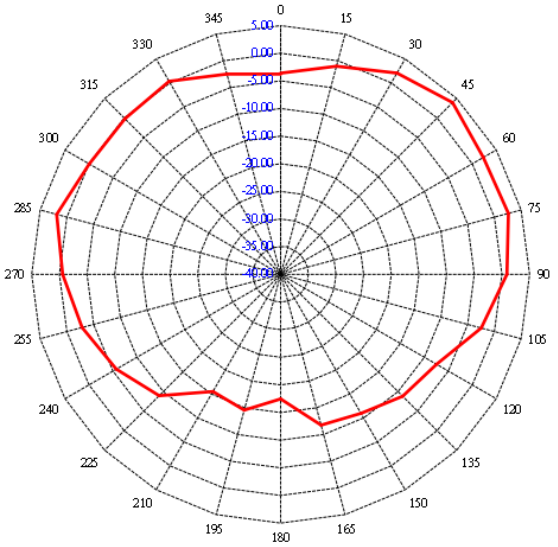


6 GHz antenna pattern (AP-655)

Down Tilt Angle From Ceiling (degree)	15
Gain Flatness (dB)	7

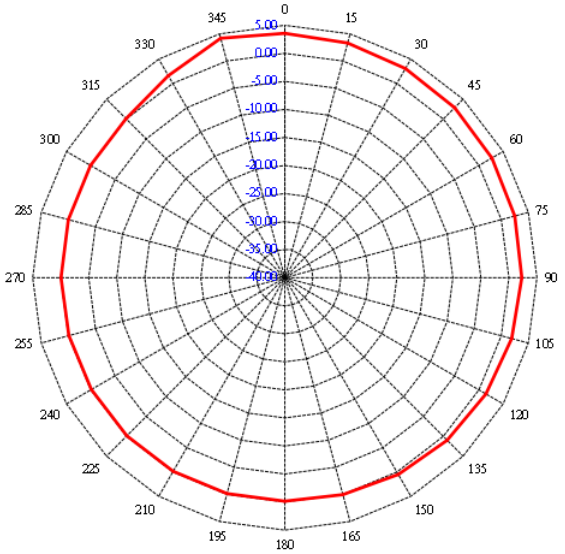
X-Z Cut

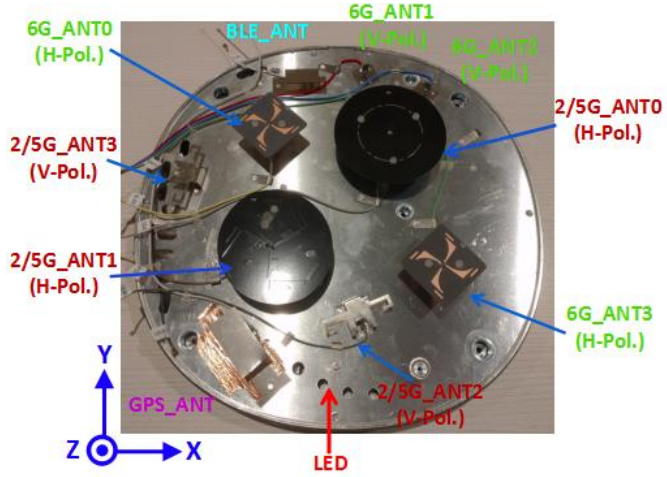




Y-Z Cut

X-Y Cut (Θ=60°)

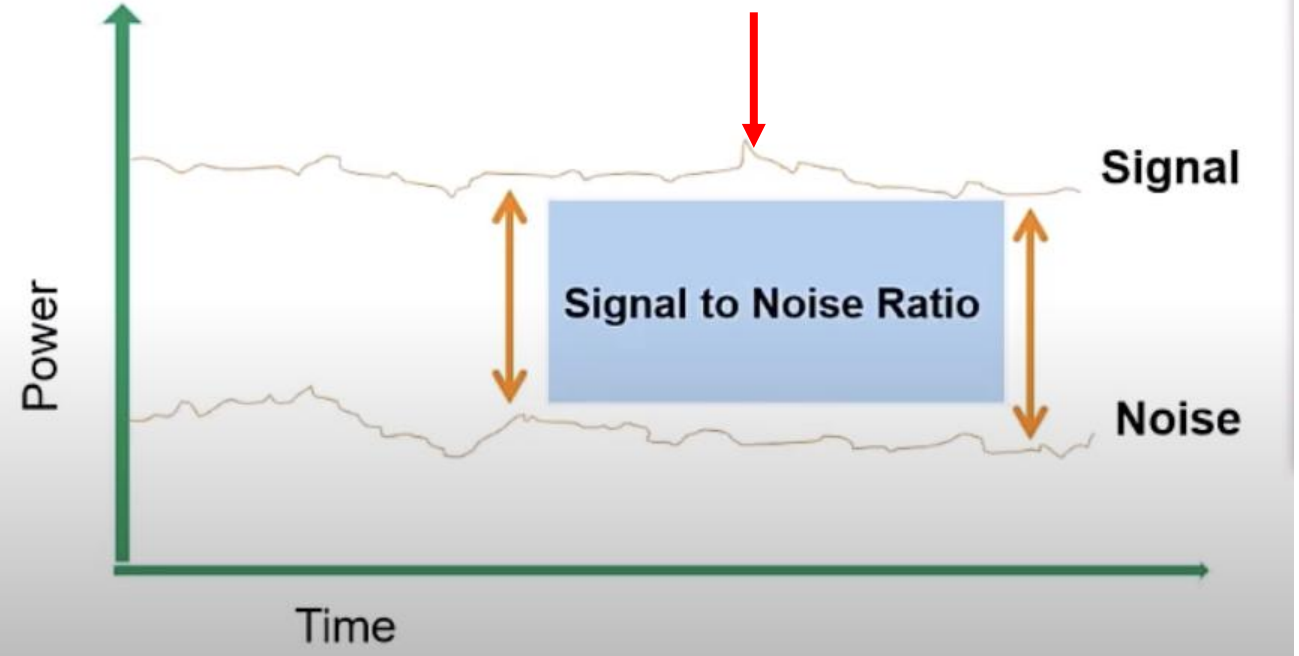




Signal to Noise Ratio (SNR)

- SNR is the difference between the wanted signal and any noise/interference
- One of the most important factors for good communications
- Higher SNR = Faster data rates
- Lower SNR = Slower data rates

RSSI, or "Received Signal Strength Indicator"



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Q&A

Should we design 1 AP to cover 100 device ?

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Q&A

Should we design 1 AP to cover 100 device in a classroom ?

NO > WiFI is working as a HUB,

Performance will drop significantly

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Q&A

What is the factor which affect to client WIFI Data Rate ?

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Q&A

What is the factor which affect to client WIFI Data Rate ?

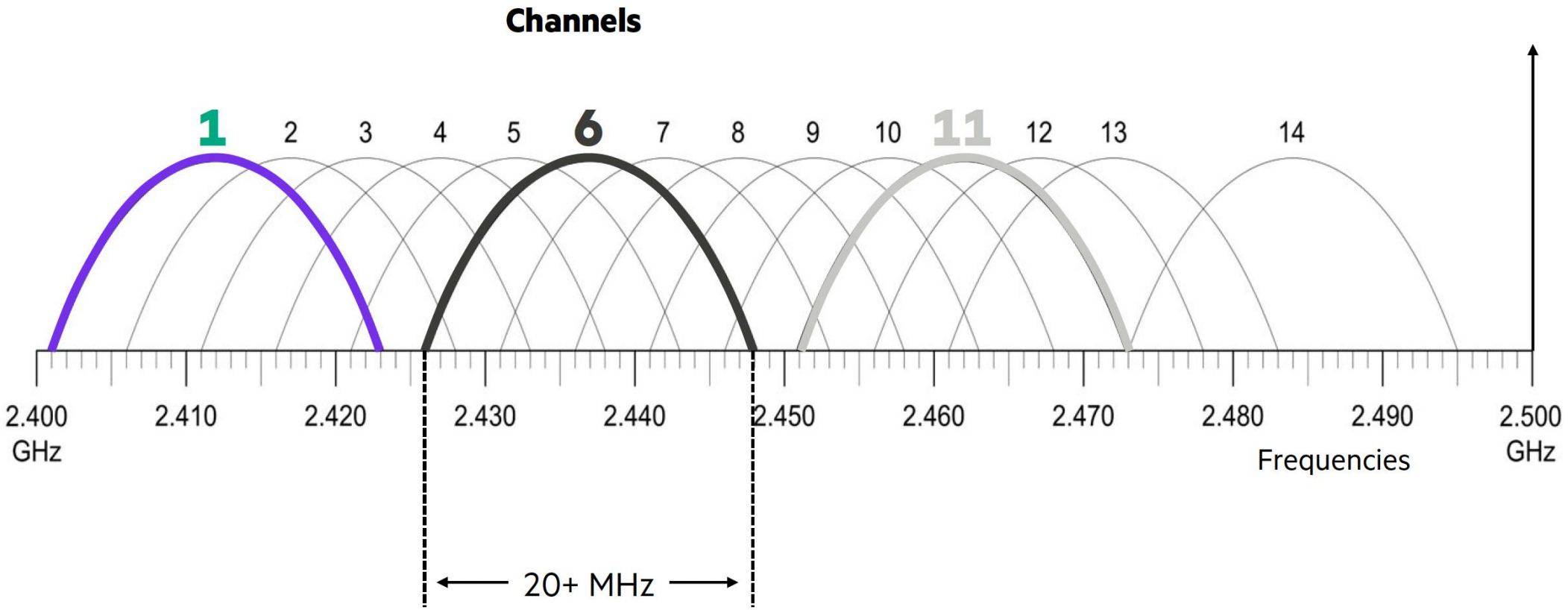
Distance
Antenna Type & Polarization
Environment (Attenuation)
Client Capability (Technology)
Client Driver
AP Power

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WIFI Foundation : Channel & Channel Width

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2.4 GHz Band and Channels



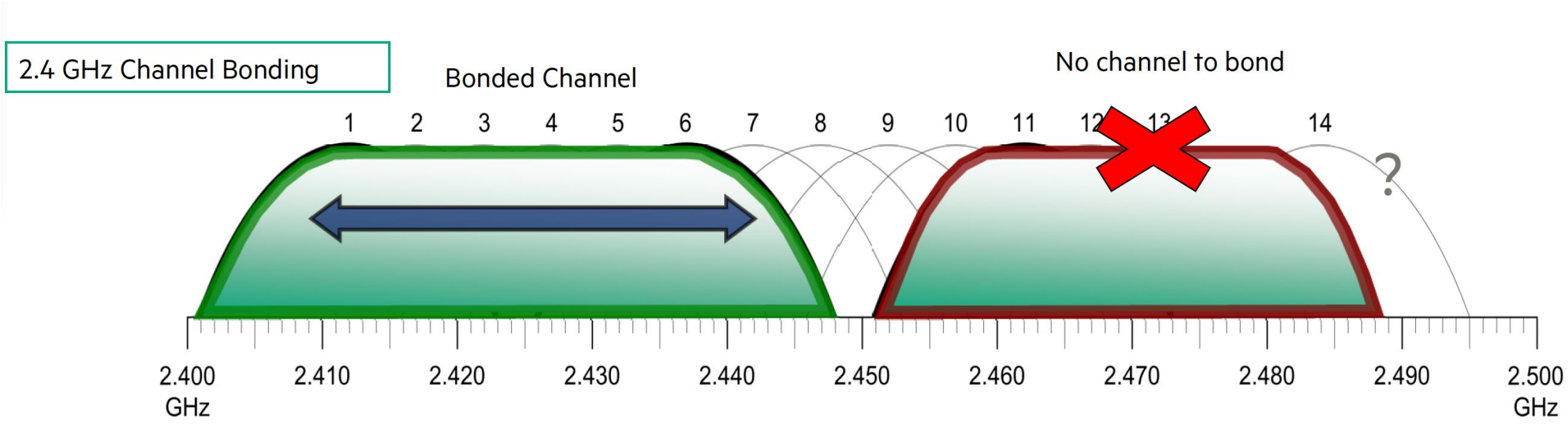
Signal spread over 20+Mhz

Adjacent channels are not usable in same physical RF area

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Channel Bonding

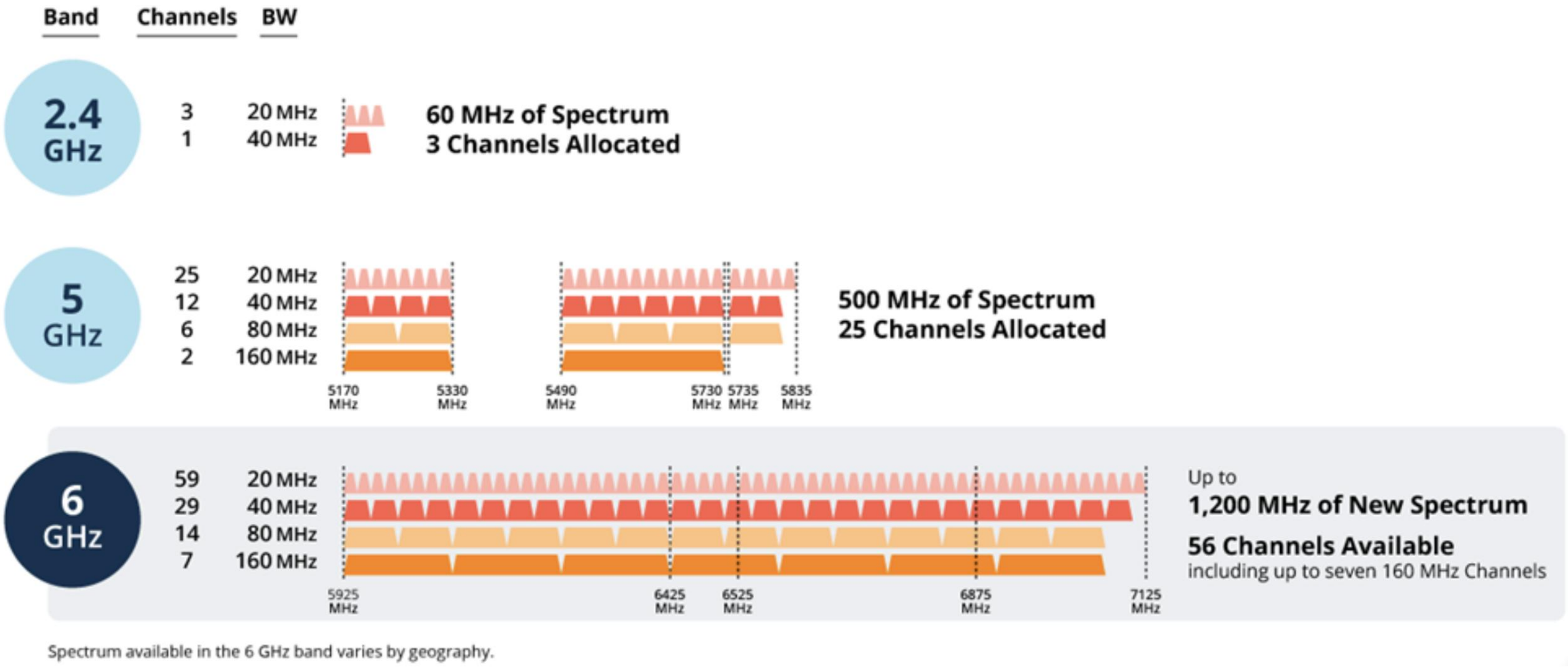
2.4 GHz รองรับ 40 MHz Channel Bonding โดยรวม 20 MHz + 20 MHz



20MHz +20Mhz

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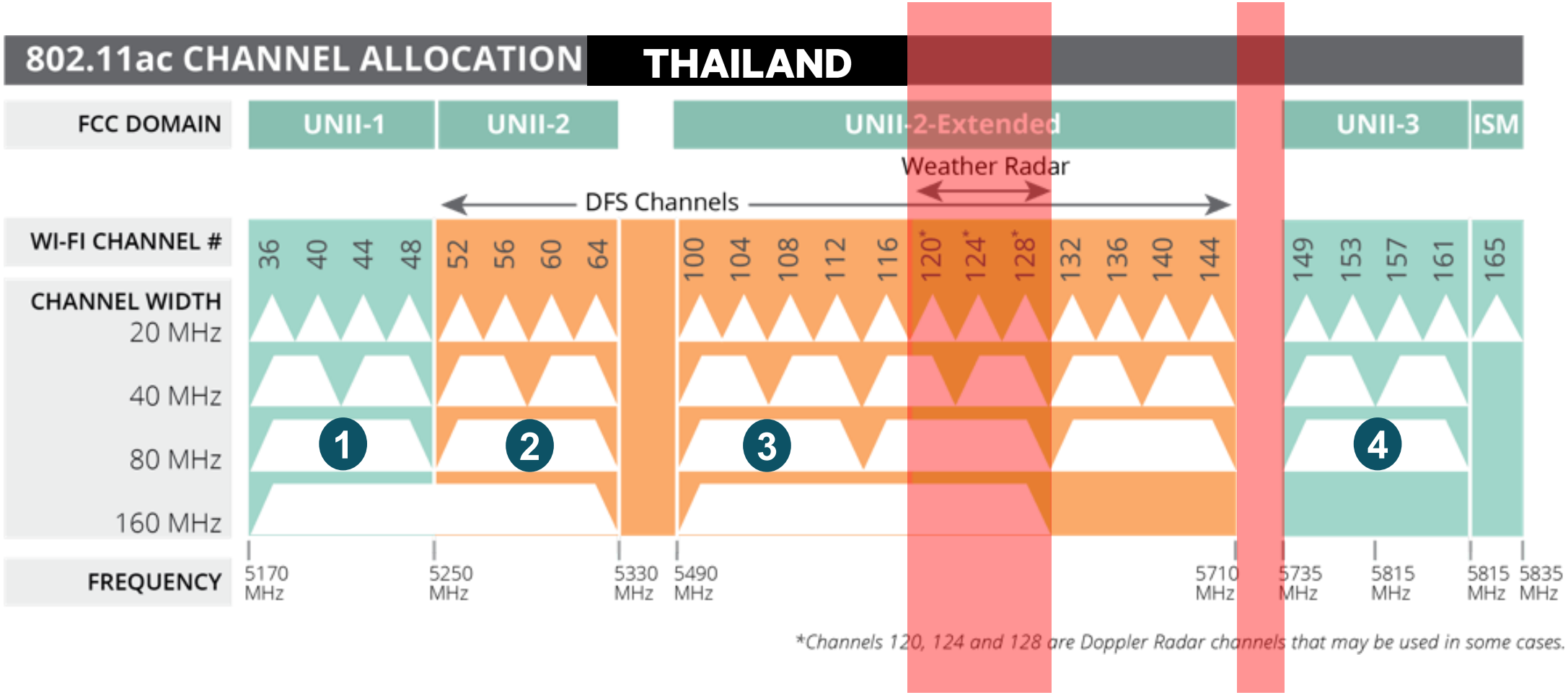
WIFI Spectrum & Channel



Spectrum available in the 6 GHz band varies by geography.

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5Ghz Channel



- https://www.uih.co.th/how-to-design-wi-fi-2/

802.11ac Channel Availability (N America)

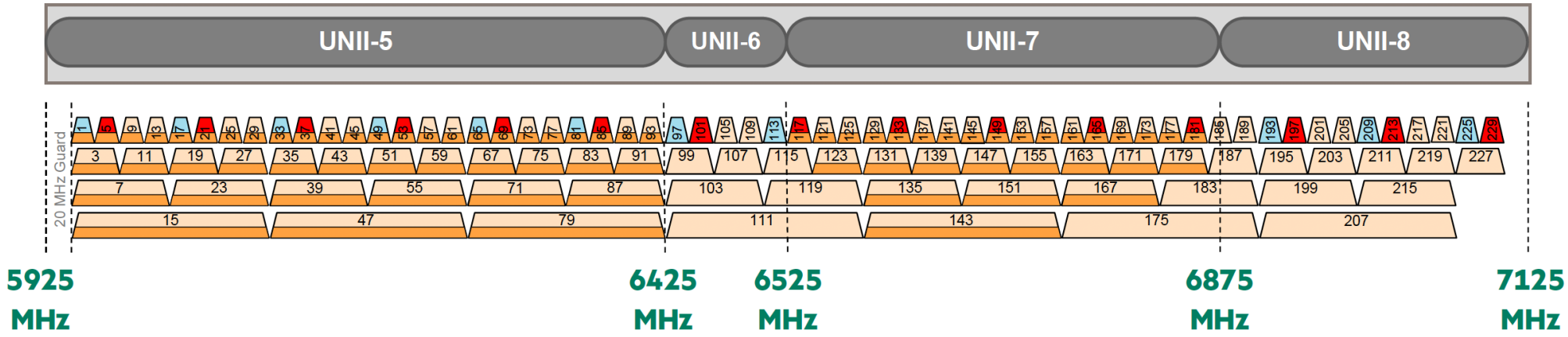
Channel Width	Number of channels available	
	Using DFS	DFS Excluded
40 MHz	9-10*	4
80 MHz	4-5*	2
160 MHz	1	0

*Channels 116 and 132 are Doppler Radar channels that may be used in some cases.

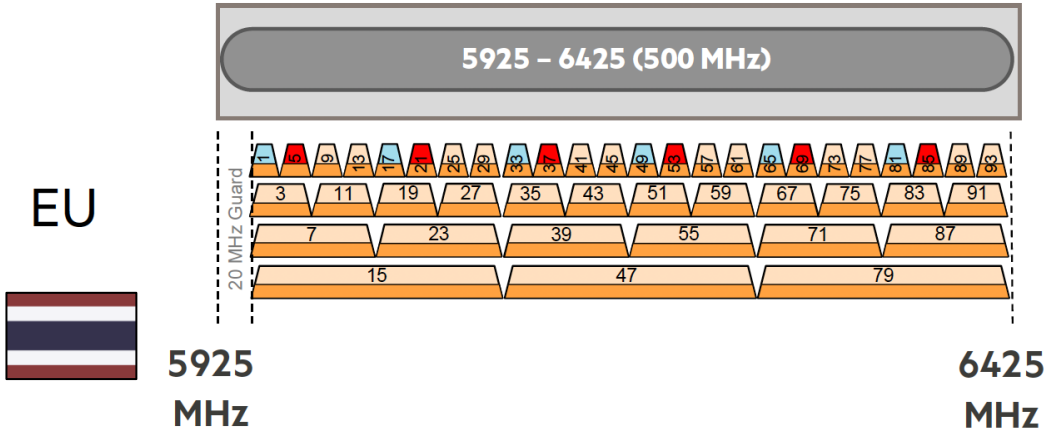
© 2013 SecurityUncorked.com

6 GHz Channels

AMS



EU



	European Model	Americas Model
20 MHz	24	58
40 MHz	12	29
80 MHz	6	14
160 MHz	3	7

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U-NII

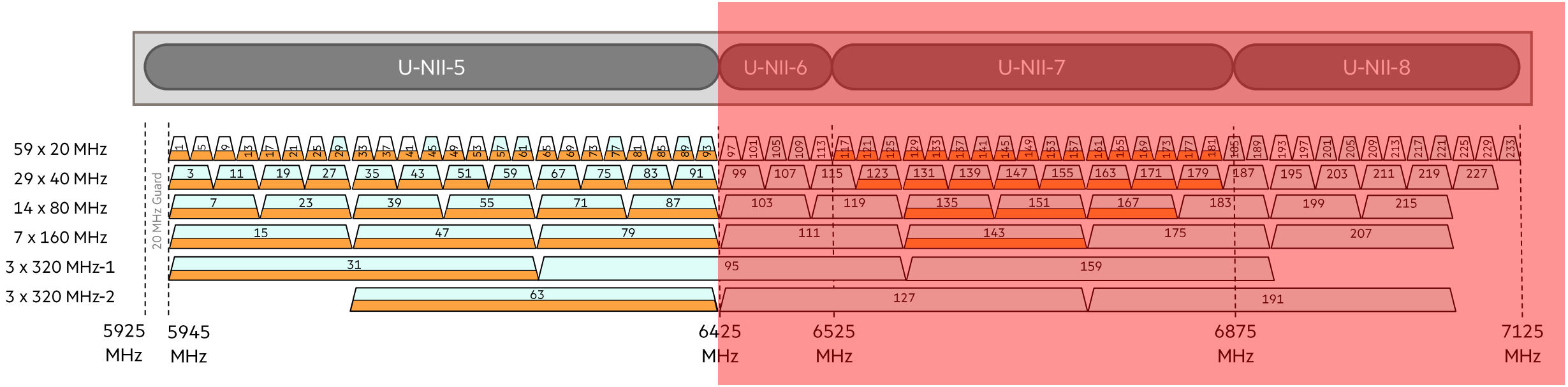
U-NII-5, U-NII-6, U-NII-7 คือการแบ่งย่านความถี่ **6 GHz** ออกเป็นช่วงย่อย ๆ ตามการกำหนดของหน่วยงานกำกับดูแล โดยเฉพาะในสหรัฐฯ

Band	Frequency	Bandwidth
U-NII-5	5.925–6.425 GHz	500 MHz
U-NII-6	6.425–6.525 GHz	100 MHz
U-NII-7	6.525–6.875 GHz	350 MHz
U-NII-8	6.875–7.125 GHz	250 MHz

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WIFI7 : 320 MHz Channel Bonding

- Wi-Fi 7 introduces optional support for 320 MHz channel width in the 6 GHz band only
- 320 MHz doubles the data transfer rates compared Wi-Fi 6 / 6E by doubling widest channel size
- Caveats: (ข้อจำกัด)
  - 320 MHz noise floor baseline starts at -89 dBm
  - Max 3 channels available with limited reuse
  - Only 1 channel available for Standard Power (U-NII-5)



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Modulation <> Channel Bonding vs WIFI Data Rate (upto 802.11ac)

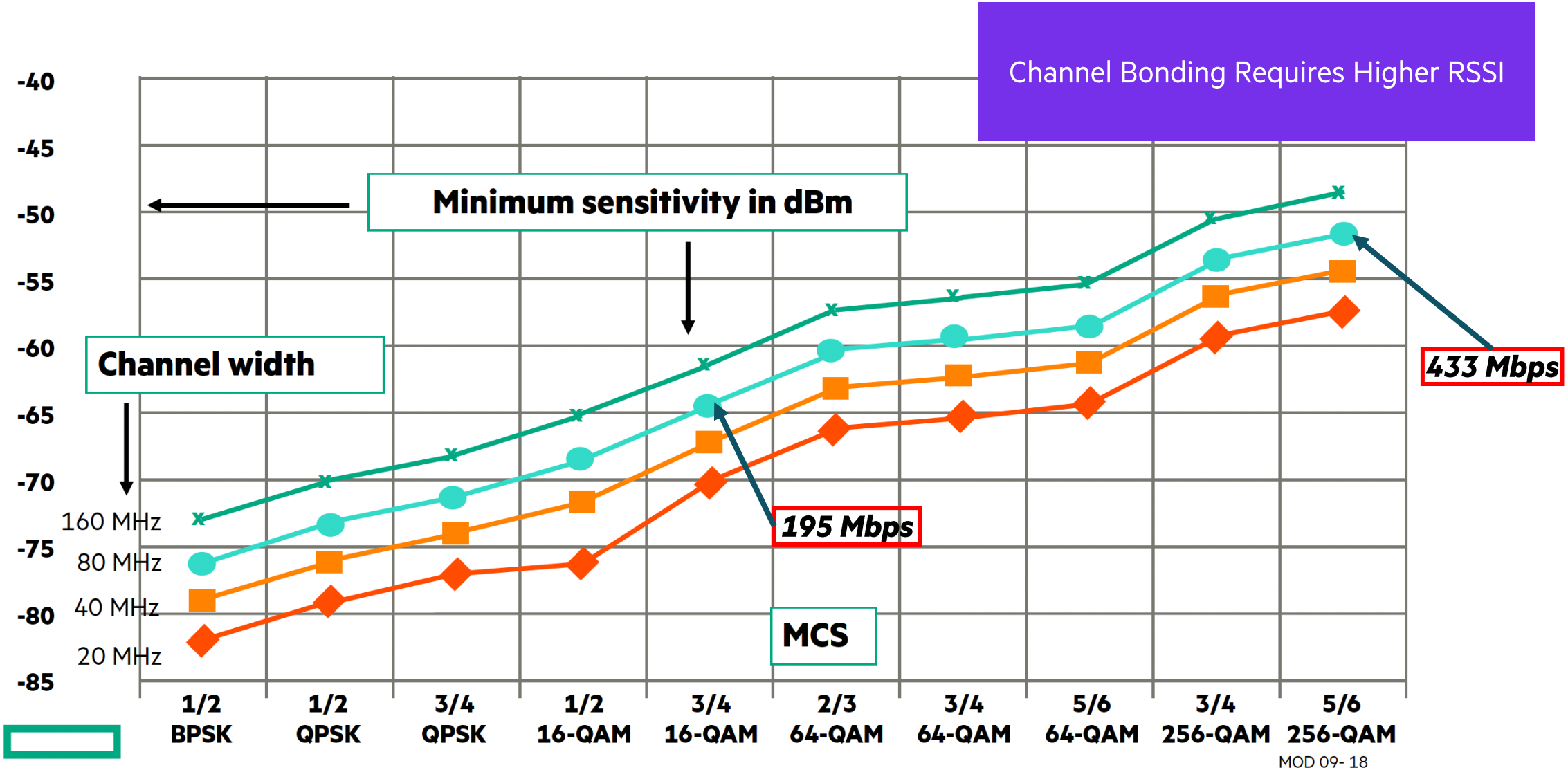
							OFDM (Prior 11ax)							
MCS Index				Spatial Stream	Modulation	Coding	20MHz		40MHz		80MHz		160MHz	
HT	VHT	HE	EHT				0.8µs GI	0.4µs GI	0.8µs GI	0.4µs GI	0.8µs GI	0.4µs GI	0.8µs GI	0.4µs GI
0	0	0	0	1	BPSQ	1/2	6.5	7.2	13.5	15	29.3	32.5	58.5	65
1	1	1	1	1	QPSK	1/2	13	14.4	27	30	58.5	65	117	130
2	2	2	2	1	QPSK	3/4	19.5	21.7	40.5	45	87.8	97.5	175.5	195
3	3	3	3	1	16-QAM	1/2	26	28.9	54	60	117	130	234	260
4	4	4	4	1	16-QAM	3/4	39	43.3	81	90	175.5	195	351	390
5	5	5	5	1	64-QAM	2/3	52	57.8	108	120	234	260	468	520
6	6	6	6	1	64-QAM	3/4	58.5	65	121.5	135	263.3	292.5	526.5	585
7	7	7	7	1	64-QAM	5/6	65	72.2	135	150	292.5	325	585	650
	8	8	8	1	256-QAM	3/4	78	86.7	162	180	351	390	702	780
	9	9	9	1	256-QAM	5/6	N/A	N/A	180	200	390	433.3	780	866.7
		10	10	1	1024-QAM	3/4								
		11	11	1	1024-QAM	5/6								
			12	1	4096-QAM	3/4								
			13	1	4096-QAM	5/6								

Modulation <> Channel Bonding vs WIFI Data Rate (802.11ax+)

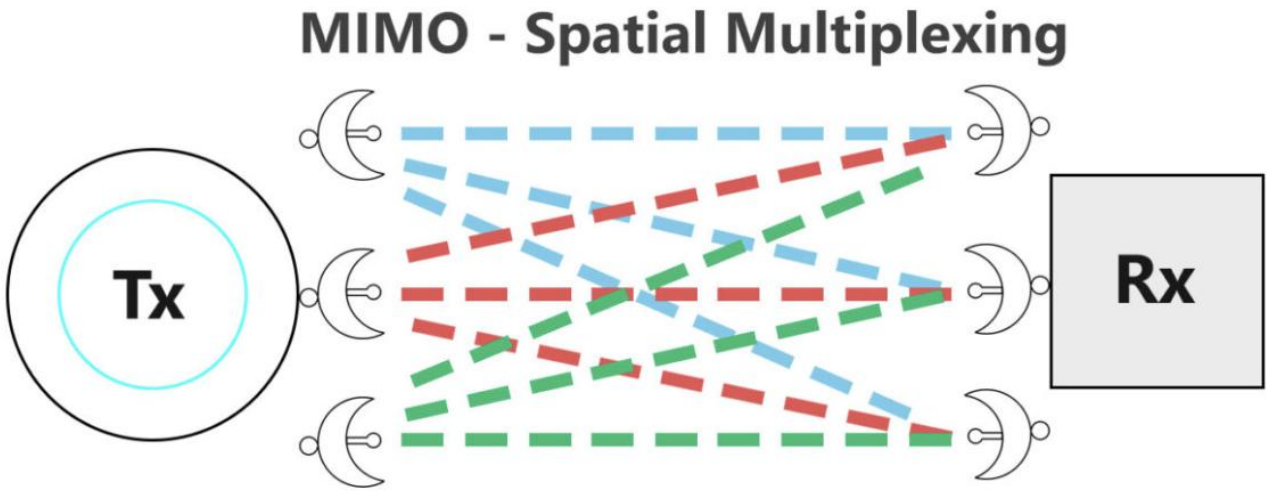
MCS Index				Spatial Stream	Modulation	Coding	OFDM & OFDMA (Starting with 11ax)															
							26-tone RU			52-tone RU			106-tone RU			242-tone RU / 20MHz			484-tone RU / 40MHz			996-tone RU / 80MHz			2x996-tone RU / 160MHz		
HT	VHT	HE	EHT				3.2µs GI	1.6µs GI	0.8µs GI	3.2µs GI	1.6µs GI	0.8µs GI	3.2µs GI	1.6µs GI	0.8µs GI	3.2µs GI	1.6µs GI	0.8µs GI	3.2µs GI	1.6µs GI	0.8µs GI	3.2µs GI	1.6µs GI	0.8µs GI	3.2µs GI	1.6µs GI	0.8µs GI
0	0	0	0	1	BPSQ	1/2	0.8	0.8	0.9	1.5	1.7	1.8	3.2	3.5	3.8	7.3	8.1	8.6	14.6	16.3	17.2	30.6	34.0	36.0	61.3	68.1	72.1
1	1	1	1	1	QPSK	1/2	1.5	1.7	1.8	3.0	3.3	3.5	6.4	7.1	7.5	14.6	16.3	17.2	29.3	32.5	34.4	61.3	68.1	72.1	122.5	136.1	144.1
2	2	2	2	1	QPSK	3/4	2.3	2.5	2.6	4.5	5.0	5.3	9.6	10.6	11.3	21.9	24.4	25.8	43.9	48.8	51.6	91.9	102.1	108.1	183.8	204.2	216.2
3	3	3	3	1	16-QAM	1/2	3.0	3.3	3.5	6.0	6.7	7.1	12.8	14.2	15.0	29.3	32.5	34.4	58.5	65.0	68.8	122.5	136.1	144.1	245.0	272.2	288.2
4	4	4	4	1	16-QAM	3/4	4.5	5.0	5.3	9.0	10.0	10.6	19.1	21.3	22.5	43.9	48.8	51.6	87.8	97.5	103.2	183.8	204.2	216.2	367.5	408.3	432.4
5	5	5	5	1	64-QAM	2/3	6.0	6.7	7.1	12.0	13.3	14.1	25.5	28.3	30.0	58.5	65.0	68.8	117.0	130.0	137.6	245.0	272.2	288.2	490.0	544.4	576.5
6	6	6	6	1	64-QAM	3/4	6.8	7.5	7.9	13.5	15.0	15.9	28.7	31.9	33.8	65.8	73.1	77.4	131.6	146.3	154.9	275.6	306.3	324.3	551.3	612.5	648.5
7	7	7	7	1	64-QAM	5/6	7.5	8.3	8.8	15.0	16.7	17.6	31.9	35.4	37.5	73.1	81.3	86.0	146.3	162.5	172.1	306.3	340.3	360.3	612.5	680.6	720.6
	8	8	8	1	256-QAM	3/4	9.0	10.0	10.6	18.0	20.0	21.2	38.3	42.5	45.0	87.8	97.5	103.2	175.5	195.0	206.5	367.5	408.3	432.4	735.0	816.7	864.7
	9	9	9	1	256-QAM	5/6	10.0	11.1	11.8	20.0	22.2	23.5	42.5	47.2	50.0	97.5	108.3	114.7	195.0	216.7	229.4	408.3	453.7	480.4	816.7	907.4	960.8
		10	10	1	1024-QAM	3/4	11.3	12.5	13.2	22.5	25.0	26.5	47.8	53.1	56.3	109.7	121.9	129.0	219.4	243.8	258.1	459.4	510.4	540.4	918.8	1020.8	1080.9
		11	11	1	1024-QAM	5/6	12.5	13.9	14.7	25.0	27.8	29.4	53.1	59.0	62.5	121.9	135.4	143.4	243.8	270.8	286.8	510.4	567.1	600.5	1020.8	1134.3	1201.0
			12	1	4096-QAM	3/4	13.5	15.0	15.9	27.0	30.0	31.8	57.4	63.8	67.5	131.6	146.3	154.9	263.3	292.5	309.7	551.3	612.5	648.5	1102.5	1225.0	1297.1
			13	1	4096-QAM	5/6	15.0	16.7	17.6	30.0	33.3	35.3	63.8	70.8	75.0	146.3	162.5	172.1	292.5	325.0	344.1	612.5	680.6	720.6	1225.0	1361.1	1441.2

https://mcsindex.net/

Channel width and Minimum Sensitivity



MIMO (Multiple-Input Multiple-Output) & Spatial Multiplexing



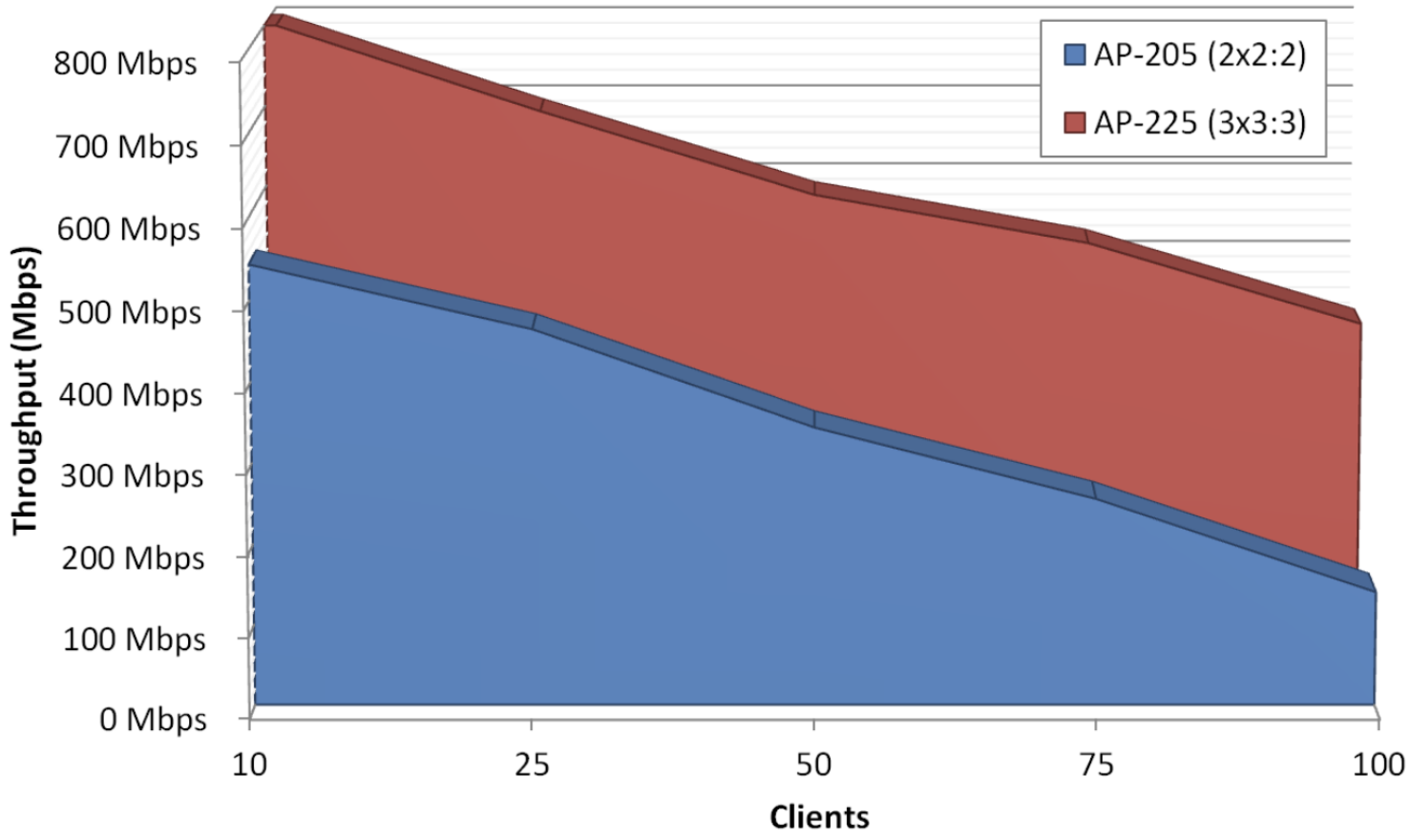
Spatial multiplexing where the idea is to increase the data transfer rate since more than one independent stream of data is transmitted via multiple streams. The risk comes from interference which is why the data streams aren't transmitted at the same time, but are phased out at different points in time. Another method that helps move data without risking collision or interference is by dividing the bandwidth into multiple frequency bands, each used to stream an independent and separate signal.

MIMO vs WIFI Data Rate

MCS Index				Spatial Stream	Modulation	Coding	OFDM (Prior 11ax)							
							20MHz		40MHz		80MHz		160MHz	
HT	VHT	HE	EHT				0.8µs GI	0.4µs GI	0.8µs GI	0.4µs GI	0.8µs GI	0.4µs GI	0.8µs GI	0.4µs GI
0	0	0	0	1	BPSQ	1/2	6.5	7.2	13.5	15	29.3	32.5	58.5	65
1	1	1	1	1	QPSK	1/2	13	14.4	27	30	58.5	65	117	130
2	2	2	2	1	QPSK	3/4	19.5	21.7	40.5	45	87.8	97.5	175.5	195
3	3	3	3	1	16-QAM	1/2	26	28.9	54	60	117	130	234	260
4	4	4	4	1	16-QAM	3/4	39	43.3	81	90	175.5	195	351	390
5	5	5	5	1	64-QAM	2/3	52	57.8	108	120	234	260	468	520
6	6	6	6	1	64-QAM	3/4	58.5	65	121.5	135	263.3	292.5	526.5	585
7	7	7	7	1	64-QAM	5/6	65	72.2	135	150	292.5	325	585	650
	8	8	8	1	256-QAM	3/4	78	86.7	162	180	351	390	702	780
	9	9	9	1	256-QAM	5/6	N/A	N/A	180	200	390	433.3	780	866.7
		10	10	1	1024-QAM	3/4								
		11	11	1	1024-QAM	5/6								
			12	1	4096-QAM	3/4								
			13	1	4096-QAM	5/6								
8	0	0	0	2	BPSQ	1/2	13	14.4	27	30	58.5	65	117	130
9	1	1	1	2	QPSK	1/2	26	28.9	54	60	117	130	234	260
10	2	2	2	2	QPSK	3/4	39	43.3	81	90	175.5	195	351	390
11	3	3	3	2	16-QAM	1/2	52	57.8	108	120	234	260	468	520
12	4	4	4	2	16-QAM	3/4	78	86.7	162	180	351	390	702	780
13	5	5	5	2	64-QAM	2/3	104	115.6	216	240	468	520	936	1040
14	6	6	6	2	64-QAM	3/4	117	130	243	270	526.5	585	1053	1170
15	7	7	7	2	64-QAM	5/6	130	144.4	270	300	585	650	1170	1300
	8	8	8	2	256-QAM	3/4	156	173.3	324	360	702	780	1404	1560
	9	9	9	2	256-QAM	5/6	N/A	N/A	360	400	780	866.7	1560	1733.3
		10	10	2	1024-QAM	3/4								
		11	11	2	1024-QAM	5/6								
			12	2	4096-QAM	3/4								
			13	2	4096-QAM	5/6								

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WIFI Spatial Stream (~MIMO)



Consideration

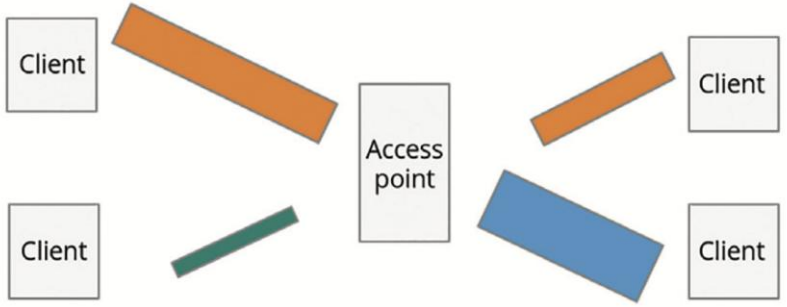
- Access Point
- Client

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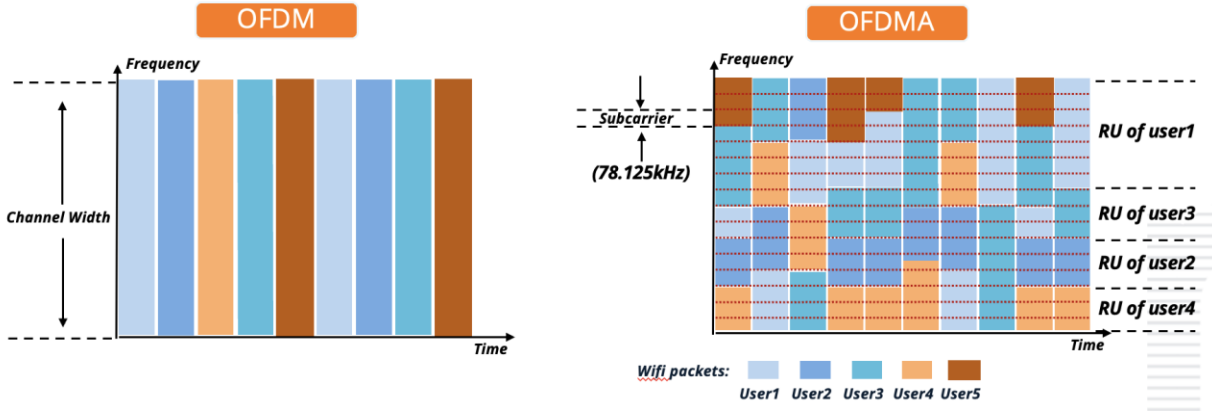
WIFI6 Multi-User Technology

Wi-Fi 6 สามารถให้บริการหลาย Client พร้อมกัน
มี 2 เทคโนโลยีหลักคือ

1. MU-MIMO
2. OFDMA



4 clients, 3 groups



- **Downlink and uplink multi-user MIMO:** The downlink version extends an existing 802.11ac feature where an access point determines that multipath conditions allow it to send, in a single time-interval, frames to different client devices. 802.11ax increases the size of downlink MU-MIMO groups, allowing more efficient operation.

Uplink multi-user MIMO is a new addition to 802.11ax, but is deferred to wave 2: like uplink OFDMA, the access point must coordinate the simultaneous transmissions of multiple clients.

มีหลายเสาอากาศ และสามารถสร้าง **หลาย Spatial Streams** เพื่อส่งข้อมูลให้ Client หลายตัวพร้อมกัน

- **Downlink and Uplink OFDMA:** OFDMA is one of the more complex features in 802.11ax. It allows a single transmission (for downlink OFDMA, the access point transmits) to be split by frequency within a channel, such that different frames addressed to different client devices use groups of subcarriers. Uplink OFDMA is equivalent to downlink OFDMA, but in this case multiple client devices transmit simultaneously, on different groups of subcarriers within the same channel. Uplink OFDMA is more difficult to manage than the downlink variety, as many different clients must be coordinated: the access point transmits trigger frames to indicate which sub-channels each client can use.

แทนที่จะให้ Client หนึ่งคนใช้ทั้ง Channel ก็แบ่ง Channel ออกเป็นกลุ่มของ Subcarriers หรือ **Resource Units (RU) แล้วให้หลาย User ใช้คนละ RU ในเวลาเดียวกัน**

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OFDMA

OFDM : ผู้ใช้หนึ่งคนใช้ Channel ทั้งหมดในช่วงเวลาหนึ่ง

OFDMA : เป็นเทคนิค Multiple Access ที่แบ่งความถี่ (Channel) ออกเป็นกลุ่มของ Subcarriers (RU) เพื่อให้ผู้ใช้หลายคนสามารถใช้ Channel เดียวกันได้ โดย**ใช้คนละ RU ในเวลาเดียวกัน**

A 4G cellular technology to combine different users in the same channel frequency.

One channel is divided into smaller sub-channels, called Resource Unit – RU.

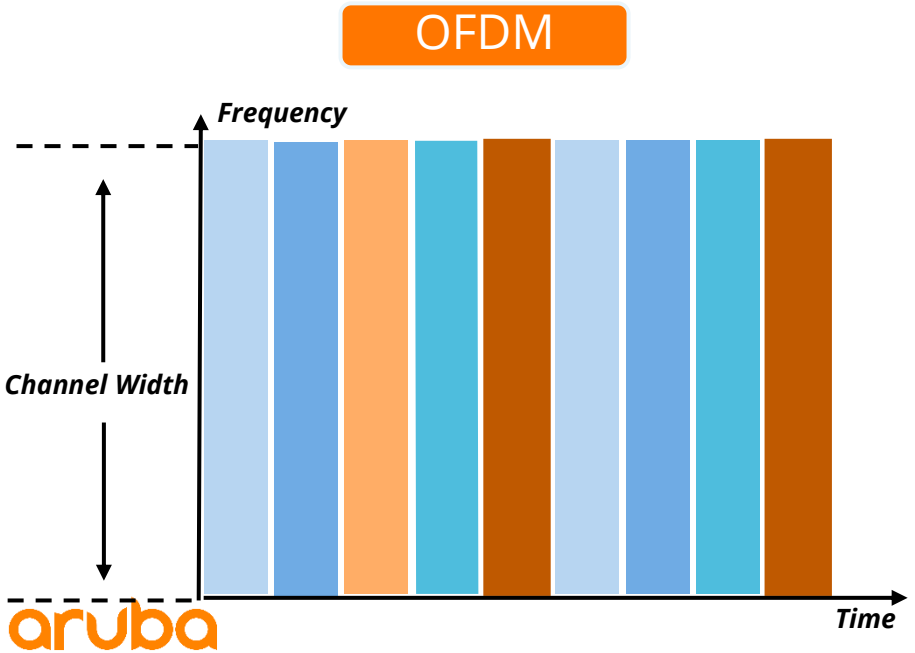
AP assigns RUs to each user per TXOP to maximize UL & DL transmission efficiency

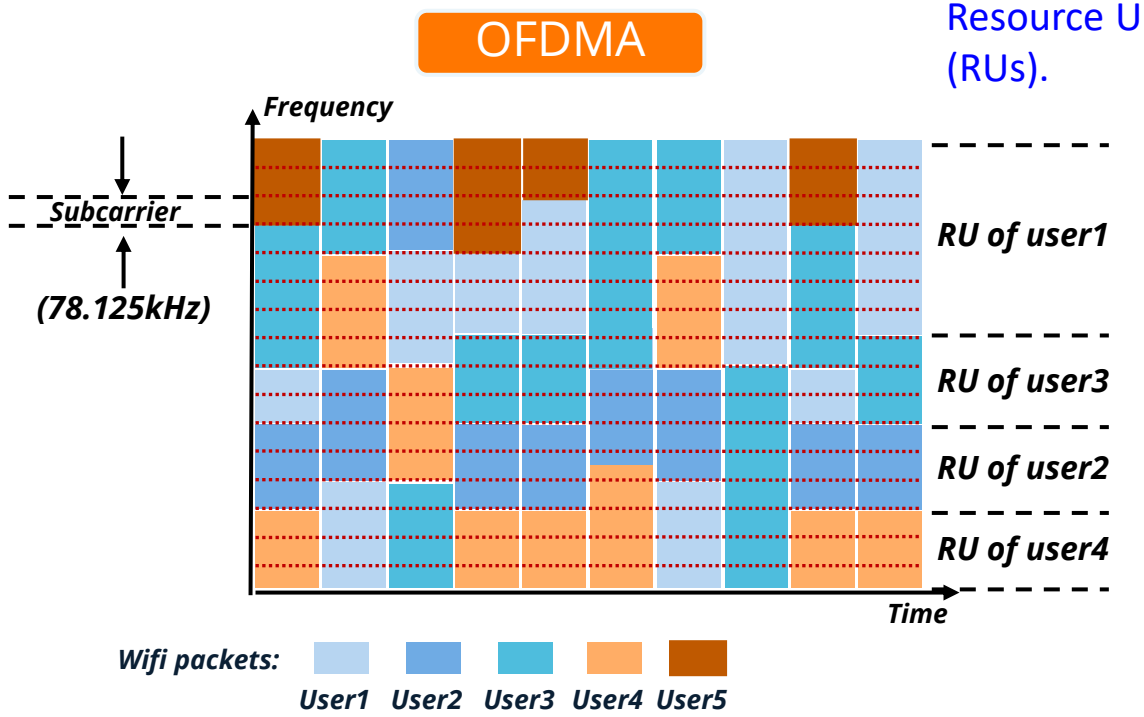
Transmit power is adjusted per RU to improve SINR for specific user

Increase average end user throughputs & overall efficiency, decrease inefficient contention for shared resources.

OFDMA
Multiple users share the same channel at the same time using different Resource Units (RUs).

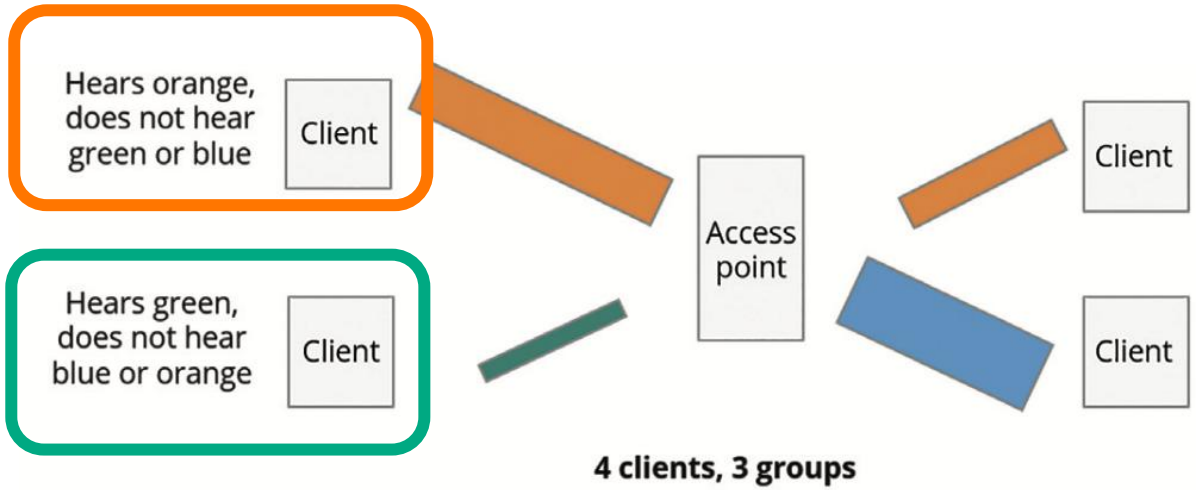
OFDM
One user uses the whole channel at a time.



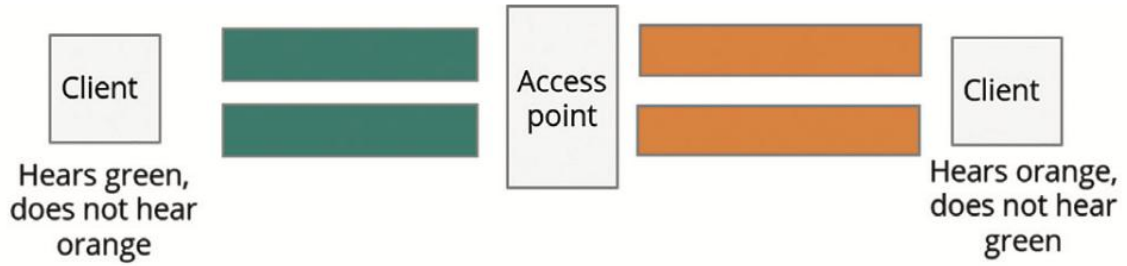


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Mu-MIMO Limitation in Real World Application



4 clients, 3 groups



2 clients, 2 groups (dual-stream clients)

MU-MIMO is only possible where propagation characteristics allow the AP to identify that a transmission optimized for one client or group of clients ***will not be heard at a significant signal strength by another client***, and vice versa. These are the conditions that allow it to build separate data frames for each client group, and transmit them simultaneously.

Simulations indicate that MU-MIMO is most effective with long packets and high SNR, while ***OFDMA is effective over the whole range of client densities and packet length distributions.***

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WIFI Foundation : Channel Re-Use

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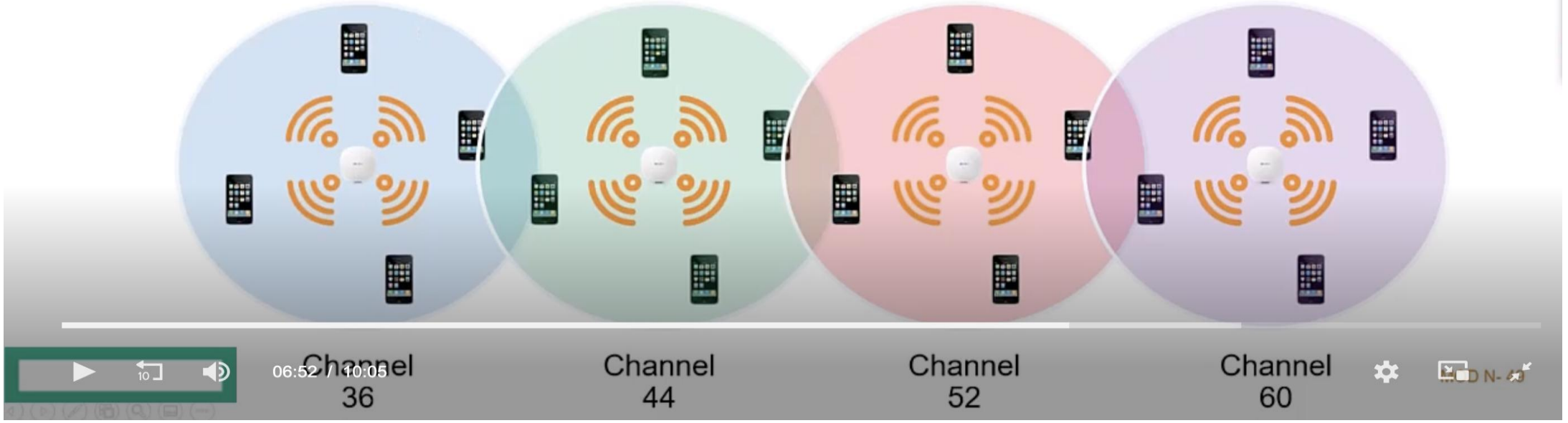
CSMA/CA & Co-Channel Interference (CCI)

– Minimizing CCI is an important concept when designing a network

– Before a radio transmits, it listens for other radios transmitting on its channel

– In the illustration, each AP is on a different channel, there is no CCI between the RF cells

– Each RF cell can have a transmission happening at the same time (4 simultaneous transmissions)



HPE aruba networking

CSMA/CA & Co-Channel Interference (CCI)

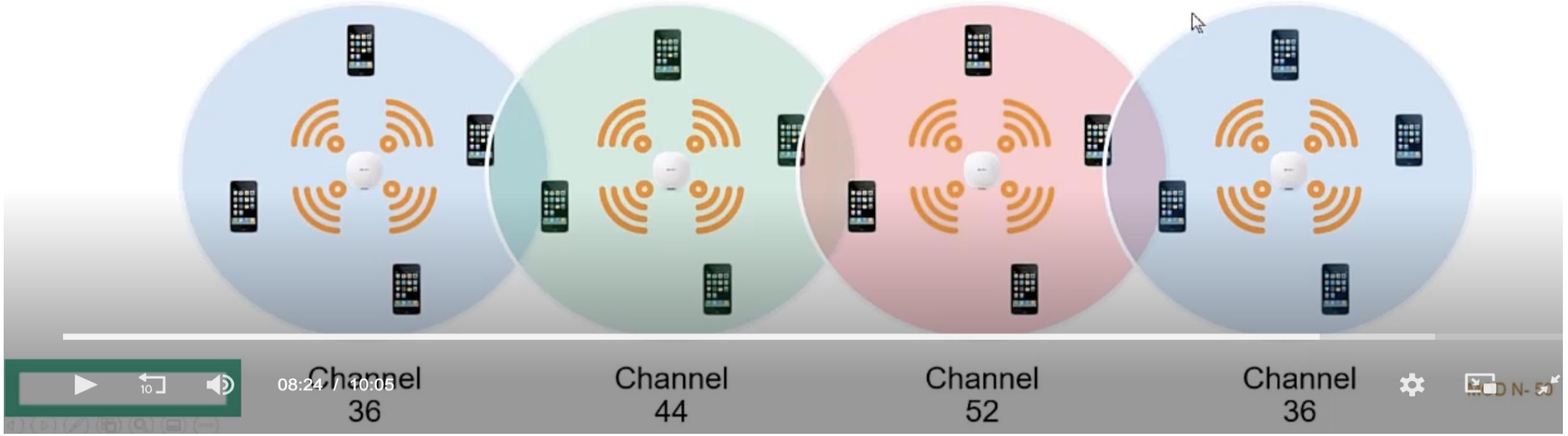
– In this illustration, two RF cells are using channel 36

– Co-Channel Interference between the two RF cells is possible

– The radios from the one cell cannot hear the radios from the other cell

– Since the two cells are far apart, CCI does not occur

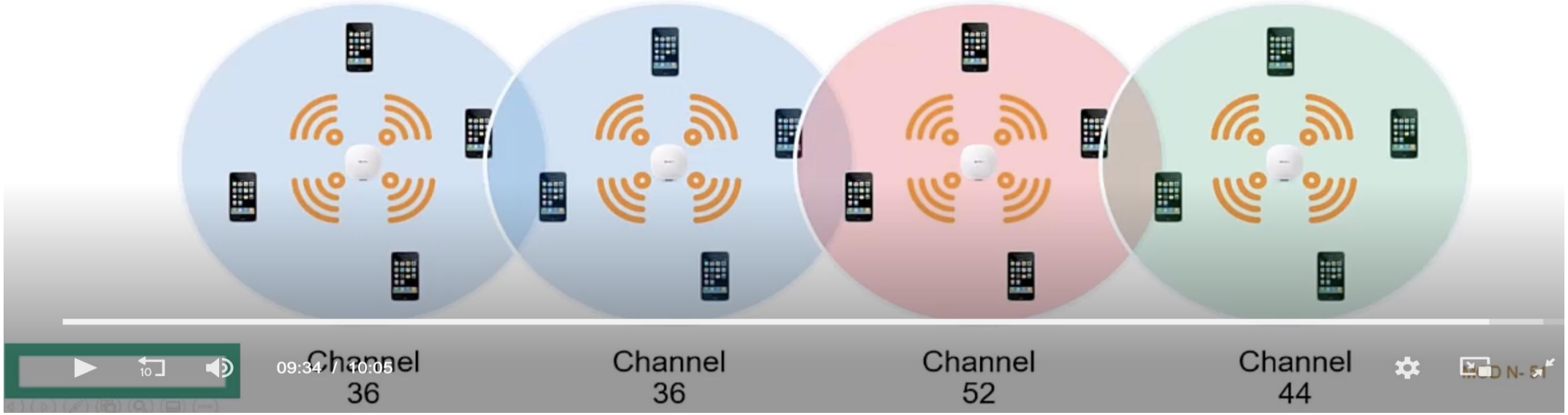
– Each RF cell can have a transmission happening at the same time (4 simultaneous transmissions)



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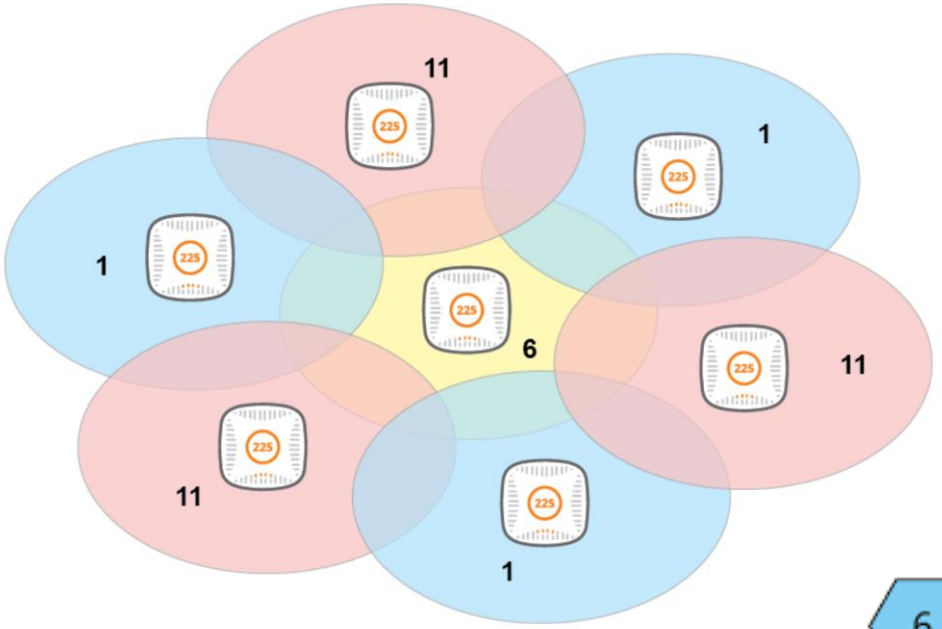
CSMA/CA & Co-Channel Interference (CCI)

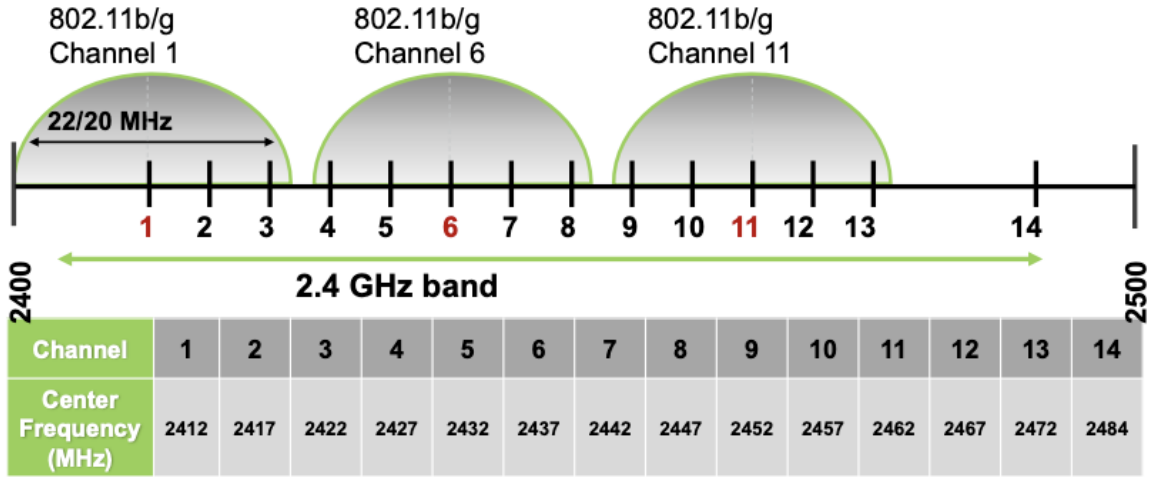
- In this illustration, two RF cells are using channel 36 and are next to each other
- When a radio from one RF cell hears a radio on another RF cell, CCI exists
- Both RF cells share channel 36, and only one device can transmit at a time on channel 36
- Since the two RF cells are experiencing CCI, only 3 simultaneous transmissions can occur
- CCI = less throughput
- More channels = less chance of CCI



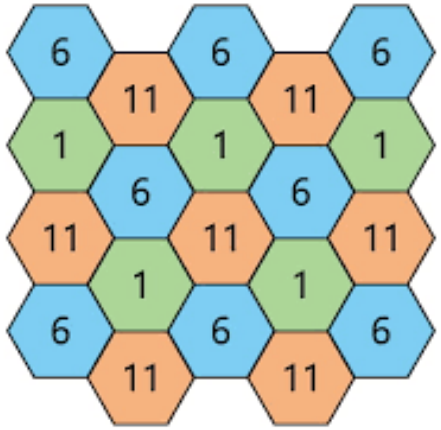
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2.4G Channel Re-Use (Best Practice)





Channel	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Center Frequency (MHz)	2412	2417	2422	2427	2432	2437	2442	2447	2452	2457	2462	2467	2472	2484

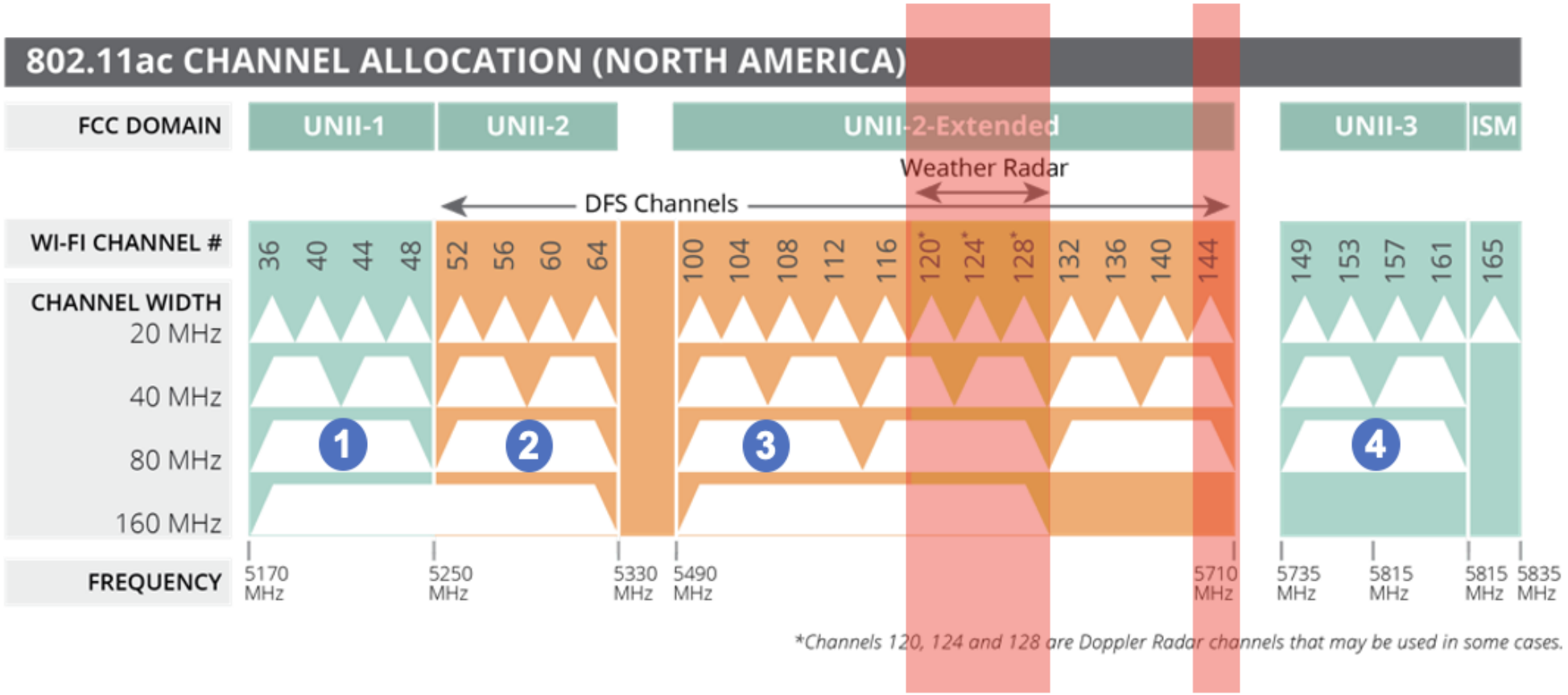


Goal = Minimize CCI

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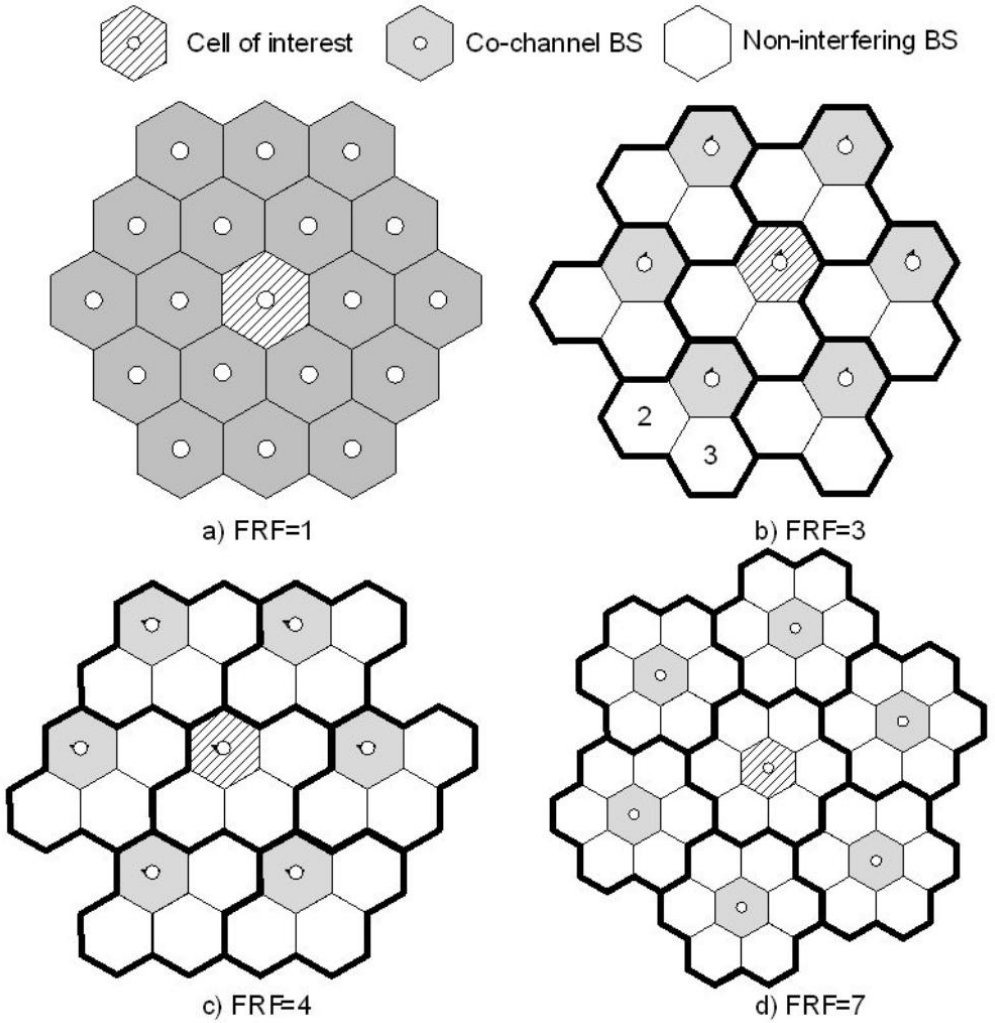
5G Channel Re-Use

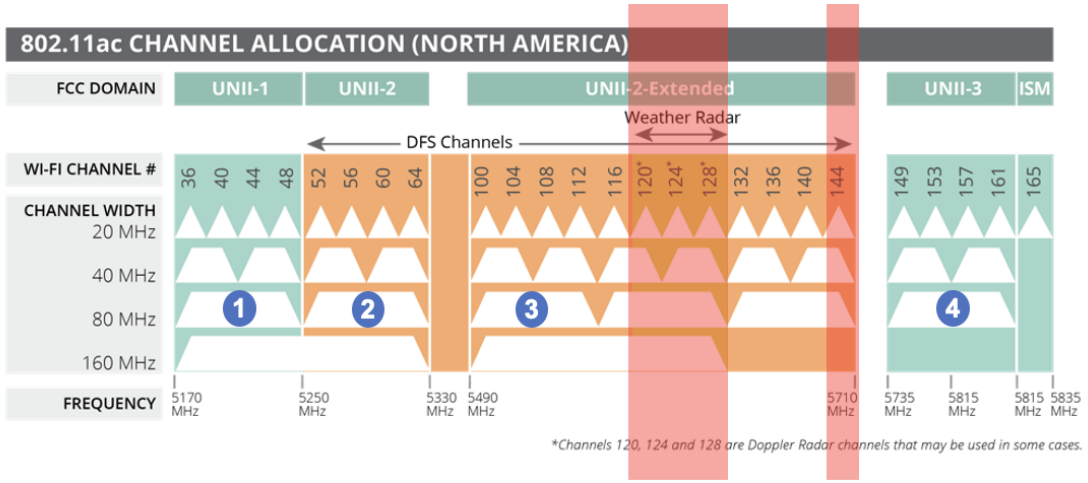
How many channel should we use ?



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5G Channel Re-Use

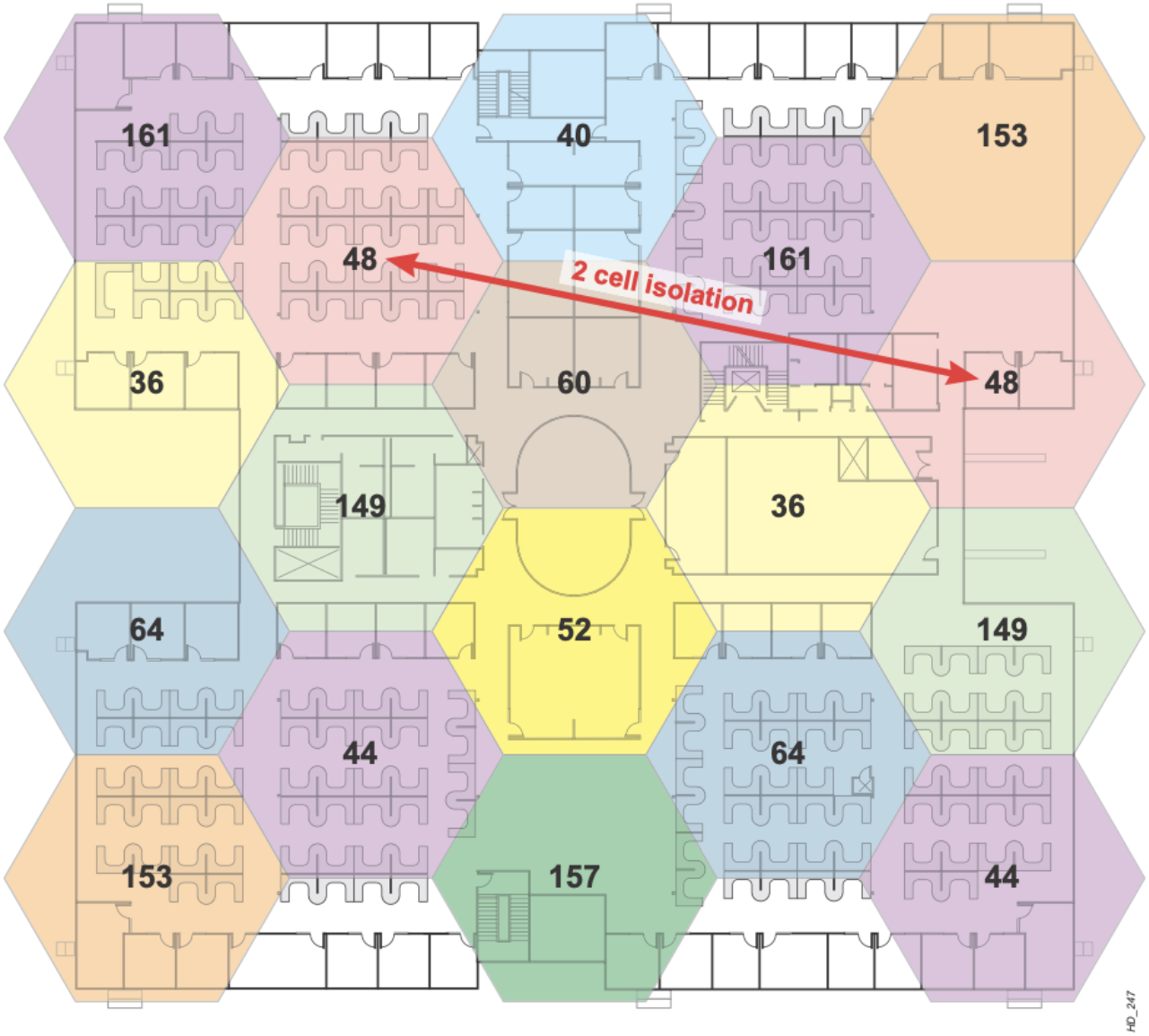




Goal = Minimize CCI

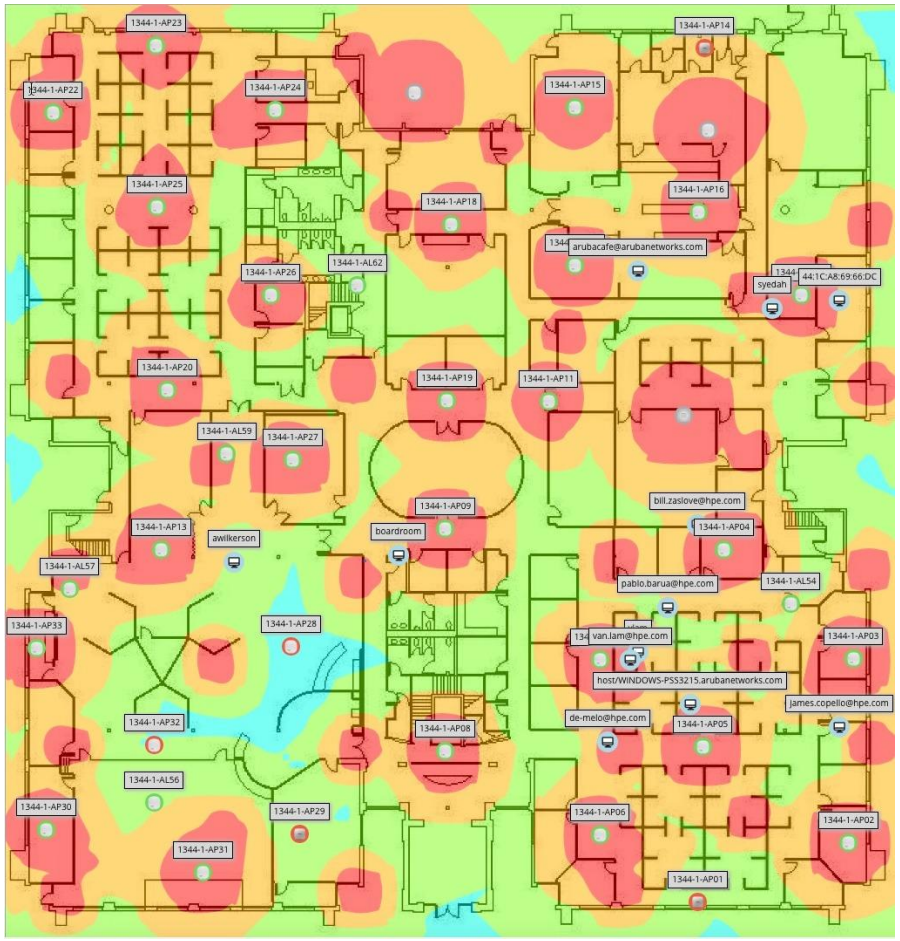
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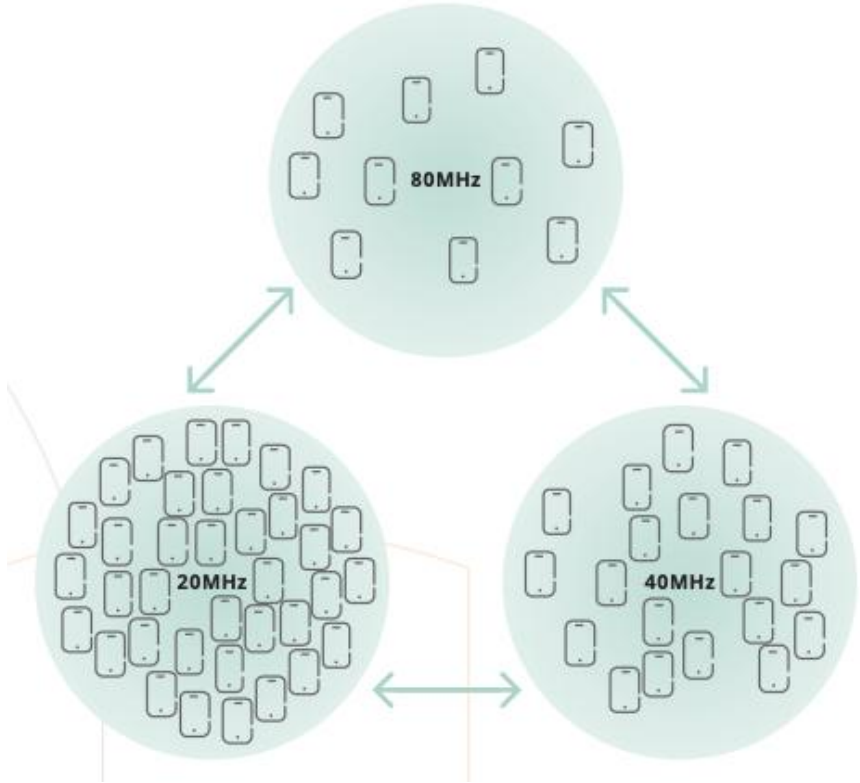
Good Co-Channel Re-Use



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How can we assign the channel ?





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Airmatch : Intelligent RF Resource Management

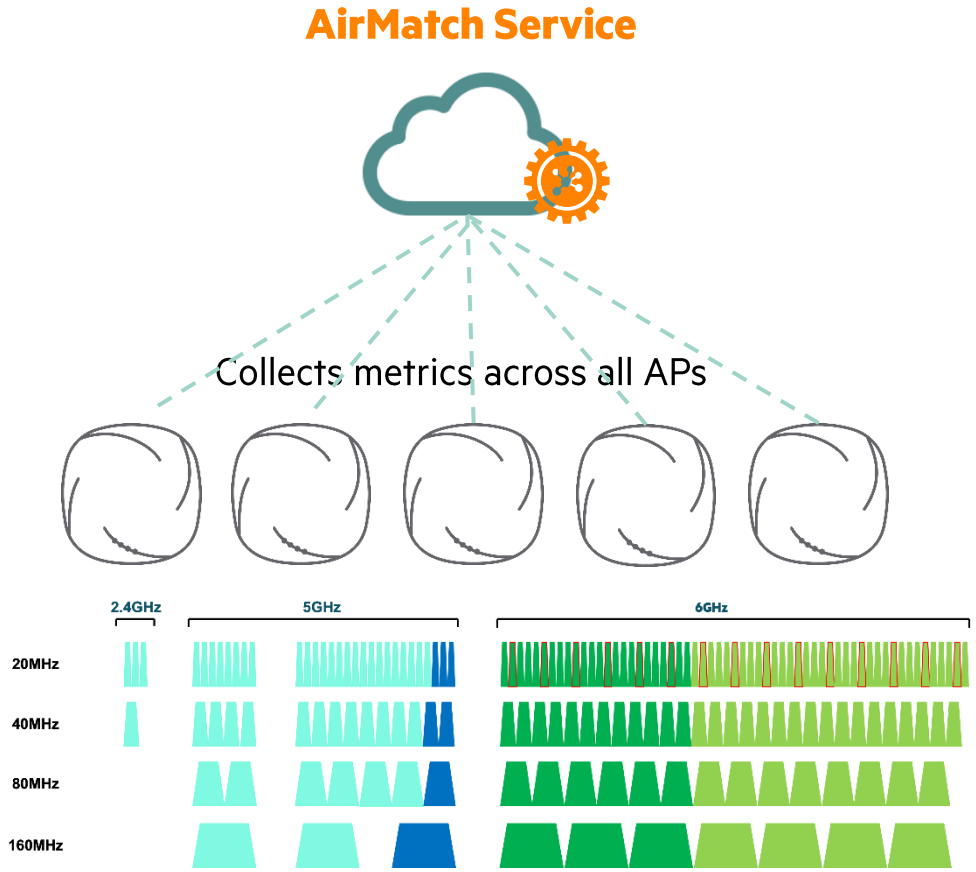
Airmatch to self-optimize channels and RF

Mitigates co-channel interference, high noise, and radar events by proactive channel assignment

Dynamic bandwidth adjustments as per changing RF environment

Enhanced roaming as a result of even distribution of EIRP across the RF partition

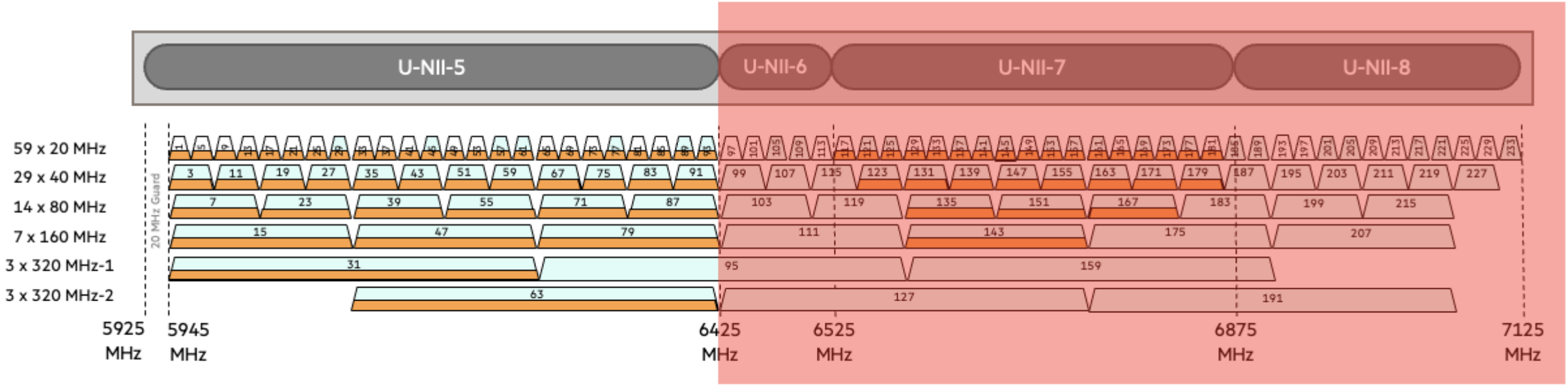
Supports automated configuration for 6GHz



Dynamically optimizes channel width and RF per AP

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6G Channel Re-Use

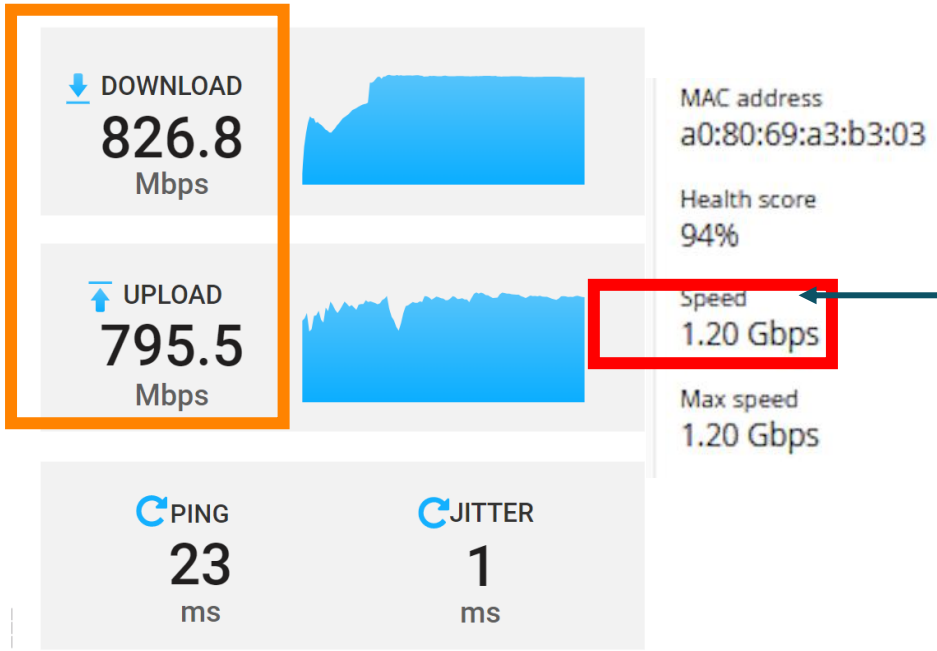


How many channel should we use ?

1. ***US : 7x160 MHz , 3x320 MHz***
2. ***Thailand : 3x160 MHz , 1x320 MHz***

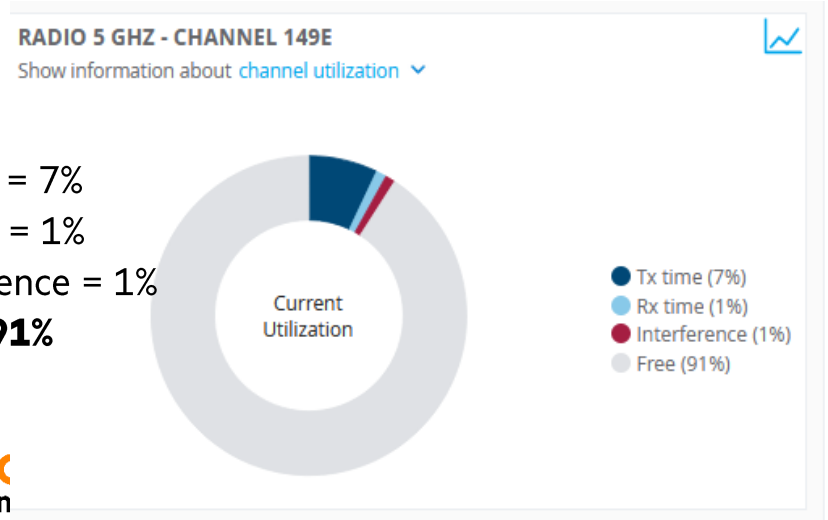
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Throughout vs Physical Rate



Client Phy Rate @ 1200Mbps

Maximum Throughput 50-70%



- Tx time = 7%
- Rx time = 1%
- Interference = 1%
- Free = **91%**

Low Channel Utilization

ถึงแม้ว่า Channel จะว่างมาก ก็ไม่ได้หมายความว่า Throughput = Physical Rate

Coverage vs Capacity design

Coverage designs:

- Fewer APs
- APs set to higher power
- Typical: Few clients
- Low Throughput needs

Capacity design:

- More APs
- APs set to lower power
- Typical: Many clients
- High aggregate throughput needs

Coverage Design

สัญญาณถึงหรือไม่

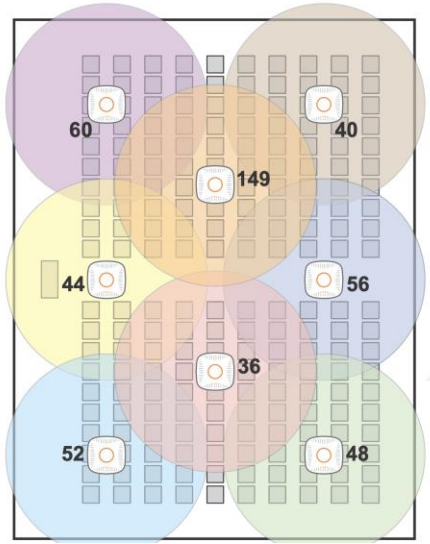
สนใจ RSSI

มี Wi-Fi ใช้งานได้

เน้นพื้นที่ครอบคลุม

AP อาจห่างกันได้

“มีสัญญาณ”



Capacity Design

สัญญาณแรงพอไหม

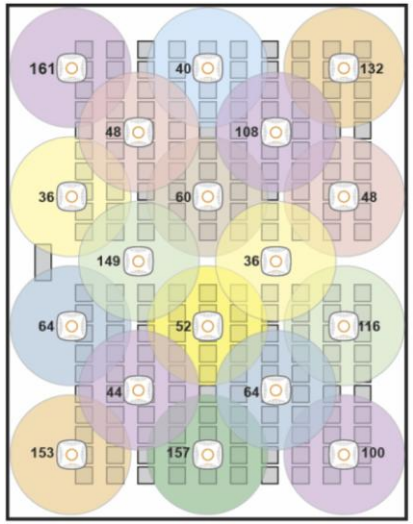
สนใจ SNR

ใช้ MCS สูงได้หรือไม่

เน้นประสิทธิภาพ/จำนวน User

อาจต้องเพิ่ม AP

“มีความจุเพียงพอ”



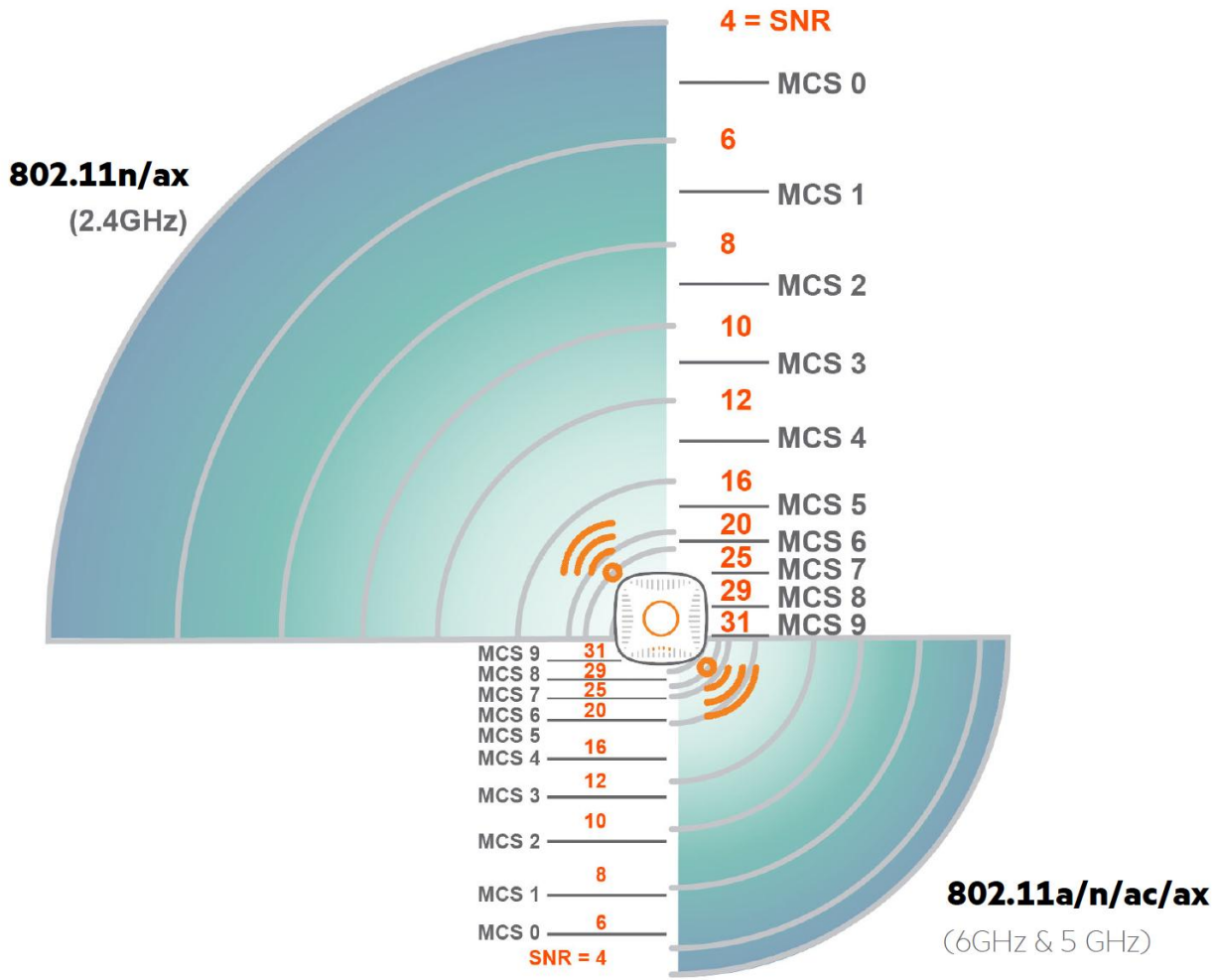
RSSI ⟶ SNR ⟶ MCS ⟶ PHY Rate ⟶ Airtime ⟶ Throughput

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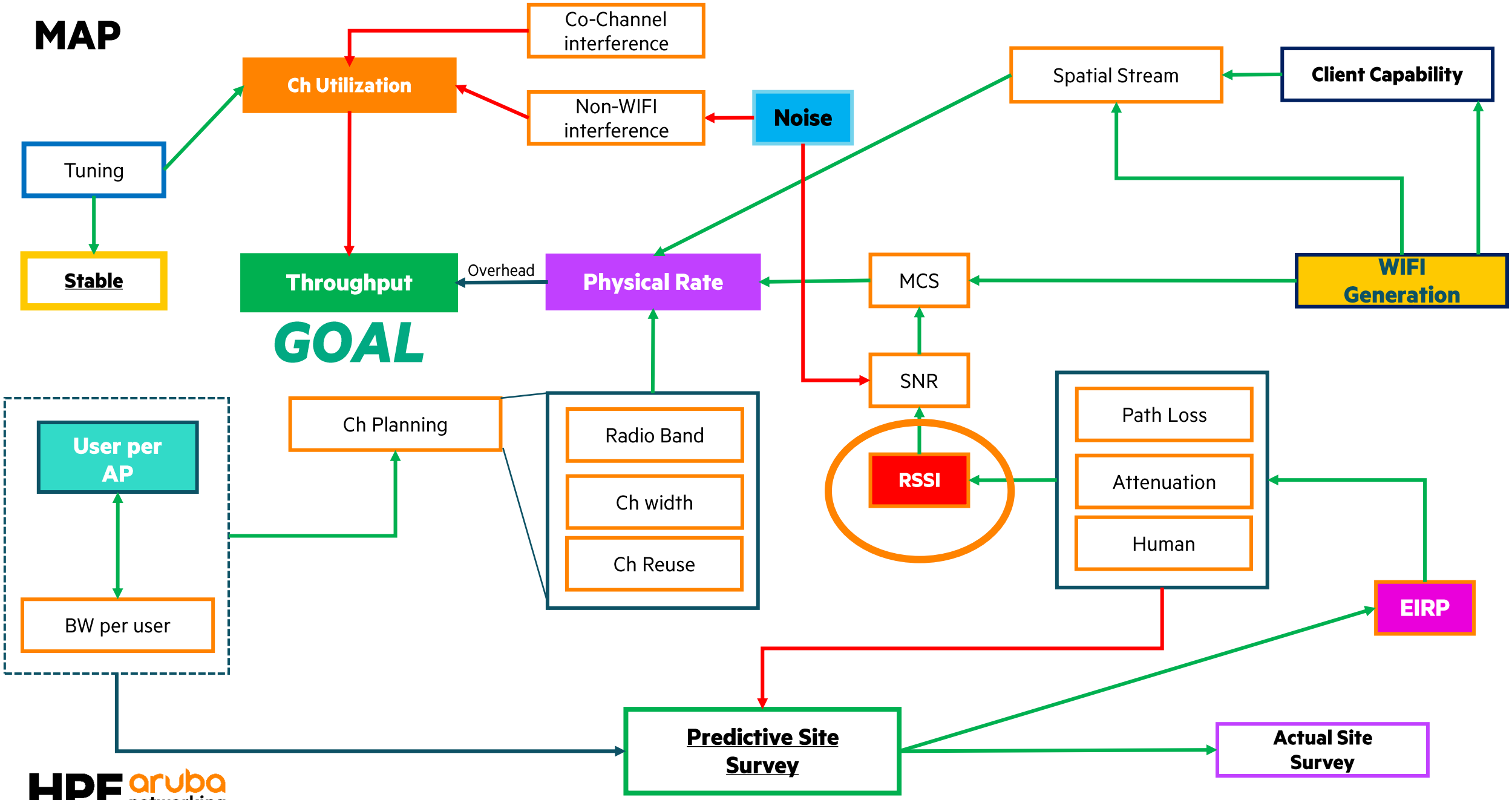
Coverage vs Capacity design

Goal : Highest MCS Possible

Why ?



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MAP
Co-Channel interference
Ch Utilization
Non-WIFI interference
Noise
Spatial Stream
Client Capability
Tuning
Stable
Throughput
Overhead
Physical Rate
MCS
WIFI Generation
GOAL
SNR
Ch Planning
Radio Band
Ch width
Ch Reuse
Path Loss
Attenuation
Human
RSSI
User per AP
BW per user
EIRP
Predictive Site Survey
Actual Site Survey
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Client Capability & Monitoring

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Iphone 16 Pro

Wi-Fi specification details for iPhone models

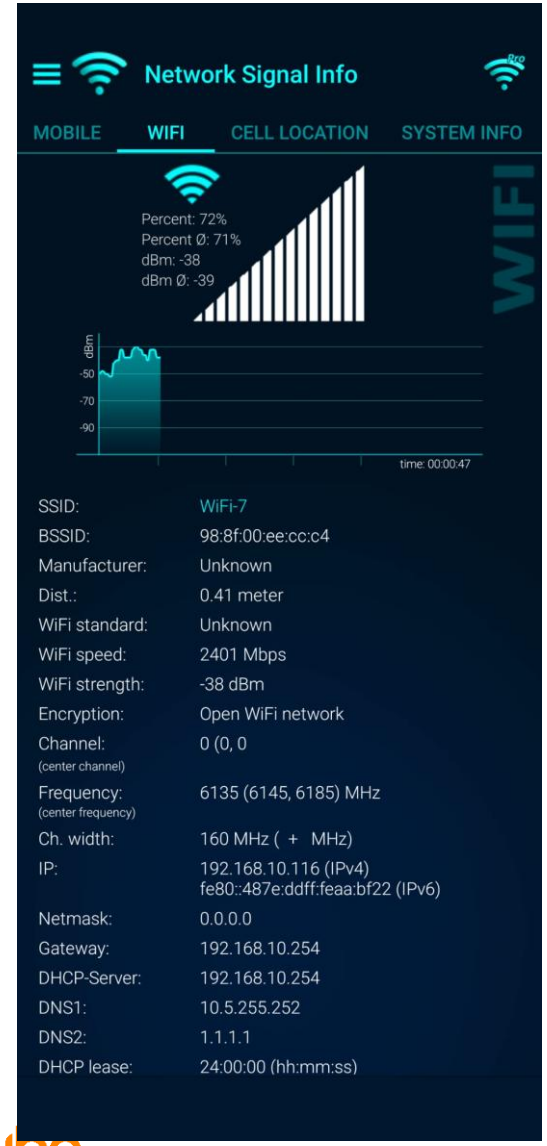
Wi-Fi specifications for all iPhone 16 models are detailed in the table below. All iPhone 16 models support Wi-Fi 7 in the 2.4 GHz, 5 GHz, and 6 GHz bands and Multi-Link Operation (MLO).

802.11 standard, name, frequency	Maximum PHY data rate	Maximum channel bandwidth	Maximum MCS index	Maximum spatial streams / Type
be@6 GHz	2400 Mbps	160 MHz	11 (EHT)	2/MIMO
be@5 GHz	2400 Mbps	160 MHz	11 (EHT)	2/MIMO
be@2.4 GHz	229 Mbps	20 MHz	9 (EHT)	2/MIMO
ax@6 GHz	2400 Mbps	160 MHz	11 (HE)	2/MIMO
ax@5 GHz	2400 Mbps	160 MHz	11 (HE)	2/MIMO
ac@5 GHz	866 Mbps	80 MHz	9 (VHT)	2/MIMO
a/n@5 GHz	300 Mbps	40 MHz	7 (HT)	2/MIMO
ax@2.4 GHz	229 Mbps	20 MHz	9 (HE)	2/MIMO
b/g/n@2.4 GHz	144 Mbps	20 MHz	7 (HT)	2/MIMO

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Tools and Command

Know You connection



Android

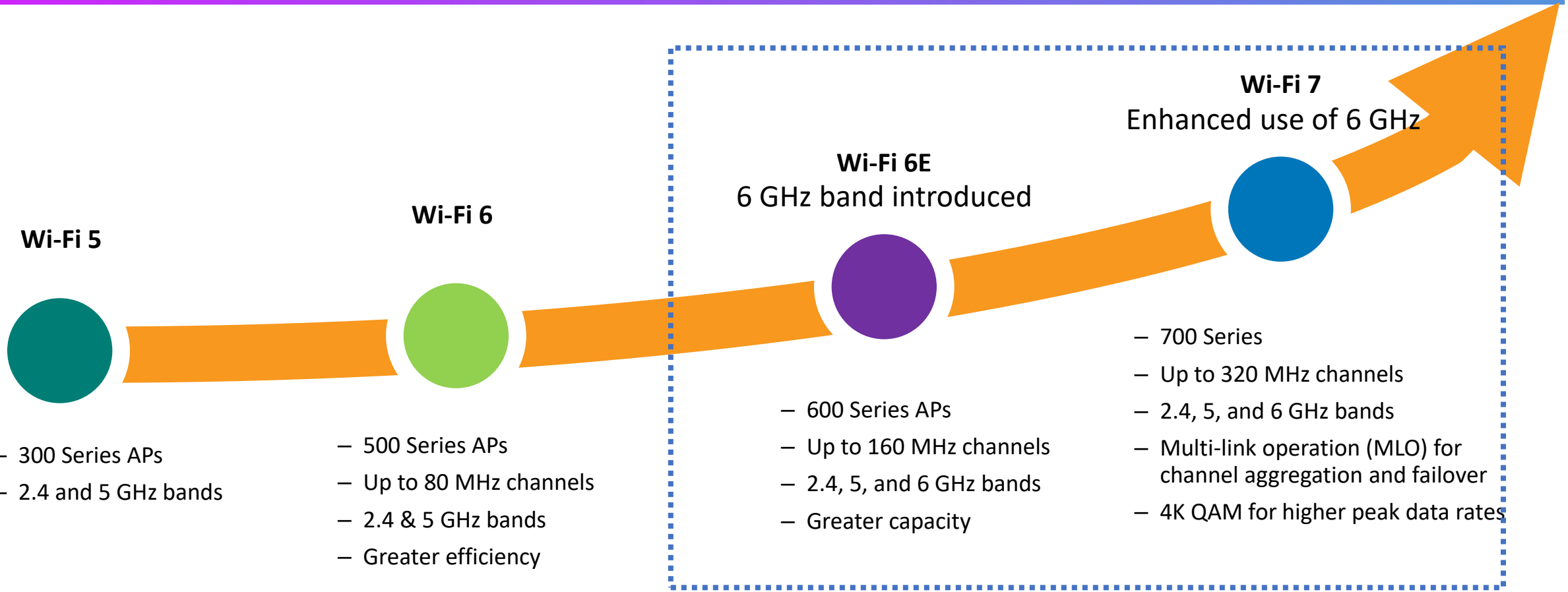


IOS

WIFI Foundation : WIFI7 Technology

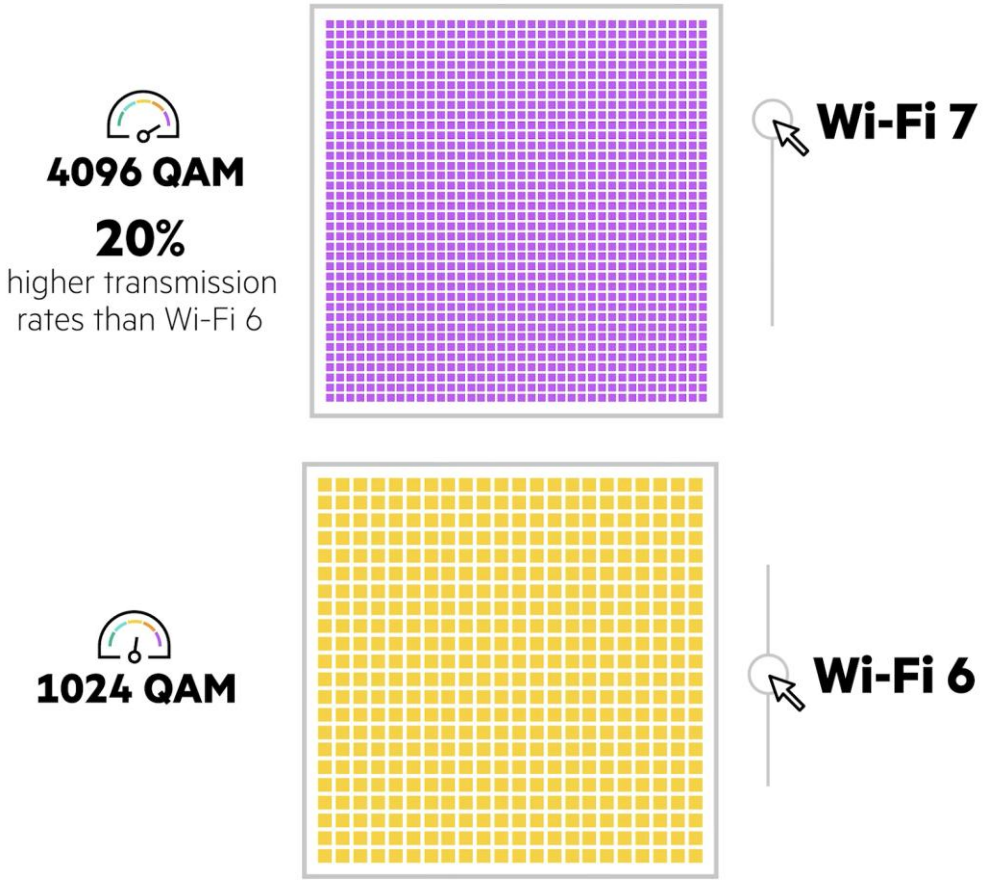
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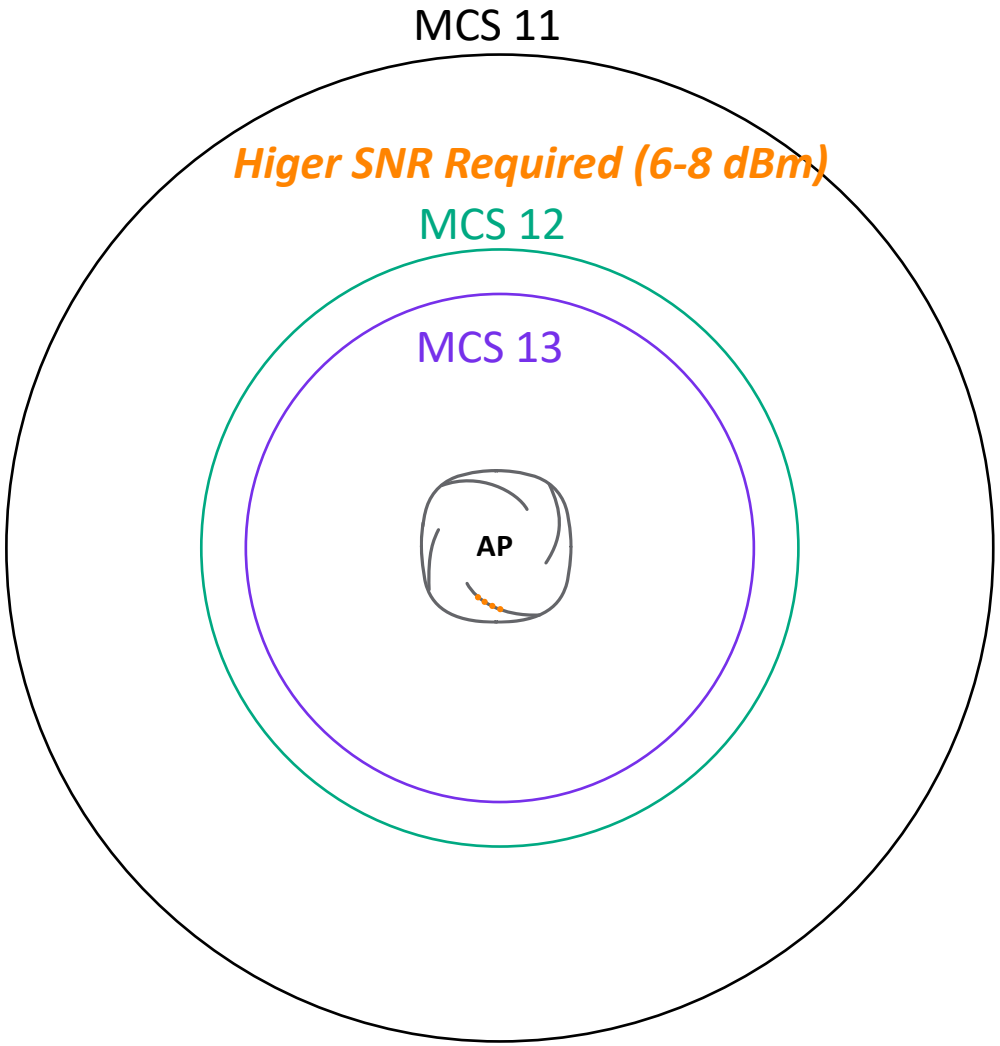
Evolution of Wireless Standards
Greater efficiency, capacity, and higher speeds



4096 QAM

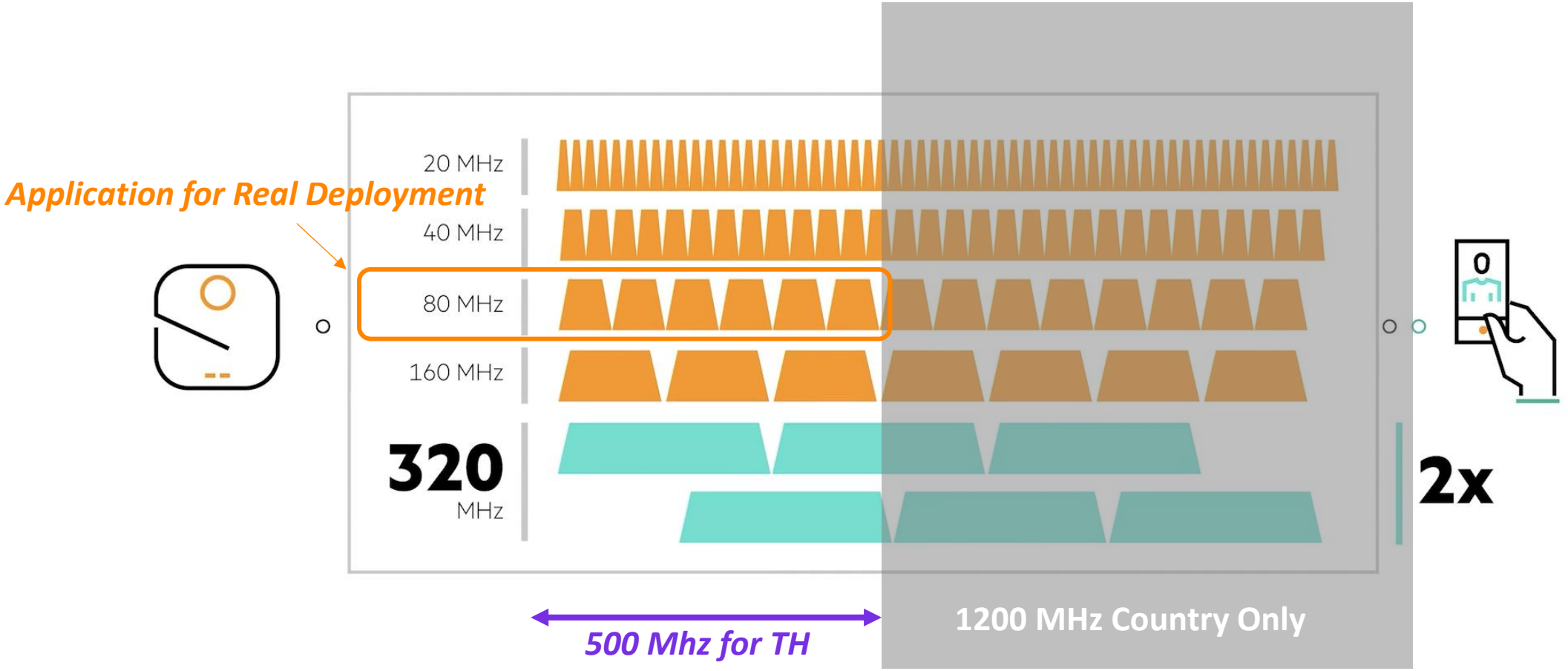
Functional or Not Functional





320 MHz Channel Bonding

Functional or Not Functional

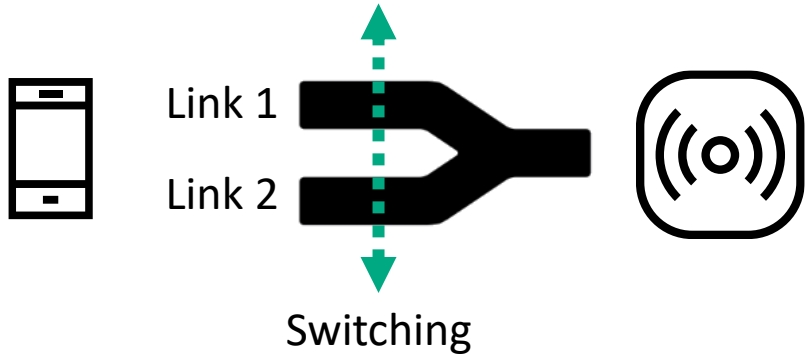


MLO : Multi Link Operation

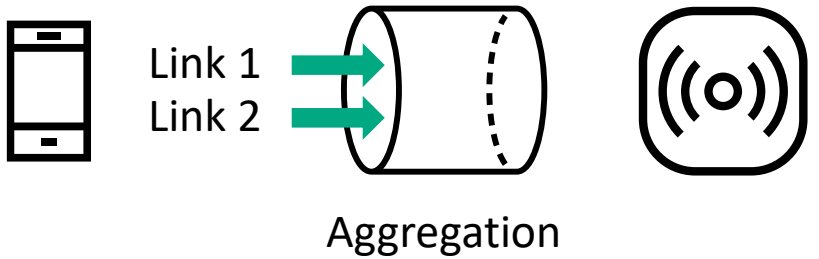
Multi-Link Operation (MLO) is a key marketed Wi-Fi 7 feature bringing improvements in throughput, latency, and reliability offering the client more control over link/band selection.

Two primary types:

1. Alternating Multi-Link Operation dynamically switches traffic between links to avoid wireless interference.
   Aims to provide deterministic and lower latency even in congested environments.



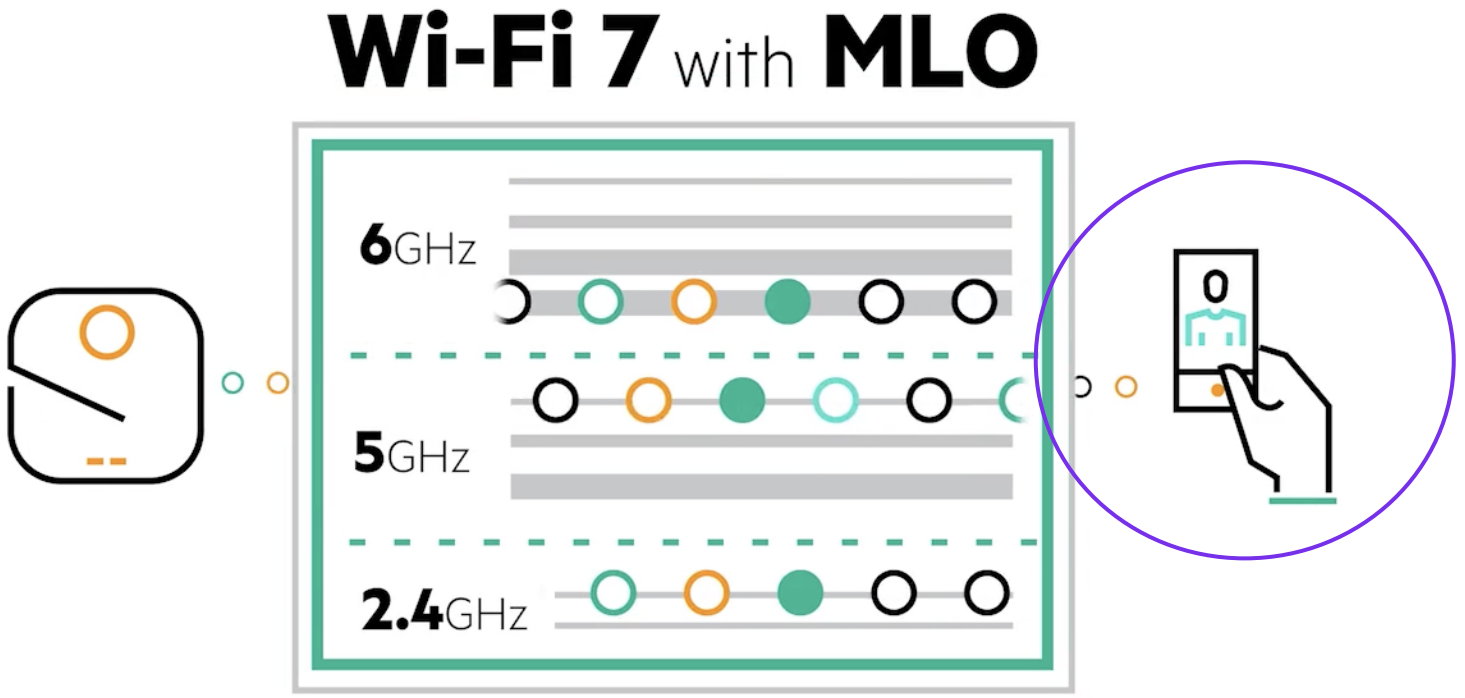
2. Concurrent Multi-Link Operation allows traffic to be aggregated across multiple bands.
   Enabling higher peak throughput and lower latency.



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Multi-Link Operation (MLO)

Functional or Not Functional



- Single Radio / Triband
- Battery Concerned
- 5G/2.4G Congestion

Multi-Link Operation (MLO) – Device Types

- Single-radio (SR) clients are limited to link switching and do not support concurrent Tx/Rx (STR)
  - Fast failover between links/bands
- Multi-Radio (MR) clients can support concurrent Tx/Rx (STR)
  - Aggregate performance from multiple links

Number of radios	Concurrent data transmission on multiple links	Concurrent data reception on multiple links	Transmit on link(s) and receive on other link(s) simultaneously	Simultaneous carrier sense on multiple links	Spatial stream switching capability between links	Device type
1 (Single Radio (SR))	NO			NO	NO	Multi-Link Single Radio (MLSR)
				YES	YES (= Enhanced)	**Enhanced Multi-Link Single Radio (EMLSR)**
>=2 (Multi Radio(MR))	YES		NO (= Non-Simultaneous Transmit Receive (NSTR))	YES*	NO	Non-Simultaneous Transmit Receive Multi-Link Multi-Radio (NSTR MLMR)
				YES*	YES (= Enhanced)	NSTR EMLMR
			YES (= Simultaneous Transmit Receive (STR))	YES	NO	**Simultaneous Transmit Receive Multi-Link Multi-Radio (STR MLMR)**
				YES	YES (= Enhanced)	STR EMLMR

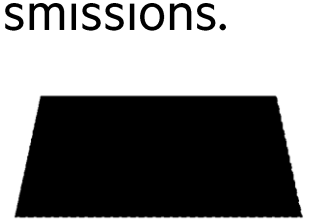
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Flexible Channel Utilization with Puncturing and Multiple RUs

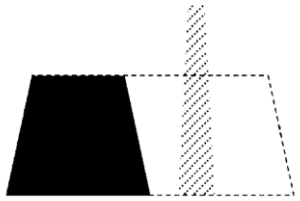
- **Puncturing** enables use of wider channels in the face of interference.
- Puncturing isolates the impacted portion of the bonded channel set.
- Resolution of 20 MHz increments.
- **Multi Resource Unit (MRU)** builds on the Wi-Fi 6 Resource Unit (RU) and allows of multiple non-contiguous RUs.
- MRU is what allows the remainder of the “punctured” channel to be utilized for data transmissions.

Static Puncturing Example
Client: Google Pixel 8 (Android 14)





Desired bandwidth

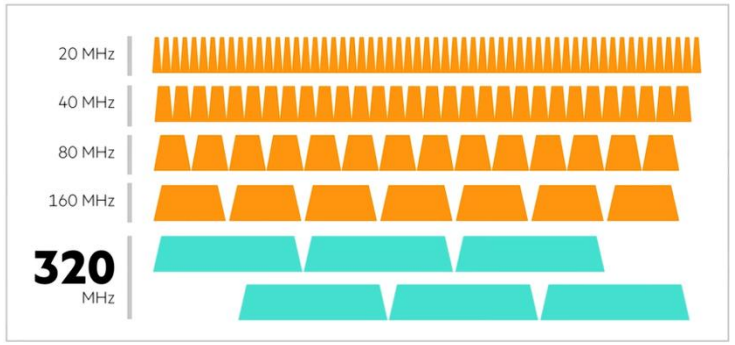


Interference impact

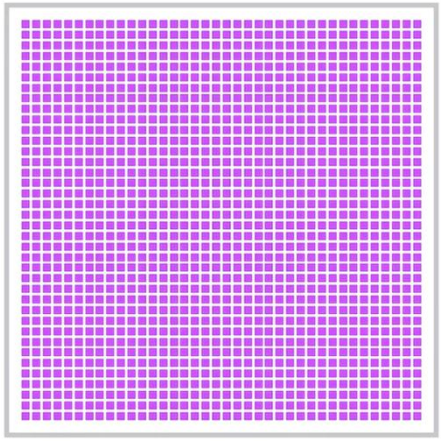


Puncturing + MRU

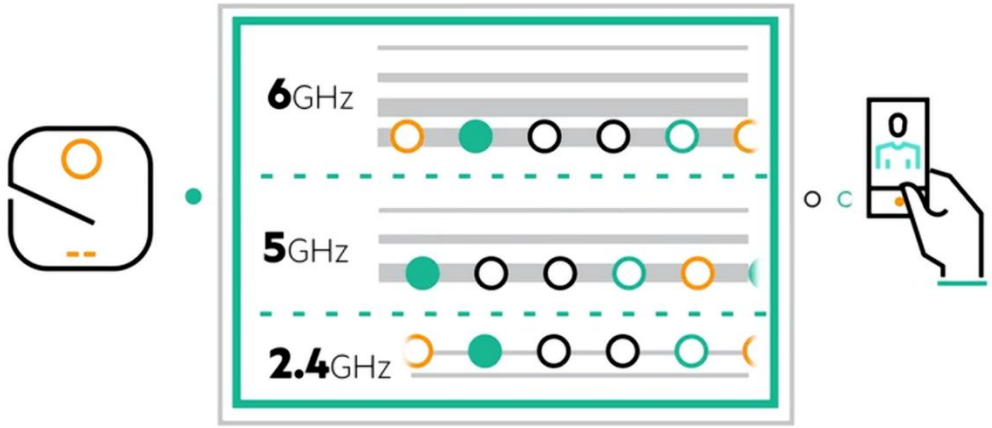
Wi-Fi6E vs Wi-Fi 7: 4 Major Difference



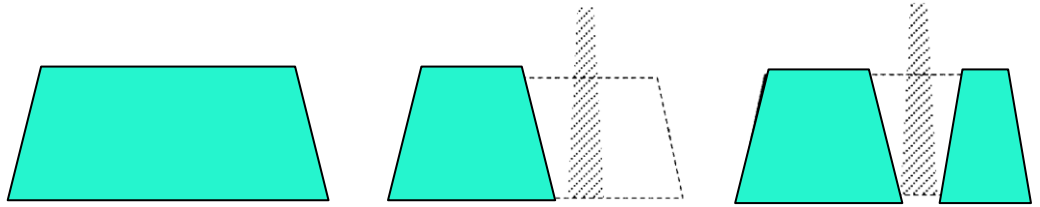
320 MHz channel



4K-QAM Spectrum



Multi-Link Operation (MLO)



Spectrum Punctuting

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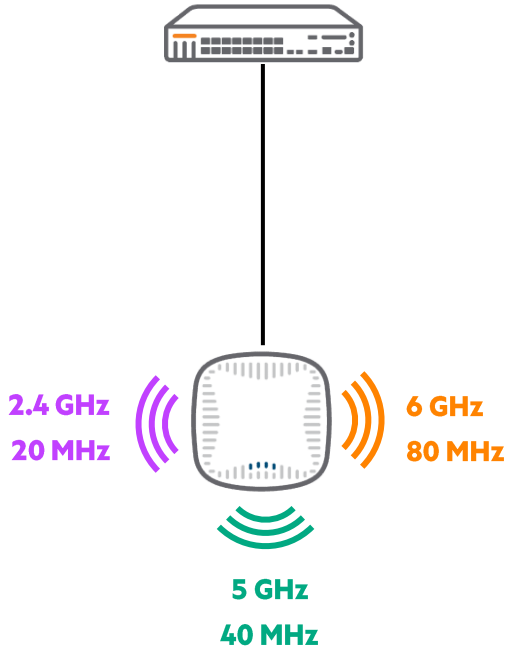
WIFI7 – 46 Gbps vs WIFI6E 9.6 Gbps

	WIFI6E Ideal 160Mhz 3 Non Overlapped Channel	WIFI6E Real 80Mhz 6 Non Overlapped Channel	WIFI7 Ideal 320Mhz 3 Non Overlapped Channel	WIFI7 Real 80 Mhz 6 Non Overlapped Channel
1 Spartial Stream	1.2Gbps	600 Mbps	2.88 Gbps	720 Mbps
2 Spartial Steam	2.4 Gbps	**1.2 Gbps**	5.76 Gbps	**1.44 Gbps**
8 Spartial Stream	***9.6Gbps***	4.8 Gbps	23 Gbps	5.76 Gbps
***** 16 Spartial Stream***	-	-	***46 Gbps***	11.5 Gbps

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Dimensioning the Enterprise Edge for Wi-Fi 6E & WIFI 7

Spatial Streams	20 MHz
1SS	143/172 Mbps
2SS	**286/344 Mbps**
3SS	430/516 Mbps
4SS	573/688 Mbps



Spatial Streams	80 MHz	160 MHz
1SS	600/720 Mbps	1.2/1.44 Gbps
2SS	**1.2/1.44 Gbps**	**2.4/2.88 Gbps**
3SS	1.8/2.16 Gbps	3.6/4.32 Gbps
4SS	2.4/2.88 Gbps	4.8/5.76 Gbps

Spatial Streams	40 MHz	80 MHz
1SS	286/344 Mbps	600/720 Mbps
2SS	**573/688 Mbps**	**1.2/1.44 Gbps**
3SS	860/1032 Mbps	1.8/2.16 Gbps
4SS	1.14/1.37 Gbps	2.4/2.88 Gbps

ถ้าเพิ่ม Spatial Stream
หรือเพิ่ม Channel Width
ทำให้ Physical Rate จะเพิ่มขึ้น

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How to archive a “Good WIFI Experience” ?

HPE aruba networking

How to archive a Good WIFI Experience ?

- Everyone Connect @ highest WIFI Data Rate possible

  >> Small Cell Designed

  Why ?

  >> WIFI is working like a HUB

  >> One low speed client will make all client slow

  Current Practice

  >> Deploy AP to AP distance ~ 8-12 meter

How to archive a Good WIFI Experience ?

- Minimize Co-Channel Interference

  >> Channel Planning, Channel Bonding

  Why ?

  >> Airtime is the key resources on WIFI

  >> Worst impact of WIFI come from CCI

  Current Practice

  >> Using AI to calculate and assign the best Channel, Power, Channel Width

How to archive a Good WIFI Experience ?

- Push Client to 5Ghz / 6GHz

  >> Feature on each product : Best Practice Deployment

  Why ?

  >> 2.4G band has very high CCI due to limited 3 Channels

  >> a lot of Interference from Bluetooth, Microwave Oven, etc.

How to archive a Good WIFI Experience ?

- Upgrading Client WIFI Driver / Firmware to the latest version

  Why ?

  >> Compatible concern and Bug fixed

Design Best Practices

- **Capacity design**, not just Coverage design
- 5 GHz and 6 GHz bands
- 6 GHz can replace APs 1:1 for previous 5 GHz deployments
- Use site-survey tools to plan/verify
- Wi-Fi 6/6E/7target a cell size of -55 dBm to -65 dBm
- ***Minimize Co-Channel Interference***



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